Seminar : $e^+e^- \longrightarrow 3j$ (3-jet events), app. 3.5.

po 23.4.

Notacija :

$x^\mu = (x^0, \vec{x})$

$x_\mu = (x_0, -\vec{x})$; $x_0 = ct \equiv t$ $(c=1)$ $\boxed{x_\mu = g_{\mu\nu} x^\nu}$

$x^2 = x^\mu x_\mu = g_{\alpha\beta} x^\alpha x^\beta = \delta_\alpha{}^\beta x^\alpha x_\beta$

$g_{\alpha\beta} = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$, $g_{\mu\nu} g^{\nu\lambda} = \delta_\mu{}^\lambda$

$\frac{\partial}{\partial x^\mu} = \partial_\mu = (\partial_0, \vec{\nabla})$ $\quad p^\mu = (i\frac{\hbar}{c}\frac{\partial}{\partial t}, -i\hbar\vec{\nabla}) = (E, p)$

$\frac{\partial}{\partial x_\mu} = \partial^\mu = (\partial^0, -\vec{\nabla})$ $\quad p_\mu = (E, -\vec{p})$

Enote : $\hbar = 6{,}6 \cdot 10^{-22}$ MeV s

$c = 3 \cdot 10^{23}$ fm/s

$\hbar c = 200$ MeV fm

$\hbar, c = 1 \Rightarrow [E] = [t^{-1}]$

$\tau = \frac{\hbar}{\Gamma} = \frac{1}{\Gamma}$

Planckova skala :

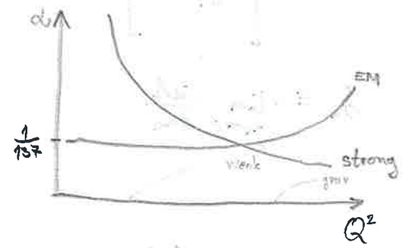
$G_{Newton} = 6{,}7 \cdot 10^{-39}$ GeV^{-2}

$L_{Plank} \approx \sqrt{G_{Newton}} = 1{,}6 \cdot 10^{-20}$ fm

$E_{Plank} \sim \frac{1}{\sqrt{G_N}} \approx 1{,}25 \times 10^{19}$ GeV

SKLOPITVENA KONSTANTA, RENORMALIZACIJSKA GRUPA

$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137}$



POINCARÉJEVA SIMETRIJA

Poglejmo transformacije četverca

$$x'^\mu = \underbrace{\Lambda^\mu{}_\nu x^\nu}_{\text{Homogena Lorentz. grupa, ohranja četvercov kvadrate } x'_\mu x'^\mu = x_\nu x^\nu} + a^\mu \quad \leftarrow (a^0, \vec{a})\ 4 \text{ param} \leftrightarrow 4 \text{ gener.}$$

$$= g_{\mu\nu} x'^\mu x'^\nu = g_{\alpha\beta} x^\alpha x^\beta = g_{\mu\nu}\Lambda^\mu{}_\alpha x^\alpha \Lambda^\nu{}_\beta x^\beta = g_{\alpha\beta} x^\alpha x^\beta \quad \Rightarrow \quad g_{\mu\nu}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta = g_{\alpha\beta}$$

INFINITEZIMALNE TRANSFORMACIJE:

$$\Lambda^\mu{}_\alpha = \delta^\mu{}_\alpha + \omega^\mu{}_\alpha$$

$$g_{\mu\nu}\cdot(\delta + \omega)(\delta^\nu{}_\beta + \omega^\nu{}_\beta) = g_{\alpha\beta}$$

$$(g_{\nu\alpha} + \omega_{\alpha\nu})(\delta^\nu{}_\beta + \omega^\nu{}_\beta) = g_{\alpha\beta}$$

$$g_{\beta\alpha} + \underbrace{\omega_{\beta\alpha} + \omega_{\alpha\beta}}_{0} + \underbrace{\omega_{\nu\alpha}\omega^\nu{}_\beta}_{0} = g_{\alpha\beta}$$

$\omega_{\alpha\beta} = -\omega_{\beta\alpha} \Rightarrow$ 4×4 matrika, realna; 6 parametrov. antisim.

i) ROTACIJE (3)

okoli z osi za kot $d\varphi_z$

$$\Lambda^\mu{}_\nu = \begin{bmatrix} 1 & & & \\ & \cos d\varphi_z & -\sin d\varphi_z & \\ & \sin d\varphi_z & \cos d\varphi_z & \\ & & & 1 \end{bmatrix} \xrightarrow{\text{inf}} \omega^\mu{}_\nu = d\varphi_z \begin{bmatrix} & & & \\ & & 1 & \\ & -1 & & \\ & & & \end{bmatrix}$$

$$\Lambda'^\mu{}_\nu \xrightarrow{\text{inf}} 1 + i\,d\varphi_z (J_z)^\mu{}_\nu \qquad J_z \equiv -\lim_{d\varphi_z\to 0}\frac{\partial\Lambda}{\partial d\varphi_z}$$

$$(J_z)^\mu{}_\nu = -i\begin{bmatrix} & & & \\ & -1 & & \\ & & 1 & \\ & & & \end{bmatrix}$$

$$(J_x)^\mu{}_\nu = -i\begin{bmatrix} & & & \\ & & & \\ & & -1 & \\ & & & 1 \end{bmatrix}$$

$$(J_y)^\mu{}_\nu = -i\begin{bmatrix} & & & \\ & & -1 & \\ & 1 & & \\ & & & \end{bmatrix}$$

$$\Lambda^\mu{}_\nu = e^{i\vec{\alpha}\cdot\vec{J}}$$

ii) POTISK (3)

x smeri: $(\Lambda_x)^\mu{}_\nu = \begin{bmatrix} \cosh\xi_x & -\sinh\xi_x & & \\ -\sinh\xi_x & \cosh\xi_x & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$; $\beta_x = \frac{v_x}{c} = \tanh\xi_x$; $\gamma_x = (1-\beta_x^2)^{-\frac{1}{2}} = \cosh\xi_x$

Preverimo: $t' = x^{0'} = \cosh\xi_x\,(x^0 - x'\tanh\xi_x) = \gamma_x(x^0 - v_x x') = \gamma(t - vx)$

$x' = x^{1'} = \cosh\xi_x(-x^0\tanh\xi_x + x^1) = \gamma_x(-v_x x^0 + x^1) = \gamma(x - vt)$

$y' = x^{2'} = x^2 = y$

$z' = x^{3'} = x^3 = z$

$$K^x \equiv +i\frac{\partial\Lambda}{\partial\xi_x}: \quad K^1 = K^x = -i\begin{bmatrix} & 1 & & \\ 1 & & & \\ & & & \\ & & & \end{bmatrix}$$

$$K^2 = K^y = -i\begin{bmatrix} & & 1 & \\ & & & \\ 1 & & & \\ & & & \end{bmatrix}$$

$$K^3 = K^z = -i\begin{bmatrix} & & & \\ & & 1 & \\ & 1 & & \\ & & & \end{bmatrix}$$

LORENTZOVA ALGEBRA $SO(1,3)$

$$\begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$[J^i, J^j] = i\varepsilon^{ijk}J^k \quad ; \quad SO(3) \cong SU(2) \Rightarrow \underset{1 \text{ Casimirjev}}{\overset{\text{rank } 1}{1 \text{ cartanov}}}$$

$$[J^i, K^j] = i\varepsilon^{ijk}K^k \quad ; \quad [K^i, K^j] = i\varepsilon^{ijk}J^k$$

K^i tvorijo $j=1$ = spin 1 adjung. up. $SO(3)$

$J^2 = J^{1^2} + J^{2^2} + J^{3^2}$, $\langle J^2\rangle = j(j+1)$

$\Rightarrow$ potiski (K^i) ne tvorijo zaključene podalgebre! $\Rightarrow$ Nerazcepna in nehermitska podalgebra

Poiščemo novo bazo: $J^i_\pm = \frac{1}{2}(J^i \pm iK^i)$

2 ločene $SU(2)$ algebre $\rightarrow$ $[J^i_\pm, J^j_\pm] = i\varepsilon^{ijk}J^k_\pm$, $[J^i_\pm, J^j_\mp] = 0$

J_+ J_- dynkinovi indeksi (j_1, j_2) $\Rightarrow$ nova baza

$SO(1,3) \simeq SL(2,\mathbb{R}) \times SL(2,\mathbb{R}) \simeq SL(2,\mathbb{C})$

$\Rightarrow$ $SO(1,3)$ polenostavne lahko upodobitve tvorim kr preko $SU(2)$ upodobitev. To je deloma res - lahko klasificiram preko dynkinovih indexov (označujejo največ večkratnike fund. upodobitev, ki so največje uteži v dani upodobitvi). Št. fund. upod = rang, $\text{rang}(SU(2)) = 1 \Rightarrow$ vsak $SU(2)$ ma 1 dynkinov index, 1 fund. up. in največje uteži v poljubni nerazcepni upodobitvi bodo večkratniki največje fund. uteži. Na nivoju algebre & klasifikacije $SO(1,3) \simeq SU(2)\times SU(2)$. Kar se tiče lastnosti upodobitev pa NE. Ker so upod. nekompaktne: $SU(2) \to SL(2)$. Nerazcepne upodobitve $SL(2,\mathbb{C})$ pa niso unitarne, tako kot pri Lorentzovih.

Lorentzov tenzor:

$$M^{\mu\nu} \neq J^i = \varepsilon^{ijk} M^{jk}, \qquad K^i = M^{0i}, \quad M^{\mu\nu} = -M^{\nu\mu}$$

splošna Lor. transf. → $\Lambda = e^{\frac{1}{2}\omega^{\mu\nu} M_{\mu\nu}}$; $\alpha^i = \varepsilon^{ijk}\omega^{jk}$, $\xi^i = \omega^{0i}$

upodobitev četvercev: $(\frac{1}{2}, \frac{1}{2})$, v tej upodobitvi smo matrike že razpisali

dinkinovi (J_+, J_-)

Nad prostorom funkcij četvercev koordinat

$M^{\mu\nu} = i(x^\mu \partial^\nu - x^\nu \partial^\mu) \equiv L^{\mu\nu}$ „posplošitev vrtilne" M je splošno, L je nad tem prostorom, poseben primer

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\nu\rho} M^{\mu\sigma} - g^{\mu\rho} M^{\nu\sigma} - g^{\nu\sigma} M^{\mu\rho} + g^{\mu\sigma} M^{\nu\rho})$$

Druge pomembne upodobitve:

$(0,0)$ ... trivialna upodobitev

$(\frac{1}{2},0), (0,\frac{1}{2})$ .... spinorske (Weylova) spinorji (brezmasni)

$(\frac{1}{2},0)$ levo ročni, $(0,\frac{1}{2})$ desno ročni

$(\frac{1}{2},0) \oplus (0,\frac{1}{2})$ .... spinorski (Diracovi), masni

Dodamo zrcaljenje prostora in časa

$$P_\mu{}^\nu = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \qquad T_\mu{}^\nu = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

to ni metrični tenzor, numerično pa ravno

Vrtilna količina: $P\vec{J}P^{-1} = \vec{J}$, J = pseudovektor/axialni

Potiski: $P\vec{K}P^{-1} = -\vec{K}$

$\Rightarrow P\vec{J}_\pm P^{-1} = \vec{J}_\mp$,

$(j_1, j_2) \xrightarrow{P} (j_2, j_1)$: upodobitve ki niso simetrične na dynkinove indexe bodo krsle parnost (maximalno)

Gremo pogledat v prostor spinorjev

$M_{\mu\nu} = \frac{1}{2}\sigma_{\mu\nu}$

$\sigma_{\mu\nu} = \frac{1}{2}[\gamma_\mu, \gamma_\nu]$, $\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}$

WEYLOVA BAZA:

$$\gamma_0 = \begin{bmatrix} & I_{2\times2} \\ I_{2\times2} & \end{bmatrix}, \qquad \gamma_i = \begin{bmatrix} & -\sigma_i \\ \sigma_i & \end{bmatrix}$$

$$J^i = \begin{bmatrix} \frac{1}{2}\sigma^i & \\ & \frac{1}{2}\sigma^i \end{bmatrix}, \qquad K^i = \begin{bmatrix} \frac{i}{2}\sigma^i & \\ & -\frac{i}{2}\sigma^i \end{bmatrix}$$

$(\frac{1}{2},0)$: $\dot{\alpha}$ $\dot{\alpha}$; $(0,\frac{1}{2})$: α α

$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \frac{1}{4!}\varepsilon^{\mu\nu\rho\sigma}\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma$

$\gamma_5^\dagger = \gamma_5$, $\gamma_5^2 = I$, $\{\gamma_5, \gamma_\mu\} = 0$

$$\gamma^5 = \begin{bmatrix} -I & \\ & I \end{bmatrix}$$

$P_{R,L} = \frac{1}{2}(1 \pm \gamma_5^*)$, $P_R^2 = P_R$, $P_L^2 = P_L$, $P_R P_L = 0$, $P_R + P_L = 1$

P_R : proj. na $(\frac{1}{2},0)$

P_L : proj. na $(0,\frac{1}{2})$

$\gamma_0 \vec{J} \gamma_0 = \vec{J}$

$\gamma_0 \vec{K} \gamma_0 = -\vec{K}$, $P^\mu{}_\nu = \gamma_0$

iii) TRANSLACIJE $x'^\mu = x^\mu + a^\mu$

Upodobitev nad prostorom funkcij

$\phi'(x) = \phi(x-a) \overset{\text{lim.}}{\Rightarrow} \phi(x) - a^\mu \underbrace{\partial_\mu \phi}_{-iP_\mu \text{ generator gibanje}} = (1 + ia^\mu P_\mu)\phi$

$\phi'(x) = e^{ia_\mu P^\mu}\phi(x)$

$[P^\mu, P^\nu] = 0$

$[M^{\mu\nu}, P^\rho] = -i g^{\mu\rho} P^\nu + i g^{\nu\rho} P^\mu$

↳ $J^i = \varepsilon^{ijk} M^{jk}$, $K^i = M^{0i}$:

$[P_0, J_i] = 0$

$[P_i, J_j] = i\varepsilon_{ijk} P_k$ .... vektor pod rotacijami

$[P_0, K_i] =$

$[P_i, K_j] =$

UPODOBITVE POINCAREJEVE

→ CASIMIRJEVI OPERATORJI

$\langle p_\mu p^\mu \rangle \equiv m^2$

$W^\lambda \equiv \frac{1}{2} \varepsilon^{\mu\nu\sigma\tau} p_\nu M_{\sigma\tau}$; $W_\mu W^\mu = W^2$ je inv. (Pauli-Lubanski)

V mirovnem sistemu: $p_\mu = (m, \vec{0})$

$\langle W^2 \rangle = m \cdot j(j+1)$

Pauli-Lubanski bolj podrobno:

$W^0 = \frac{1}{2} \varepsilon^{0ijk} p_i M_{jk}$

$= \vec{p} \cdot \vec{J}$

$\lambda = \frac{\vec{p} \cdot \vec{J}}{|\vec{p}|}$ „kiralnost"

Stanje znotraj upodobitve: $|m, j, \vec{p}, \lambda\rangle$ ($|m^2, j, p, \lambda\rangle$; Casimirji)

To nam definira upodobitev Poinc. grupe.

Zahteva: $\underline{\underline{m \neq 0}}$.

→ BREZMASNE UPODOBITVE $\langle p^2 \rangle = 0$

a) $\langle W^2 \rangle = 0 \implies$ edina možnost $W^\mu = \lambda p^\mu$, $W^2 \propto p^2$ in $\frac{W^2}{p^2} = \lambda^2$

b) $\langle W^2 \rangle \neq 0 \implies$ upodobitve z

⇓

stanja znotraj $\pm\sqrt{\lambda^2}$

Vse up. so max 2

Primer: $A'^\alpha(x') = \Lambda^\alpha{}_\beta A^\beta(x)$. Kateri upodobitvi $A^\alpha(x)$ pripada in kaj je Λ?

$A'^\alpha(x') = \Lambda^\alpha{}_\beta A^\beta(\Lambda x) \underset{\text{inf.}}{\approx} (\delta^\alpha{}_\beta + \omega^\alpha{}_\beta)\, A^\beta((\delta^\sigma{}_\rho + \omega^\sigma{}_\rho) x^\rho)$

$= (\delta^\alpha{}_\beta + \omega^\alpha{}_\beta)\left[A^\beta(x) + \omega^\mu{}_\nu x^\nu \partial_\mu A^\beta(x)\right]$

$= \left[\delta^\alpha{}_\beta - i\frac{1}{2}\omega^{\mu\nu}(\Sigma_{\mu\nu})^\alpha{}_\beta\right]\left[A^\beta(x) - \frac{i}{2}\omega^{\mu\nu} L_{\mu\nu} A^\beta(x)\right]$

$= A^\alpha(x) - \frac{i}{2}\omega^{\mu\nu}\underbrace{\left(L_{\mu\nu} + \Sigma_{\mu\nu}\right)^\alpha{}_\beta}_{(M_{\mu\nu})^\alpha{}_\beta} A^\beta(x)$

$(\Sigma_{\mu\nu})^\alpha{}_\beta = i\left(\delta^\alpha{}_\mu g_{\nu\beta} - \delta^\alpha{}_\nu g_{\mu\beta}\right)$

$J^i = i\varepsilon^{ijk} M_{jk} = L^i + S^i$ ← $(S^i)^a{}_b = -i\varepsilon^{iab}$ ⇒ $A^\mu(x)$ troni spin = 1, adj. upodobitev.

TEOREM COLEMAN - MANDELA

Velja za lokalne teorije, ki ubogajo Lorenzovo simetrijo.

Trditev: Lijeva algebra generatorjev simetrij lahko sestoji le iz poincarejeve algebre ⊕ generatorjev notranjih simetrij.

Posledica: Unifikacija spina in ohranjenih nabojev ni možna.

Izjema: To so supersimetrične algebre, kjer bozonskim generatorjem (komutatorji) dodamo fermionske (antikomutatorje). To pomeni, da Lorentzovo algebro razširimo s $\hat{Q}_{ar}$ v $(0, \frac{1}{2})$ in Q^*_{ar} v $(\frac{1}{2}, 0)$. Potem

$$\{Q_{ar}, Q^*_{bs}\} = 2\sigma^\mu_{ab} \delta_{rs} P_\mu$$

$\uparrow$

$(I, \vec{\sigma})$

POVEZAVA SIMETRIJ IN OHRANITVENIH.

Noetherjin teorem: $\delta S = 0$

Klasično: $S = \int_{t_1}^{t_2} L(q, \dot{q})\, dt$

Kvantno: $S = \int\int d^3x\, dt\, \mathcal{L}(\phi(x), \partial_\mu \phi(x))$

$\delta S = \int \delta\mathcal{L}\, d^4x + \underset{\text{integracij}}{\text{variacije}} = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) + \partial_\mu \mathcal{L}\, \delta x^\mu\right]$

$d^4x' = \left|\frac{\partial x'}{\partial x}\right| d^4x$ (= 1)

$\delta\phi = \phi'(x) - \phi(x)$

$\delta\partial_\mu\phi = \partial_\mu\phi' - \partial_\mu\phi = \partial_\mu(\delta\phi)$

$= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial\phi}\delta\phi + \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right] - (\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)})\delta\phi\right]$

$\int d\Sigma_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi \to 0$

$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial\phi} - \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right] = 0}$ E-L enačbe.

1) Invariance akcije na translacijo

i) $x' = x + a \longrightarrow \partial x_\mu = a_\mu$

$\phi'(x) = \phi(x-a) \simeq \phi(x) - \underbrace{a^\alpha \partial_\alpha \phi}_{\delta\phi}$

$\delta S = 0$:

$$\delta S = \int d^4x \left[\delta \mathcal{L} + \partial_\mu \mathcal{L}\, \delta x^\mu \right] = 0$$

$$\underbrace{\frac{\partial \mathcal{L}}{\partial \phi}}_{\partial_\mu\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right]} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \underbrace{\delta(\partial_\mu\phi)}_{\partial_\mu \delta\phi} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} (-a^\alpha \partial_\alpha \phi) \right)$$

$$= \int d^4x \left[a_\beta \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \partial^\beta \phi - g^{\mu\beta} \mathcal{L} \right] \right] = 0$$

$$= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right] - \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) \right]$$

2) Invariance na Lorentzove transformacije

$$x'^\alpha = \Lambda^\alpha{}_\beta x^\beta = x^\alpha + \underbrace{\delta x^\alpha}_{\omega^\alpha{}_\beta x^\beta}$$

$\phi'(x) = \phi(x - \delta x)$

$$\delta\phi = -\delta x^\alpha \partial_\alpha \phi = -\omega^\alpha{}_\beta x^\beta \partial_\alpha \phi$$

$$\delta S = \int d^4x \left[\partial_\alpha \left(\frac{\partial \mathcal{L}}{\partial(\partial_\alpha\phi)} \cdot (-\omega^\sigma{}_\beta x^\beta \partial_\sigma \phi) \right) + \partial_\alpha (\omega^\alpha{}_\beta x^\beta \mathcal{L}) \right]$$

$(\partial_\mu \mathcal{L})\, \omega^\mu{}_\beta x^\beta = \partial_\alpha [\omega^\alpha{}_\beta x^\beta \mathcal{L}] - \omega^\alpha{}_\beta \underbrace{\partial_\alpha x^\beta}_{\delta_\alpha{}^\beta} \mathcal{L}$

$$\tfrac{1}{2} \omega^\sigma{}_\beta \partial_\alpha \left[\frac{\partial \mathcal{L}}{\partial(\partial_\alpha\phi)} x^\beta g_{\sigma\rho} \phi - \delta_\sigma^\alpha x^\beta \mathcal{L} \right] = 0$$

$$\partial_\alpha \underbrace{\left[T^\alpha_\sigma x_\beta - T^\alpha_\beta x_\sigma \right]}_{M^\alpha_{\sigma\beta}} = 0 \quad ; \quad \int M^0$$

3) Interne simetrije (na prostoru funkcij, ne koordinat)

$\phi'(x) = e^{i\varepsilon} \phi(x) \simeq (1 + i\varepsilon)\, \phi(x)$

$\delta\phi = i\varepsilon\phi$

$$\delta S = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) = i\varepsilon \partial_\mu \underbrace{\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \phi \right)}_{j^\mu}$$

j^μ ... ohranjen tok $\quad \partial_\mu j^\mu = 0$

$\int j^0 d^3x = Q$ naboj $\quad \partial_0 Q = 0$

Posplositev na (ne)abelske simetrije

$\phi_i'(x) = \left(e^{i\varepsilon_A F^A} \right)_{ij} \phi_j(x)$

$j^\mu_A = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} (T_A)_{ik} \phi_k$

QFT RECAP

SKALARNI DELCI

$\Phi'(x') = \phi(x)$; $\phi^* = \phi$ realno polje brez naboja

$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) \xrightarrow{\text{E-L. enačba}} \partial_\mu \partial^\mu \phi + m^2 \phi = 0$ Klein-Gordan

KANONIČNA KVANTIZACIJA

$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot\phi}$; $[\phi(x), \pi(x')] = i\,\delta(x-x')$

Uvedemo Hamiltonjan (Lastne vrednosti so

$$\mathcal{H} = \frac{\partial \mathcal{L}}{\partial \dot\phi} \dot\phi - \mathcal{L} = \frac{1}{2}\left[\pi^2 + (\nabla\phi)^2 + m^2\phi^2\right]$$

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3 2E_k} \left[a(\vec k) \cdot e^{-ikx} + a^\dagger(\vec k) \cdot e^{ikx} \right] = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E_k}} \left[a(k) f_k(x) + a^\dagger(k) f_k^*(x) \right]$$

$$f_k(x) = \frac{1}{\sqrt{(2\pi)^3 2E_k}} e^{-ikx}$$

Uvedemo odvod po

$$A(t) \overleftrightarrow{\partial_t} B(t) \equiv A(t)(\partial_t B(t)) - A(t) \overleftarrow{\partial_t} B(t)$$
$$= A(t)(\partial_t B(t)) - (\partial_t A(t)) B(t)$$

$$\int f_k^*(x) i \overleftrightarrow{\partial_0} f_{k'}(x)\, d^3x = \int \frac{d^3x}{(2\pi)^3 \sqrt{2E_k 2E_{k'}}} \left[e^{ikx} i(-ik'_0) - i(ik_0) e^{ikx} e^{-ik'x} \right]$$
$$= \delta(k-k')$$

($-ik'_0 = E_{k'}$)

$$\Rightarrow a(\vec k) = \int d^3x \sqrt{(2\pi)^3 2E_k}\, f_k^*(x) i \overleftrightarrow{\partial_0} \phi(x)$$
$$a^\dagger(\vec k) = \int d^3x \sqrt{(2\pi)^3 2E_k}\, f_k(x) \overleftrightarrow{\partial_0} f_k(x)$$

$$[a(k), a^\dagger(k')] = (2\pi)^3 \int d^3x \int d^3x' \sqrt{4 E_k E_{k'}}\, f_{k'}^*(t,\vec x) \overleftrightarrow{\partial_0} f_k(t,\vec x') \cdot [\phi(t,\vec x'), \pi(t,\vec x')]$$
$$\vdots$$
$$= (2\pi)^3 2E_k\, \delta(k-k')$$

$[a(k), a(k')] = 0$

$[a^\dagger(k), a^\dagger(k')] = 0$

$a^\dagger(k)|0\rangle = |\vec k\rangle$

(vakuum; enodelčno stanje; $a(k)|0\rangle = 0 \quad \forall k$)

$\langle k|k'\rangle = \langle 0|a(\vec k) a(\vec k')^\dagger|0\rangle = (2\pi)^3 2E_k\, \delta(k-k')$

($\langle x| = (|x\rangle)^\dagger$)

OPERATOR ŠTETJA

$N(\vec k) \equiv a^\dagger(\vec k) a(\vec k)$

$[N(k), a(k)] = -a(\vec k)$;

$[N(k), a^\dagger(k)] = a^\dagger(\vec k)$

$$[N(\vec{k}), N(\vec{k}')] = 0$$

$$N(\vec{k})\,|n(\vec{k}')\rangle = n(\vec{k})\,|n(\vec{k})\rangle$$

$$N(\vec{k})\,a^\dagger(\vec{k})\,|n(\vec{k}')\rangle = (1+n(\vec{k}))\,\underbrace{a^\dagger(\vec{k})\,|n(\vec{k})\rangle}_{|n(\vec{k})+1\rangle}$$

$$a(\vec{k})\,|0\rangle = 0$$

$$\langle 0|\phi(x)|\vec{k}\rangle = \langle 0|\int \frac{d^3k}{(2\pi)^3 2E_{k'}}\left[a(k')\,e^{-ik'x} + a^\dagger(k')\,e^{ik'x}\right]a^\dagger(\vec{k})|0\rangle$$
$$= \int \frac{d^3k}{(2\pi)^3 2E_{k'}}\,(2\pi)^3\,2E_{\vec{k}}\,\delta(k-k')\,e^{-ik'x} =$$
$$= e^{-ikx} \qquad E_{k'} = \sqrt{m^2+|\vec{k}'|^2}$$

Energija in gibalna

$$H = \int \frac{d^3k}{(2\pi)^3 2E_k}\,\frac{E_k}{2}\left(a^\dagger(\vec{k})a(\vec{k}) + a(\vec{k})a^\dagger(\vec{k})\right)$$
$$= \int \frac{d^3k}{(2\pi)^3 2E_k}\,E_k\left(N(\vec{k})+\tfrac{1}{2}\right) \quad ; \quad \langle 0|H|0\rangle = \infty$$

$$\vec{P} = \int \frac{d^3k}{(2\pi)^3 2E_k}\,\vec{k}\left(N(\vec{k})+\tfrac{1}{2}\right) \quad ; \quad \langle 0|\vec{P}|0\rangle = \infty$$

Vrejanje

$$:a^\dagger a: = a^\dagger a$$
$$:aa^\dagger: = a^\dagger a$$
$$:H: = \int \frac{d^3k}{(2\pi)^3 2E_k}\,E_k\,N(\vec{k}) \quad ; \quad \langle 0|:H:|0\rangle = 0$$
$$:\vec{P}: = \int \frac{d^3k}{(2\pi)^3 2E_k}\,\vec{k}\,N(\vec{k}) \quad ; \quad \langle 0|:\vec{P}:|0\rangle = 0$$

Razcep na (+) in (-) Fourierove načine

$$\phi^{(+)}(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E_k}}\,a(k)\cdot g_k(x)$$
$$\phi^{(-)}(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E_k}}\,a^\dagger(k)\cdot g_k^*(x)$$
$$\phi(x) = \phi^{(+)}(x) + \phi^{(-)}(x)$$

KOMPLEKSNO POLJE

$$\phi^*(x) \neq \phi(x) \quad \Longrightarrow \quad \phi(x) = Re(\phi(x)) + i\,Im(\phi)$$

$$\mathcal{L} = \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi$$
$$\Downarrow$$
$$\partial_\mu\partial^\mu\phi + m^2\phi = 0$$
$$\partial_\mu\partial^\mu\phi^* + m^2\phi^* = 0$$

$$\phi(x) = \int \frac{d^3\vec{k}}{(2\pi)^3 2E_k}\left[\underbrace{a(k)\,e^{-ikx}}_{\phi^{(+)}} + \underbrace{b^\dagger(k)\,e^{ikx}}_{\phi^{(-)}}\right]$$
$$\phi^*(x) = \int \frac{d^3k}{(2\pi)^3 2E_k}\left[\underbrace{a^\dagger(k)\,e^{ikx}}_{\phi^{*(-)}} + \underbrace{b(k)\,e^{-ikx}}_{\phi^{*(+)}}\right]$$

$$[a(\vec{k}), a^\dagger(\vec{k}')] = (2\pi)^3 2E_k\,\delta(k-k')$$
$$[b(\vec{k}), b^\dagger(\vec{k}')] = (2\pi)^3 2E_k\,\delta(k-k')$$

$$a^\dagger(k)\,|0\rangle = |\vec{k}\rangle$$
$$b^\dagger(k)\,|0\rangle = |\vec{k}\rangle$$

$$\langle 0|\phi^{(-)}|\vec{k}\rangle = e^{-ikx} \qquad \langle k|\phi^{*(-)}|0\rangle = e^{ikx}$$
$$\langle 0|\phi^{*(+)}|\vec{k}\rangle = e^{ikx} \qquad \langle k|\phi^{(+)}|0\rangle = e^{-ikx}$$

SPINORSKO POLJE

$$\mathcal{L} = \underbrace{\bar\psi_A}\,(i\underbrace{\gamma^\mu\partial_\mu}_{\not\partial} - m)_{AB}\,\psi_B$$
$$(\psi^\dagger\gamma^0)_A$$

$$\psi'(x') = e^{i\frac{1}{2}\omega_{\mu\nu}\sigma^{\mu\nu}}\psi(x) \quad ; \quad x' = \Lambda x\,, \quad \sigma^{\mu\nu} = \tfrac{i}{2}[\gamma^\mu,\gamma^\nu]$$

$$\Rightarrow (i\not\partial - m)\,\psi(x) = 0$$
$$\psi^\dagger(i\gamma_\mu^\dagger\overleftarrow{\partial}^\mu + m) = 0 \quad \text{ali} \quad \bar\psi(i\overleftarrow{\not\partial} + m)$$

Invariante (po členih):

$$\bar\psi'\psi' = \bar\psi\psi$$
$$(\bar\psi\not\partial\psi)' = \psi\not\partial\psi$$
$$\hookrightarrow \partial_\mu \longrightarrow (\Lambda_\mu{}^\nu)\,\partial_\nu \quad \Rightarrow \quad \bar\psi'\gamma^\mu\psi' \longrightarrow \Lambda^\mu{}_\nu\,\bar\psi\gamma^\nu\psi$$

$$\Pi_A \equiv \frac{\partial\mathcal{L}}{\partial\dot\psi_A} = i\psi_A^\dagger$$
$$\psi_A(x) = \int \frac{d^3p}{(2\pi)^3 2E_p}\sum_{s=-\frac{1}{2}}^{+\frac{1}{2}}\left[a(\vec{p},s)\cdot u_A(\vec{p},s)e^{-ipx} + b^\dagger(\vec{p},s)\,v_A(\vec{p},s)\,e^{ipx}\right]$$
$$= \psi^{(+)} + \psi^{(-)}$$

$$\{b(p,s),\ b^+(p',s')\} = (2\pi)^3 2E_{\vec p}\,\delta_{ss'}\delta(\vec p-\vec p')$$
$$\{a(p,s),\ a^+(p',s')\} = (2\pi)^3 2E_{\vec p}\,\delta_{ss'}\delta(\vec p-\vec p')$$
$$\Rightarrow \{\psi_A(t,x),\ \psi_B(t,x')\} = \delta_{AB}\,\delta(x-x')$$

Energija in gibalna količina

$$H = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\vec p}}\, E_{\vec p}\,\left(a^+(p,s)a(p,s) - a(p,s)a(p,s)\right)$$
$$:H: = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_p}\, E_p\,\left(a^+(p,s)a(p,s) + a^+(p,s)a(p,s)\right)$$

$:a^+a: = a^+a$ $\quad :b^+b: = b^+b$
$:aa^+: = -a^+a$ $\quad :bb^+: = -b^+b$

Fermionsko število

$$:Q: = \int \frac{d^3k}{(2\pi)^3 2E_k} \sum_s \left[a^+(k,s)a(k,s) - b^+(k,s)b(k,s)\right]$$

$\bar u_A(p,s)\,u_A(p,s) = 2m\,\delta_{ss'}$; $\quad u_A^+(p,s)\,u_A(p,s') = 2E_{\vec p}\,\delta_{ss'}$
$\bar v_A(p,s)\,v_A(p,s') = -2m\,\delta_{ss'}$ $\quad v_A^+(p,s)\,v_A(p,s') = 2E_p\,\delta_{ss'}$

$$u(\vec p,s) = \sqrt{E_p+m}\begin{bmatrix} \chi_s \\ \frac{\vec\sigma\cdot\vec p}{E_p+m}\chi_s \end{bmatrix}$$
$$v(p,s) = \sqrt{E_p+m}\begin{bmatrix} \frac{\vec\sigma\cdot\vec p}{E_p+m}\chi_s \\ \chi_s \end{bmatrix}$$

$$\sum_s u_A(\vec p,s)\,\bar u_B(\vec p,s) = (\not p+m)_{AB}$$
$$\sum_s v_A(\vec p,s)\,\bar v_B(\vec p,s) = (\not p-m)_{AB}$$

$$a^+(p,s)|0\rangle = |p,s\rangle$$
$$\langle 0|\psi(x)|p,s\rangle = \langle 0|\psi^{(+)}(x)|p,s\rangle = \sum_{s'}\langle 0|\int \frac{d^3p'}{(2\pi)^3 2E_{p'}}\, a(p',s')\,e^{-ip'x}\,\bar u(p',s')\,a^+(p,s)|0\rangle$$
$$= e^{-ipx}\,u(p,s)$$

$$\langle p,s|\bar\psi|0\rangle = e^{ipx}\,\bar u(p,s)$$
$$\langle 0|\psi|\overline{p,s}\rangle = e^{-ipx}\,\bar v(p,s)$$
$$\langle \overline{p,s}|\psi|0\rangle = e^{ipx}\,v(p,s)$$

B-L

ELEKTROMAGNETNO POLJE

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}\ ; \qquad F^{\mu\nu}(x) = \partial^\mu A^\nu(x) - \partial^\nu A^\mu(x)$$

$A^\mu(x) = (\varphi(x), \vec A(x)) \Rightarrow \underbrace{\vec E, \vec B}$ ne tvorijo neskončne upod. Poin. grupe.

$$\frac{\partial\mathcal{L}}{\partial A_\mu} - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\nu A_\mu} = 0 \quad \text{oz.} \quad \partial_\mu F^{\mu\nu} = 0 \quad \text{ali} \quad \Box A^\nu - \partial^\nu A_\mu A^\mu = 0$$

Umeritvena invarianca

$$A'^\mu(x) = A^\mu(x) + \partial_\mu\Lambda(x)$$

Na praktičnem nivoju fiksiramo umeritev:

1) Lorentzova umeritev:
$$\partial_\mu A'^\mu = \partial_\mu A^\mu + \Box\Lambda(x) = 0$$

2) $\frac{\partial\Lambda}{\partial t} = -\varphi \Rightarrow \varphi' = 0$ oz $\vec\nabla\cdot\vec A' = 0$ „Coulombska um."

Kvantizacija : Coulombska umeritev

$$\pi^0 = \frac{\partial\mathcal{L}}{\partial\dot A_0} = 0$$
$$\pi^i = \frac{\partial\mathcal{L}}{\partial\dot A_i} = -\partial^0 A^i - \partial^i A^0 = -E^i$$
$$[A^i(t,\vec x),\ \pi^j(t,\vec x')] = i\,\delta_{ij}\,\delta(\vec x-\vec x')$$

$\rightarrow \vec\nabla\cdot\vec A = 0$:
$$\partial_i[A^i(t,\vec x),\ \pi^j(t,\vec x')] = 0 + \partial_i\,\delta(\vec x-\vec x')$$

Nekaj bolj regularnega:
nastavek $[A^i(t,\vec x),\ \pi^j(t,\vec x')] = i\int\frac{d^3k}{(2\pi)^3}\,e^{-ik(\vec x-\vec x')}\,\Delta_{ij}$
dobimo pogoj $\Delta_{ij}k^j = 0 \Rightarrow \Delta_{ij} = \delta_{ij} - \frac{k_ik_j}{|\vec k|^2}$

$$A_\mu(x) = \int\frac{d^3k}{(2\pi)^3 2\varepsilon_0}\sum_{\lambda=0}^{3}\varepsilon_\mu(\lambda,\vec k)\left[a(\lambda,\vec k)e^{-ikx} + a^+(\lambda,\vec k)e^{ikx}\right]$$

upoštevamo $\vec\nabla\cdot\vec A = 0 \Rightarrow \vec k\cdot\vec\varepsilon(\lambda,k) = 0$
časovna komp.
$A^0 = 0 \Rightarrow \varepsilon^0(\lambda,\vec k) = 0$
in $\vec\varepsilon\perp\vec k$ ← longit. komp. = 0.

Ortogonalnost:
$\varepsilon_\mu(\lambda,\vec k)\,\varepsilon^\mu(\lambda',\vec k) = \delta_{\lambda\lambda'}$ } imamo le dva
$\vec\varepsilon(\lambda,\vec k)\cdot\vec\varepsilon(\lambda',\vec k) =$ } $\lambda = 1,2$

$$a(\lambda,\vec k) = \int d^3x\,\sqrt{(2\pi)^3 2k_0}\,f_k^*(x)\,i\overleftrightarrow{\partial_0}\,[\vec\varepsilon(\lambda,\vec k),\vec A(x)]$$
$$a^+(\lambda,\vec k) = \int d^3x\,\sqrt{(2\pi)^3 2k_0}\,f_{\vec k}(x)\,i\overleftrightarrow{\partial_0}\,[\vec\varepsilon(\lambda,\vec k),\vec A(x)]$$
$$[a(\lambda,k),\ a^+(\lambda',\vec k')] = (2\pi)^3 2k_0\,\delta_{\lambda\lambda'}\,\delta(k-k')$$

osnovno stanje : $a(\lambda,\vec{k})\,|0\rangle = 0$

Hamiltonyan : $H = \frac{1}{2}\int d^3x\,(\vec{E}^2+\vec{B}^2)$

$$:H: = \int \frac{d^3k}{(2\pi)^3 2k_0} \sum_\lambda k_0 \underbrace{N(\lambda,\vec{k})}_{a^+(\lambda,\vec{k})a(\lambda,\vec{k})}$$

Kvantizacija : Lorentzova umeritev

$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2$

$\pi^0 = \frac{\partial\mathcal{L}}{\partial \dot{A}_0} = \partial_\mu A^\mu$ ← vez, Lagrangeev multiplik.

$[A_\mu(t,\vec{x}),\ \pi_\nu(t,\vec{x}')] = i\cdot g_{\mu\nu}\,\delta(\vec{x}-\vec{x}')$

$[a(\lambda,\vec{k}),\ a^+(\lambda',\vec{k}')] = -g^{\lambda\lambda'}\cdot 2k_0(2\pi)^3\,\delta(\vec{k}-\vec{k}')$

$a^+(0,\vec{k})$ so nefizikalni ⟷ Enosovna polarizacija
$a^+(3,\vec{k})$ longitudinalno = nefizikalno

$\langle\lambda=0,k|\,\lambda=0,k\rangle = 0$

$\langle H\rangle \geq 0$

Upoštevanje umeritve :

1) $\partial_\mu A^\mu = 0$ →←

2) $\partial_\mu A^\mu\,|\psi\rangle_{fiz} = 0$ →←

3) $\boxed{\partial_\mu A^{(+)\mu}\,|\psi\rangle_{fiz} = 0}$ ✓ $\iff$ $\langle\psi|\,\partial_\mu A^\mu\,|\psi\rangle_{fiz} = 0$

Fermi - Gubta - Belulan ?

$$\Rightarrow \sum_{\lambda=0}^{3} \underbrace{k^\mu \varepsilon_\mu(\lambda,\vec{k})}\, a(\lambda,\vec{k})\,|\psi\rangle_{fiz} = 0$$

$k^\mu\varepsilon_\mu(\lambda=1,2,\vec{k}) = 0$

$\Rightarrow [a(0,\vec{k}) - a(3,\vec{k})]\,|\psi\rangle_{fiz} = 0$

↳ $|\psi\rangle = |\psi\rangle_{fiz}\underbrace{|\psi\rangle_{nefiz.}}_{\text{BRST simetrija}}$

$\Rightarrow \boxed{k^\mu M_{0\mu} \equiv 0}$ za QED ; $M_0 = M_{0\mu}\varepsilon^\mu(\lambda,\vec{k})$

PROPAGATORJI

K-G : $[\phi(x),\phi(x')] = \int \frac{d^3k}{(2\pi)^3 2E_k}\int \frac{d^3k'}{(2\pi)^3 2E_{k'}}\left\{[a(k),a^+(k')]e^{-ikx}e^{ikx} + [a^+(k),a(k')]e^{ikx}e^{-ikx}\right\}$

$= \int \frac{d^3k}{(2\pi)^3 2E_k}\left(e^{-ik(x-x')} - e^{ik(x-x')}\right)$

$\equiv i\Delta(x-x')$

$\partial_0\Delta(x-x')\big|_{x_0=x_0'} = -\delta(\vec{x}-\vec{x}')$

$G(x,x') = \theta(t'-t)\,\langle 0|\phi(x')\phi(x)|0\rangle + \theta(t-t')\langle 0|\phi(x)\phi(x')|0\rangle$

$\equiv \langle 0|T[\phi(x')\phi(x)]|0\rangle$

$= \int \frac{d^3k}{(2\pi)^3 2E_k}\left[\theta(t'-t)\,e^{-ik(x-x)} + \theta(t-t')\,e^{+ik(x'-x)}\right]$

$\theta(t) = \lim_{\varepsilon\to 0^+}\int \frac{d\omega}{2\pi i}\,\frac{e^{i\omega t}}{\omega - i\varepsilon}$

$= i\int \frac{d^4k}{(2\pi)^4}\,\frac{e^{-ik(x-x')}}{k^2 - m^2 + i\varepsilon}$

$k^{02} - |\vec{k}|^2 = \omega^2 - |\vec{k}|^2$

$(\Box + m^2)\,G(x,x') = -\delta^{(4)}(x-x')$ v klasični teoriji

DIRAC : $iS_F(x-y) = \langle 0|T[\psi_\alpha(x)\bar{\psi}_\beta(y)]|0\rangle$

$\vdots$

$= \int \frac{d^4p}{(2\pi)^4}\,\frac{\not{p}+m}{p^2-m^2+i\varepsilon}\,e^{-ip(x-y)}$

$(i\not{\partial} - m)\,S_F(x-y) = -\delta^{(4)}(x-y)$

E-M : a) Coulombska umeritev:

$D(x-y)_{\mu\nu} = \langle 0|T[A_\mu(x)A_\nu(y)]|0\rangle$

$= i\int \frac{d^4k}{(2\pi)^4}\,\frac{e^{-ik(x-y)}}{k^2+i\varepsilon}\sum_{\lambda=1}^{3}\varepsilon_\mu(\lambda,k)\varepsilon_\nu(\lambda,k)$

$= -g_{\mu\nu} - \frac{k_\mu k_\nu}{(k\cdot\eta)^2 - k^2} + \frac{(k\cdot\eta)(k_\mu\eta_\nu + \eta_\mu k_\nu)}{(\quad)^2 - k^2} - \text{———}$; $\eta = (1,\vec{0})$

b) Lorentzova umeritev :

$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2$

$D_{\mu\nu}(k)_{xy} = \frac{-1}{k^2+i\varepsilon}\left[g_{\mu\nu} + (\xi-1)\frac{k_\mu k_\nu}{k^2}\right]$

$D_{\mu\nu}(x) = \int \frac{d^4k}{(2\pi)^4}\,e^{-ikx}\,D_{\mu\nu}(k)$

$\begin{cases}\xi = 1 ; \text{ Feynman}\\ \xi = 0 ; \text{ Landau}\end{cases}$

(3. teden 5.3.)

Schrödinger VS Heisenberg

$\frac{\partial}{\partial t}|\psi_s\rangle = H_s|\psi_s\rangle$

$\frac{\partial}{\partial t}|\psi_H\rangle = 0$; $|\psi_H\rangle = e^{iH_0 t}|\psi_s(t)\rangle$

$A_H = e^{iH_0 t} A_s e^{-iH_0 t}$

$[A_H, H_s] = i\frac{\partial}{\partial t} A_H$

Interakcijska slika

$H_0 = H_s^0 + H'$

$|\psi_I(t)\rangle = e^{iH_s^0 t}|\psi_s(t)\rangle$

$|\psi_I(t)\rangle = U(t,t_0)|\psi_I(t_0)\rangle$

$A_I(t) = e^{iH_s^0 t} A_s e^{-iH_s^0 t}$

$i\frac{\partial}{\partial t} U(t,t_0) = H_I' U(t,t_0)$; $\frac{\partial}{\partial t} A_I(t) = [A_I(t), H_s^0]$

$\Rightarrow U(t,t_0) = T\{e^{-i\int dt' H_I'(t')}\}$; $\int d^3x' \mathcal{H}_I'(t',x')$ — gostota drug način

Sipalna matrika

= matrika, ki podaja propagacijo iz začetnega stanja ki je lastno $H_0(x)$ v končno ki je tudi lastno $H_0(x)$

$S \equiv U(\infty,-\infty) = T\{e^{-i\int d^4x' \mathcal{H}_I'(x')}\}$; $\int_{-\infty}^{\infty} dt \int dx$

matrični elementi sipalne matrike: $S_{fi} = \langle f|S|i\rangle = \delta_{fi} + iT_{fi}$

$T_{fi} = (2\pi)^4 \delta^{(4)}(p_f - p_i)\, \mathcal{M}_{fi}$

her celoten δ ne bi ohranja pojm. simetrije $\Rightarrow$ vsota četvercev p se ohr.

netrivialen del: sipalna amplituda (izračun preko razvoja v časovno vrsto, int. hamiltonjana.)

DISKRETNE SIMETRIJE PROSTORA IN ČASA

„neorazcepne upodobitve Poincarejeve grupe niso nujno skladne z operacijo Parnosti"

PARNOST : $\vec{x} \longrightarrow -\vec{x}$

op. na prostu četver.: $\hat{P}$: $\Lambda^\mu{}_\nu = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$ in $x'^\mu = \Lambda^\mu{}_\nu x^\nu$

KLASIČNO:
$\vec{x} \longrightarrow -\vec{x}$
$\vec{p} \longrightarrow -\vec{p}$
$\vec{L} \longrightarrow \vec{L}$

KVANTNO:
$P|lm\rangle = (-1)^{l-m}|lm\rangle$
$Y(\zeta,\varphi)$: lastna stanja v prostoru funkcij, ki majo vrtilno količino. Taka stanja opisujejo sferični harmoniki.
$\Rightarrow$ lastne vrednosti P so $(-1)^{l-m}$

QTP:
$P|\vec{p}\rangle = \eta_P|\hat{p}\rangle$
fazni faktor, intrinzična parnost
skalar $\rightarrow +1$
psevdoskal $\rightarrow -1$
ker želimo da je P unitarna.
tukaj smo predp. 1-delčno stanje. Če se lahko opišemo samo s $\vec{p}$ $\Rightarrow$ spin = 0.

Skalarno polje:

$P a^\dagger(\vec{p})|0\rangle = \eta_a a^\dagger(\vec{p})|0\rangle$; vzdimo da konvencijo $P|0\rangle = |0\rangle$, $P^{-1} = P^\dagger$

$P a^\dagger(\vec{p})|0\rangle = \eta_a a^\dagger(\vec{p})$

$\Rightarrow P a^\dagger(\vec{p}) P^{-1} = \eta_a a^\dagger(-\vec{p})$

$P a(\vec{p}) P^{-1} = \eta_a a(-\vec{p})$

„parnost je pri skalarjih opazljiva količina, pri fermionih ne bo.

vstavimo v Def.: Skalarno polje: $P\phi(t,x)P^{-1} = \eta_\phi \phi(t,-\vec{x})$

$j^\mu = i[\phi^+ \partial_\mu \phi - (\partial^\mu \phi^+)\phi]$

$P j^0(t,\vec{x}) P^{-1} = j^0(t,-\vec{x})$

$P \vec{j}(t,\vec{x}) P^{-1} = -\vec{j}(t,-\vec{x})$

Diracovo polje: (enodelčno)

$P|\vec{p},s\rangle = \eta_F|-\vec{p},s\rangle$ (smer spina ostane, hitrost se obrne)

proj. spina na to gib.

„tu se malo zakomplicira, ker mamo spin. = vrtilna količina je axialni vektor (ne spremeni predzn) to moramo konsistentno vpeljat v definicijo operacije parnosti."

intrinzična parnost: ni opazljiva količina (pri fermionih) (razen pri Majorana fenn, ampak tudi tam ne moreš ločit te faze od drugih...)

Konvencija $\eta_F(p) = 1$ ← proton, od tod izhaja

$P|\overline{p,s}\rangle = -\eta_F|\overline{-p,s}\rangle$ rel. iz antisim. valovne funkc. oz. iz povezave spina in stat.

$P a(p,s) P^{-1} = \eta_F a(-\vec{p},s)$; $P b(p,s) P^{-1} = -\eta_F b(-p,s)$

$P a^\dagger(p,s) P^{-1} = \eta_F a^\dagger(-p,s)$; $P b^\dagger(p,s) P^{-1} = -\eta_F b^\dagger(-p,s)$

} od tod sledi transf. polja $\hat{\Psi}$

$\Rightarrow P\Psi(t,x)P^{-1} = \eta_F \gamma^0 \Psi(t,\vec{x})$

↑ to konsist. rečemo z γ^0

$\gamma^0 u(p,s) = u(-p,s)$

$\gamma^0 v(p,s) = -v(p,s)$

ko mamo to lahko gradimo kompozite, npr bilinearne forme_

Bilinearne forme:

$P\bar{\Psi}(x)\Gamma\Psi(x)P^{-1} = \bar{\Psi}(t,-\vec{x})\underbrace{\gamma^0\Gamma\gamma^0}\Psi(t,-\vec{x})$

Γ: $\Gamma = I \longrightarrow I$
$\gamma^5 \longrightarrow -\gamma^5$ ta tvori psevdoskalar
$\gamma^\mu \longrightarrow (-1)^{\mu\neq 0}\gamma_\mu$ vektorski tok
$\gamma^\mu\gamma^5 \longrightarrow (-1)^{\mu=0}\gamma_\mu\gamma^5$ axialni tok: t komp. se transf. kot psevdoskalar krajevna kot axialni vektor

Vektorsko polje:

$\mathcal{E}^\mu_\lambda(-\vec{p}) = P^\mu{}_\nu \mathcal{E}^\nu_\lambda(\vec{p})$
index polariz.

Primer: Intrinzična parnost piona $\pi^0 \to 2\gamma$ (preko razpada, ga zmerimo)

Merljive količine (v tem razpadu): $\mathcal{E}_1, \mathcal{E}_2, \vec{k} \equiv \vec{k}_1 = -\vec{k}_2$, v težiščnem sistemu (ena neodvisna glabala)

$\vec{k}_1 \cdot \vec{\mathcal{E}}_1 = \vec{k}_2 \cdot \vec{\mathcal{E}}_2 = 0$ Wardove identitete: ni k čisto neodvisna.

Proces potekn preko EM int: $\mathcal{P} H_{int} \mathcal{P}^{-1} = H_{int}$ ker v QED ni kršitve parnosti

$|\pi^0\rangle \xrightarrow{H_{int}} |2\gamma\rangle$

$\mathcal{P}|\pi^0\rangle \longrightarrow \mathcal{P}|2\gamma\rangle$ ... to bo enako.

$\eta_{\pi^0} \longrightarrow \eta_\gamma^2 \eta_L$ (int. parnost fotona $=2$; gostor)

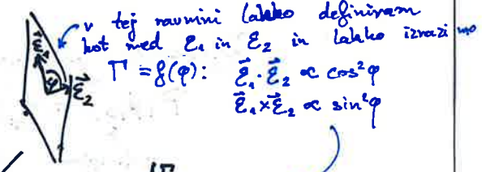
parnost začetnega stanja bo enaka parnosti končnega stanja.

$\eta_L = +1$ $\quad$ $\eta_L = -1$ $\Leftarrow$ iz merljivih količin lahko tvorimo samo dve invariante

$\Rightarrow$ Parametriziramo $\mathcal{M}$ v splošni obl.: $\mathcal{M} = a\, \vec{\mathcal{E}}_1 \cdot \vec{\mathcal{E}}_2 + b\, \vec{k} \cdot (\vec{\mathcal{E}}_1 \times \vec{\mathcal{E}}_2)$

parnost kinematike, konfiguracije, mamo vrtilno (tirno)

Eksperimentalno: γ_2 — $\vec{\mathcal{E}}_2$ — π^0 — $\vec{\mathcal{E}}_1$ — γ_1



↳ eksperim.: $\frac{d\Gamma}{d\varphi} \propto \sin^2\varphi$

$\Downarrow$

$\eta_L = -1 \Rightarrow \boxed{\eta_{\pi^0} = -1}$

KONJUGACIJA NABOJA

(ne trdimo da je to simetrija, je pa dobro definirana operacija)

= zamenjava „delca" ("stanja") z antidelcem

Poglejmo na "Diracovem polju":

$C|\vec{p},s\rangle = \xi_F |\vec{p},s\rangle$

$C\, a(\vec{p},s)\, C^{-1} = \xi_F\, b(\vec{p},s)$ $\qquad$ $C\, \bar{u}^T(p,s) = v(\vec{p},s)$

$C\, b(\vec{p},s)\, C^{-1} = \xi_F^*\, a(\vec{p},s)$ $\qquad$ $C\, \bar{v}^T(p,s) = u(\vec{p},s)$

$\Rightarrow \boxed{C\psi(x)C^{-1} = i\gamma^2\gamma^0 \xi_F \bar{\psi}^T}$

Fermionski/Spinorski tok:

$C\bar{\psi}\gamma^\mu\psi C^{-1} = (-)\bar{\psi}\gamma^\mu\psi$

nabojni tok obrne predznak $\Rightarrow$ „konjugacija naboja"

obrne tok naboja: $C j^\mu C^{-1} = -j^\mu$

želimo da so EM interakcije inv. na konjugacijo naboja

$C\mathcal{L}^{QED}C^{-1} = \mathcal{L}^{QED}$

to izhaja iz zahteve inv. EM

lahko rečemo konvencija.

$(A_\mu j^\mu \longleftrightarrow \boxed{\xi_A = -1})$

$\mathcal{L}^{QED}$

to zberemo da bo $A_\mu j^\mu$ inv.

ko gremo k šibki int. ne moremo mojt konvencije da ne bi bilo to kršeno!

OBRAT ČASA

$\mathcal{T}$: $t \to -t$, $\vec{r} \to \vec{r}$, $\vec{p} \to -\vec{p}$, $\vec{L} \to -\vec{L}$, $\vec{S} \to -\vec{S}$

Poincarejeva transformacija: $U(\omega_{\mu\nu}, a_\mu) \approx I + \frac{i}{2}\omega_{\mu\nu} M^{\mu\nu} + i a_\mu P^\mu$

(generatorji; $\omega_{\mu\nu}$ lorentz., a_μ transl.)

$\mathcal{T}\, U(\omega,a)\, \mathcal{T}^{-1} = U(\mathcal{T}\omega_{\mu\nu}\mathcal{T}^{-1}, \mathcal{T}a)$

$\mathcal{T}(P^0)\mathcal{T}^{-1} = -P^0$ oz $H \to -H$ (to je energija) ∇ 0

obrne predznak energijam $\Leftrightarrow$ ni unitarna operacija

Podoben zaplet je pri kompleksni konjugaciji (sCH):

$i\frac{\partial}{\partial t}\phi(t,x) = H\phi(t,x)$ $\quad /^*$

$-i\frac{\partial}{\partial t}\phi^*(t,x) = H\phi^*(t,x)$

Rešitev: $K\phi \equiv \phi^*$ in $\mathcal{T} = KU$ (kompleksna konj., unitarni operator)

Lastna stanja: $\langle x|p\rangle = e^{-iHt + i\vec{p}\vec{x}}$

$\langle x|p\rangle^* = e^{iHt - i\vec{p}\vec{x}} = \langle (-t,\vec{x})|E,(-\vec{p})\rangle$ ✓

(to naredi tudi obrat časa)

$\mathcal{T}|\vec{p}\rangle = |-\vec{p}\rangle$

$\mathcal{T}|l,m\rangle = (-1)^{l-m}|l,-m\rangle$

$\langle x|l,m\rangle = i^l Y_{l,m}(\theta,\varphi)$

$\langle x|l,m\rangle^* = (-i)^l Y_{lm}$

$\mathcal{T}\, a_i|\phi(t)\rangle_i = (KU)\, a_i|\phi(t)\rangle_i = a_i^* U|\phi(t)\rangle_i^*$ (razvoj po bazi)

Diracovo polje: $\mathcal{T}\chi(t,x)\mathcal{T}^{-1} = \xi_F \gamma^1\gamma^3\psi(-t,x)$ (a ker $\mathcal{T} = \mathcal{T}^{-1}$ „antiunitaren": $\langle\mathcal{T}\psi|\mathcal{T}\phi\rangle = \langle\psi|\phi\rangle^*$)

$\mathcal{T}\bar{\psi}(t,\vec{x})\mathcal{T} = \xi_F^*\bar{\psi}(-t,\vec{x})\gamma^0(\gamma^1\gamma^3)^\dagger\gamma^0 = -\xi_F^*\bar{\psi}(-t,\vec{x})\gamma^1\gamma^3$ ... iz Peskina

Zakaj smo 2 ure porabili za te diskretne operacije, če nam skoraj nobena ne da simetrije v naravi?

Zato ker obstaja CPT teorem.

Simetrije $\underbrace{C, P}_{\text{maximalno}}$, $\underbrace{CP \text{ in tudi } \mathcal{T}}_{\text{parametrično (faza v CKM)}}$ so kršene v šibkih procesih

CPT TEOREM

$\Theta \equiv CP\mathcal{T}$,

Teorem: $\Theta\, \mathcal{H}(t,\vec{x})\, \Theta^{-1} = \mathcal{H}(-t,-\vec{x})$ in $\Theta|a\rangle = |\bar{a}\rangle$ $\Rightarrow \langle a|\mathcal{H}|b\rangle = \langle\bar{a}|\mathcal{H}|\bar{b}\rangle$ matrični elementi so enaki za antidelce

Pomembno: iz stanja delcev naredi stanje z antidelci

Posledice: 1) $\mathcal{H} = \mathcal{H}^0$, $a = b$. Potem $\underbrace{\langle a|\mathcal{H}^0|a\rangle}_{E_a = m_a} = \underbrace{\langle\bar{a}|\mathcal{H}^0|\bar{a}\rangle}_{E_{\bar{a}} = m_{\bar{a}}}$

$\Rightarrow$ Mase delcev in antidelcev so enake

2) $\mathcal{H} = \mathcal{H}_I'$. Prehod stanja a v katerokoli stanje:

$\Gamma_a \propto \sum_{\beta,\bar{\beta}} |\langle\beta|\mathcal{H}_I'|a\rangle|^2$ ... razpadna širina stanja $|a\rangle$, ki ni več lastno stanje tega $\mathcal{H}_I'$. $\langle\beta|$ so „vsa" stanja (možna); za t razpad „v katerokoli stanje".

$\Gamma_{\bar{a}} \propto \sum_{\beta,\bar{\beta}} |\langle\beta|\mathcal{H}_I'|\bar{a}\rangle|^2$

$\Rightarrow$ razpadne širine/časi so enaki za delce in antidelce

3) Povezava spina in statistike (to ni relevantno za nas)

Diskretne transf.: večinoma niso simetrijske operacije, tista s π^0 je simetrijska, dokler nimamo interakcij, ki jo kršijo. Še bolje: π^0 je vezano stanje močne int. $\Rightarrow \eta$ bo povezana z močno int.
Ali je π^0 lastno stanje ~~[redacted]~~ parnosti

$\Rightarrow$ Ali močna int. ohranja parnost?
- Mordá; problem močne CP: teorija dopušča možnost, da je CP kršena, ni bilo pa še opaženo. Če bi bila kršena, potem π^0 niso več lastna stanja parnosti

Za efektiven teoretičen opis nizkoenergijskih nukleonskih stanj se bomo naslanjali na exaktne simetrije (npr. Poincarejeva) in približne (npr. CP).

LIEJEVE GRUPE IN NJIH UPODOBITVE

SU(N) upodobitve $e^{-i\varepsilon_a T^a}$ (parametri, generatorji); $[T^a, T^b] = i f_{abc} T^c$, $\mathrm{Tr}[T^a] = 0$ ... brezsledni
N^2-1 parametrov / generatorjev $T_a^\dagger = T_a$

SU(2): $[I^i, I^j] = i\varepsilon_{ijk} I^k$; $I^i = T^i$ iz zgodovinskih razlogov. ; $I^i = \frac{\sigma^i}{2}$ (fund. up. dim = 2)
(ε_{ijk} antisim. Levi-Civita)
$(I^i)_{jk} = i\varepsilon_{ijk}$ (adjung. up. dim = 3)

SU(3): $[T^i, T^j] = d_{ijk} T^k$; $T^i = \frac{\lambda_i}{2}$ ("Gell-Mann") (fund. up. dim = 3)
d_{ijk}
$\mathrm{Trace}[\lambda_i \lambda_j] = 2\delta_{ij}$, $\mathrm{Tr}[\lambda_i] = 0$ (to je samo normaliz.)
$(T^i)_{jk} = i A_{ijk}$ (adj. up. dim = 8)

→ dimenzije preko Youngovih

KOMPLETNOSTNE RELACIJE

M hermitska $M = M^\dagger$ 3×3, to je lahko npr. maša

$M = \frac{1}{3}\mathrm{Tr}(M) I + \frac{1}{2}\sum_a^8 \mathrm{Tr}(M\lambda^a)\lambda^a$ razvoj po Gell-Mann matrikah (sledni del + brezsledni del)

$M_{AB} = \frac{1}{3} M_{cc}\delta_{AB} + \frac{1}{2}\sum_{a=1}^{8} M_{cd}\lambda^a_{dd}\lambda^a_{AB}$ (sled)

$M_{cd}\delta_{ac}\delta_{DB} = \frac{1}{3}(M_{cd}\delta_{dc})\delta_{AB} + \frac{1}{2}\sum_a M_{cd}\lambda^a_{dc}\lambda^a_{AB}$ ... samo drug zapis (M_{cd} izpost.)

Dobimo relacijo, ki uporablja samo permutacije indexov in produkt Gell-Mann

SU(N): $\delta_{ac}\delta_{DB} = \frac{1}{N}\delta_{cd}\delta_{AB} + \frac{1}{2}\sum_{\lambda=1}^{N^2-1}\lambda^\lambda_{dc}\lambda^\lambda_{AB}$ (N^2-1 št. generatorjev)
(dim. fund. upodobitve)

Ta relacija nam omogoča da permutiramo barvne indexe v produktih: Mamo relacijo med različnimi permutacijami barvnih indexov.

Nukleoni in izospinska SU(2) simetrija

Nukleon $N = \begin{bmatrix} p \\ n \end{bmatrix}$ (proton, nevtron), masna matrika $M_N = \begin{bmatrix} m_p & 0 \\ 0 & m_n \end{bmatrix}$ (to je vbazi p,n)
Ker sta p,n fermiona ju opišemo z Diracovem

$\mathcal{L} = \bar{N}(i\partial\!\!\!/ - M_N)N$ ← The most general free Lag. invariant under SU(2)

$\bar{N} = \begin{bmatrix} \bar{p} \\ \bar{n} \end{bmatrix}$

Pod SU(2) transformacijami:

$N \to e^{-i\alpha_a I^a} N = UN = N'$ (α parametri SU(2))
$N'^\dagger = N^\dagger e^{i\alpha_a I^a}$ ($I^{a\dagger} = I^a$)
} $\bar{N}' = \bar{N}U^\dagger$; $U \equiv e^{-i\alpha_a I^a}$

$\mathcal{L}' = \bar{N}'(i\partial\!\!\!/ - M_N)N' = \bar{N}e^{i\alpha_a I^a} i\partial\!\!\!/ e^{-i\alpha_a I^a} N - \bar{N}e^{i\alpha_a I^a} M_N e^{-i\alpha_a I^a} N$
$\simeq \bar{N}i\partial\!\!\!/ N - \bar{N}(1+i\alpha_a I^a) M_N (1-i\alpha_a I^a) N$
(inf. transf. majhni α_a)
$\bar{N} M_N N + i\alpha_a \bar{N}[-I^a M_N + M_N I^a]N + \mathcal{O}(\alpha^2)$ ← $[M_N, I^a]$
$= \mathcal{L} + i\alpha_a \bar{N}[M_N, I^a]N + \mathcal{O}(\alpha^2)$

$m_p = m_n \Longrightarrow [M, I^a] = 0$, $\mathcal{L}' = \mathcal{L}$ ← pogoj za invarianco prostih p in n na SU(2) transf. iz $\mathcal{L}$

Vsi operatorji morajo komutirat z vsemi generatorji
Zakaj? - ker

Noetherjin teorem: $j^\mu_a \equiv -\frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}\frac{\partial N}{\partial\alpha_a}$ ($\bar{N}i\gamma^\mu$; $\frac{\delta N}{\alpha_a}$ variacija polja, normir. na parameter)
$\delta N = -i\alpha_a I^a N$

"izospinski tok" ("VEKTORSKI !"): $j^\mu_a = \bar{N}\gamma^\mu I^a N$ in $\partial_\mu j^\mu_a = 0$

$Q_a = \int d^3x\, j^0_a = \int d^3x\, \bar{N}\gamma^0 I^a N$, $\frac{\partial Q_a}{\partial t} = 0$

Aksialni tok: edini del v j^μ_a ki se spreminja s transformacijami je tist $\delta N = -i\alpha_a I^a N$". Če hočemo dobit axialni tok moremo sem ↗ noter nekako γ^5 upeljat.

"Axialna SU(2) transformacija": $N' = e^{-i\alpha_a I^a \gamma^5} N$
(γ^5 deluje na prostoru kor. indexov (spinorjev)!; I^a deluje na izospinskem prostoru (p,n))
} I^a in γ^5 delujeta na ortogonalna prostora

$N'^\dagger = N^\dagger e^{i\alpha_a I^a \gamma^5} \simeq N^\dagger(1 + i\alpha_a I^a \gamma^5 + \dots)$; $\gamma^5 = \gamma^{5\dagger}$ in $\{\gamma^\mu, \gamma^5\} = 0$
$I^{a\dagger} = I^a$
$\bar{N}' = N'^\dagger\gamma^0 = \bar{N}(1 - i\alpha_a I^a \gamma^5 + \dots)$
$\simeq \bar{N}e^{-i\alpha_a I^a \gamma^5}$

$\mathcal{L}' = \bar{N}'(i\partial\!\!\!/ - M_N)N' \simeq \bar{N}(1 - i\alpha_a I^a\gamma^5)(i\partial\!\!\!/ - M_N)(1 - i\alpha_a I^a\gamma^5)N$
$= \bar{N}i\partial\!\!\!/ N - i\alpha_a(\bar{N}I^a\gamma^5\gamma_\mu\partial^\mu N + \bar{N}\gamma^5\partial^\mu I^a\gamma^5 N) -$ (*)
$- \bar{N}M_N N + i\alpha_a(\bar{N}I^a\gamma^5 M_N N + \bar{N}M_N I^a\gamma^5 N) + \mathcal{O}(\alpha^2)$ (**)

(*): $-i\alpha_a\bar{N}(I^a\gamma^5\gamma_\mu + I^a\gamma_\mu\gamma^5)\partial^\mu N =$
$= -i\alpha_a\bar{N}I^a\{\gamma^5, \gamma_\mu\}\partial^\mu N = 0$

(**): $+i\alpha_a\bar{N}(I^a M_N\gamma^5 + M I^a\gamma^5)N =$
$= i\alpha\bar{N}\{I^a, M_N\}\gamma^5 N = 0$; le če $M_N = 0$ oz $m_p = m_n = 0$

To v naravi ni res, ampak v limiti $m_p = m_n$ imamo še axialne tokove in axialne naboje, ki so ohranjeni.

$$\boxed{j^{5a}_\mu = \bar N I^a \gamma_\mu \gamma_5 N}, \qquad \partial N = -i\alpha_a I^a \gamma_5 N$$

„Axialna simetrija SU(2)" vodi do ohranitve axialnih tokov (3).

$\partial_\mu j^{\mu a} = 0$, $\partial_\mu j^{\mu 5a} = 0$ — dva tripleta tokov $\Rightarrow$ tudi njih lin. komb. je ohranjena.

↳ pod Lorentzovimi transf. se transf. kot axialni vektor.

„KIRALNA SIMETRIJA" $SU(2)_L \times SU(2)_R$
↳ namesto vektorske + axialne. Ta je bolj pripravna ker so v $m_p = m_n = 0$ limiti upodobitve Poinc. grupe Weylove kiralne up.

$$j^{(L,R)a}_\mu \equiv \left(j^a_\mu \mp j^{5a}_\mu\right)\frac{1}{2}$$

kiralna tokova

$\Rightarrow \partial_\mu j^{\mu(L,R)} = 0$

Metoda Gell-Mann-Levy (klasično)

Omogoča, da zvemo kakšen bi bil ohranjen tok, če bi bla simetrija izpolnjena in katerim pogojem more zadoščat, da je simetrija exaktna.

$\mathcal{L} = \mathcal{L}(\phi, \partial\phi)$

$\phi' = U(\varepsilon)\phi$ — neka transf. ↑ parametri transf. — npr $\phi' = e^{i\varepsilon_a T^a}\phi$

Trik te metode je da premoviramo parametre transformacij v nedinamična polja (brez kin. člena)

$\varepsilon \longrightarrow \varepsilon(x)$ „umeritvena transformacija" (ne bomo zahtevali invariance na to.)

$\partial_\mu\phi \longrightarrow U(\varepsilon)(\partial_\mu\phi) + (\partial_\mu U(\varepsilon))\phi$

Variacija Lagrangeve gostote pod inf. transformacijo $\phi' \approx (1 - i\varepsilon_a T^a)\phi$

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi)$$

$\delta\phi = -i\varepsilon_a T^a\phi$

$\partial_\mu\phi - i\varepsilon_a T^a\partial_\mu\phi - i(\partial_\mu\varepsilon_a)T^a\phi$ ← obravnavam $\delta(\partial_\mu\phi)$ kot $\delta(\varepsilon)$:

$$\delta\mathcal{L}(\varepsilon^a, \partial_\mu\varepsilon^a) = \frac{\partial\mathcal{L}}{\partial\varepsilon^a}\varepsilon^a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varepsilon^a)}\partial_\mu\varepsilon^a$$

$$\delta\mathcal{L} = \left(\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)(-i\varepsilon_a T^a)\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\phi(-i\varepsilon_a T^a\partial_\mu\phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}(-i(\partial_\mu\varepsilon)T^a\phi)$$

$$= \partial_\mu\underbrace{\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}(-iT^a)\phi\right]}_{j^{a\mu}}\varepsilon_a + \underbrace{\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}(-iT^a\phi)}_{j^{a\mu}}\partial_\mu\varepsilon^a \quad \ldots\ldots \text{Prepoznamo tok!}$$

$$= \underbrace{\partial_\mu j^{a\mu}\varepsilon_a}_{0,\ \text{če } \mathcal{L}' = \mathcal{L}} + j^{a\mu}\partial_\mu\varepsilon_a$$

ta člen nam da explicitno obliko toka, ki pripada $\phi' = U(\varepsilon)\phi$ transformaciji (Noetherjin teorem nam da samo divergenco toka)

Primer te metode: SU(2) model nukleonov

$\mathcal{L} = \bar N(i\partial\!\!\!/ - M_N)N$, $N' = e^{-i\alpha_a I^a}N$ v SU(2)

$\delta N = N' - N = -i\alpha_a I^a N$

$\delta\partial_\mu N = \partial_\mu N' - \partial_\mu N = -i\alpha_a I^a\partial_\mu N - i\partial_\mu\alpha_a I^a N$

$\delta\mathcal{L} = \left(\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}\right)(-i)\alpha_a I^a N + \frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}(-i)I^a\partial_\mu N - i(\partial_\mu\alpha_a)\frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}I^a N$

$= -i\alpha_a\bar N[I^a, M]N + \partial_\mu\alpha^a\bar N\gamma_\mu I^a N$

$\partial_\mu j^{a\mu}$; $j^{a\mu} = \bar N\gamma^\mu I^a N$

$\partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu N} = \frac{\partial\mathcal{L}}{\partial N} = \frac{\delta\mathcal{L}}{\delta N}$; $= \frac{-i\alpha_a\bar N[I^a, M]N}{-i\alpha_a\bar N}$

to pokorno enčimo $\bar N i\gamma^\nu$

axialni tok: $j^{a5}_\mu = \bar N\gamma_\mu\gamma_5 I^a N$ in $\partial^\mu j^{5a}_\mu = -i\bar N\{I^a, M\}\gamma_5 N$

GML (kvantno): Algebra operatorjev naboja

$j^a_\mu = \bar N\gamma_\mu I^a N$

$Q_a = \int d^3x\, j^a_0 = \int \bar N\gamma_0 I^a N\, d^3x$

Klasično je Q_a „številka". Kvantno so $N, \bar N \Rightarrow j^a_\mu$ in Q_a operatorji.
V QFT imamo (anti)komutacijske relacije:

$$\{N^\dagger_A(t,\vec x)\ ,\ N_B(t,\vec y)\} = \delta_{AB}\,\delta(\vec x - \vec y)$$

Do rezultata vodi dolg račun. Poglejmo najprej:

$$[Q_a\ ,\ N_A(x)] = \left[\int \underbrace{N_B(y)}\ \gamma_0 I^a N_c(y)\, d^3y\ ,\ N_A(x)\right]$$

$\bar N\gamma_0 = N^\dagger\gamma_0^2 = N^\dagger$

$$= I^a_{Bc}\int\left(N^\dagger_B(y)N_c(y)N_A(x) - N_A(x)N^\dagger_B(y)N_c(y)\right)d^3y$$

$$= -I^a_{Bc}\int\left(\underbrace{N^\dagger_B N_A + N_A N^\dagger_B}\right)N_c\, d^3y$$

$\{N^\dagger_B, N_A\} = \delta_{AB}\,\delta(\vec x - \vec y)$

$$= -I^a_{Ac}N_c(x)$$

to bo edini netrivialen antikomutator. N_c spravimo na desno z antikomutiranjem (prvi člen spremeni predznak)

$$[Q_a, N^\dagger_A(x)] = I^a_{Ac}N^\dagger_c(x)$$

$$[Q^a, Q^b] = \int d^3y\,[Q^a, N^\dagger_c(y)I^b_{cD}N_D(x)] =$$

⋮ prevedemo na prejšnja dva komutatorja, uporabimo gornji rezultat, uvidimo nov komutator ⋮

$$= [I^a_{BD}, I^b_{BD}]\cdot\int d^3y\, N^\dagger_B(y)N_D(y)$$

$$= i\varepsilon_{abc}Q^c$$

INTERAKCIJE NUKLEONOV — PIONI

(4. teden 12. 3.)

Energijska skala jedrskih reakcij: $E \leq$ nekaj MeV

To je pomembno ker:

* $E_{(na\,nivoju)} \ll m_{jedrskega}$: nukleoni v teh reakcijah bodo nerelativistični (manj pomembno)
* pri reakcijah med nukleoni moramo upoštevat vsa stanja (tudi vzdelčna), vse do nekaj MeV. Katero je prvo naslednje ležeče stanje?
  ⇒ prostim nukleonom dodamo pione in interakcije z njimi:
  $|N\rangle \longrightarrow |N\pi\rangle$ .... nukleon + pion (roper resonance so višje)

Prispevajo le najnižje ležeča stanja močne int: pioni
Približna degeneracija $\underbrace{\pi^+, \pi^0, \pi^-}_{\text{„triplet"}}$, pseudoskalarji $P\pi_i P^+ = -\pi_i$
↑ spin: 0
intrinzična parnost: −1

INTERAKCIJA MED NUKLEONI IN PIONI:

→ $\mathcal{L}_{int}$ more bit inv. na vse simetrije, ki so exaktne v naravi, predvsem na Poincarejevo simetrijo.

Vemo: nukleoni so fermioni ($s=\frac{1}{2}$), sklaplajamo jih s skalarji
⇒ iz parov „nukleon-antinukleon" moramo tvorit upodobitve, ki jih bomo lahko tvorili tudi iz produktov operatorjev pionov.

Lorentzove upodobitve, ki jih lahko tvorimo (spin 0, 1, 2, ...):

$\bar{N}N$, (skalar) $\eta_P=+1$; $\bar{N}\gamma^\mu N$ (vektor) $\eta_P=-1$; $\bar{N}\gamma^\mu\gamma^5 N$ (vektor) $\eta_P=+1$; $\bar{N}\sigma^{\mu\nu}N$ (tenzor) $\eta_P=+1$; $\bar{N}\gamma^5 N$ (skalar) $\eta_P=-1$

Ta dva prideta v upošteV

Če sklaplajamo par nukleonov s pionom lahko uporabimo tiste, ki z enim samim pionom dajo pozitivno parnost $\mathcal{L}$ — to je zahteva, ker močna int. ohranja parnost

Izospinska simetrija SU(2) (približna)

Razlog: opis nam narodi bolj ekonomičen + močna int. ne loči p^+ in n^0
Upodobitev pionov: $dim = 3$ .... to je adjungirana oz. vektorska
↑ druga možnost bi bila da tvorijo dve ločeni upodobitvi dublet + singlet

Baza: π_1, π_2, π_3 ; $(\pi^0, \pi^\pm)$ ni baza vektorske adj. upodobitve kot jo poznamo. Rabimo kanonično bazo

$$\pi^\pm = \frac{1}{\sqrt{2}}(\pi^1 \pm i\pi^2)$$
$$\pi^0 = \pi^3 \leftarrow \text{diagonalen}$$

Algo π^+, π^- nista hermitska operatorja (π^0 je, ker ja sam svoj antidelc)
⇒ pripadajoča ohranjena naboja ne moreta bit generatorja Liejeve algebre.

Adj. upodobitev tvorijo generatorji grupe. Če želimo uporabit vse teoreme mormo najt mormo najt bazo v kateri so hermitski.

a) $\mathcal{L}_{int} = -g\,\bar{N}_i \gamma_5 I^a_{ij} N_j \pi^a$ (iz zgodovinskih; sklopit mormo simetrično ⇒ generatorji SU(2); hermitski)

i,j = izospinski indexi; a = ?

↑ produkt vektorja v izospinskem (triplet (izospinski)) ; adj. triplet ; samo s tem se v sklaplja

$\bar{N} \sim \bar{2}$ (za SU(2)) ~ 2, $N \sim 2$

$\bar{N} \otimes N \sim \bar{2} \otimes 2 = 2 \otimes 2 = \square \otimes \square = \square\!\square \oplus \square = 1 \oplus 3$

ε_{ij}... dvoindeksni singlet (antisim.) ← generatorji grupe } singleta ne moremo sklepit s π^a tripletom pa dobit singlet.

Preverimo: $N' = UN = e^{-i\varepsilon_a I^a} N$

$N'^\dagger = N^\dagger U^\dagger \Rightarrow \bar{N}' = \bar{N} U^\dagger$ { γ_0 deluje na spinorski prostor ⇒ komutira z oper. v izospinskem prostoru

$\Rightarrow \mathcal{L}'_{int} = -g\bar{N}U^\dagger I^a \pi'^a U \gamma_5 N = \mathcal{L}_{int}$

$\doteq I^a\pi^a$, če hočemo $\mathcal{L}' = \mathcal{L}$

$\pi' = U\pi$ pod adj. upodobitvijo

$U^\dagger I^a \pi^a U \overset{inf}{=} [1 + i\varepsilon_b I^b]\, I^a \pi^a [1 - i\varepsilon_a I^a]$

$\approx I^a\pi^a + i\varepsilon_b \underbrace{I^b I^a}_{} \pi^a - i\varepsilon_a \underbrace{I^b I^a}_{} \pi^b$

$= I^a\pi^a + i\varepsilon_b [I^b, I^a]\pi^a$

$i\varepsilon_{abc} I^c$

$= I^a\pi^a - \varepsilon_b i\varepsilon_{abc}\pi^c = i\ldots$

$-i\varepsilon_{bac} = [I^b_{adj}]_{ac}$

tvorijo adj. upod. SU(2) točno to kar smo predp., da se π^a transf. pod adj.

Prehod med bazama (masna vs adj.):

$$\begin{bmatrix}\pi^+\\ \pi^0\\ \pi^-\end{bmatrix} = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0\\ 0 & 0 & 1\\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0\end{bmatrix}\begin{bmatrix}\pi^1\\ \pi^2\\ \pi^3\end{bmatrix}$$

masna baza (nekanonična) — R — adjungirana baza

Povezava med bazama generatorjev:

$I^a_{mass} = R\, I^a_{adj}\, R^\dagger$

↑ I^3 je diagonalen, Cartanov, vse upodobitve bomo klasificirali na podlagi njega lastnih vrednosti.

nikakor ne moremo tvorit spinorskih upod: Iz celoštev. spina v 3+1 dim ne moremo dobit polcelih. To lahko naredimo v 1+1 dimenzijah.

Če π^a tvorijo adj. upodobitev, potem $(\vec{I}\cdot\vec{\pi})' = U\,\vec{I}\cdot\vec{\pi}\,U^\dagger$ (matriki find upodob.; vektorja adj. upod.)

$$\vec{I}\cdot\vec{\pi} = \begin{bmatrix}\pi^0 & \sqrt{2}\pi^+\\ \sqrt{2}\pi^- & -\pi^0\end{bmatrix}$$

↑ generatorji izospina SU(2) → $\frac{1}{2}$ Paulijeve

$\Rightarrow \mathcal{L}_{int} = -g\,[\bar{p}, \bar{n}]\begin{bmatrix}\pi^0 & \sqrt{2}\pi^+\\ \sqrt{2}\pi^0 & -\pi^0\end{bmatrix}\gamma^5\begin{bmatrix}p\\ n\end{bmatrix}$

$= -g\,[\bar{p}, \bar{n}]\,\gamma^5\begin{bmatrix}\pi^0 p + \sqrt{2}\pi^+ n\\ \sqrt{2}\pi^- p - \pi^0 n\end{bmatrix}$

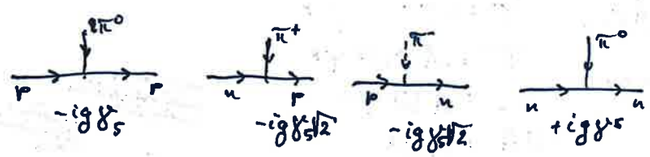
$= -g\,[\bar{p}\,\pi^0\gamma^5 p + \bar{p}\sqrt{2}\pi^+\gamma^5 n + \bar{n}\sqrt{2}\pi^-\gamma^5 p - \bar{n}\pi^0\gamma^5 n]$

Z uporabo izospinske simetrije smo napovedali predznake in velikosti inter. med p^+, n^0 in vsemi temi pioni.

FEYNMANNOVA PRAVILA

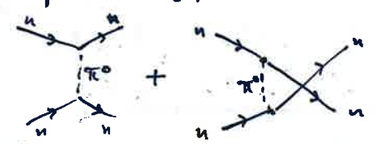
(2. vaja)

$\langle\beta|i\mathcal{L}_{int}|i\rangle$; $\langle\beta|S|i\rangle = \langle\beta|e^{i\int \mathcal{L}_{int}\,d^4x}|i\rangle \approx \delta_{\beta i} + i\int\langle\beta|\mathcal{L}_{int}|i\rangle\,d^4x$

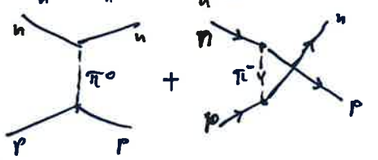


Meritve sipalnih presekov (probamo izločit ta parameter g):

$nn \longrightarrow nn$



$np \longrightarrow np$



$\pi^+ n \longrightarrow \pi^0 p$



} pioni v končnih stanjih

po primerjavi meritev z napovedmi pri zelo nizkih energijah ⇒ $g \doteq 10$.

Teorija ni perturbativna, je pa na robu

Višji redi: kvalitativno / marginalno



$\simeq \#g^2 + \#\left(\frac{1}{16\pi^2}\right) g^4 + \dots$ (zaradi integrala po zanki $\approx \frac{1}{16\pi^2}$)

konvergenca: vsak naslednji člen je manjši ?? $\frac{g^4}{16\pi^2} < g^2$

Če želimo da to konvergira more bit $g < 10$. Če nebi blo te približne konvergence, vprašanje kako bi sploh ugotovili, da je izospinska simetrija zanimiva za opis, prvi red se že nebi ujemal.

Imamo še drugo možnost za opis interakcije nukleonov:

b) $\mathcal{L}^{(2)}_{int} = g' \bar{N} \gamma_\mu \gamma_5 I^a N (\partial^\mu \pi^a) =$ ($\bar{N}\gamma_\mu\gamma_5 I^a N$ je axialni vektor.)

$= g' N (\partial \vec{I}\cdot\vec{\pi}) N$

FEYNMANNOVA PRAVILA



$i g' \not{p}_\pi \gamma_5$

Pion ma gibalno količino p_π.
Če se spomnimo za nazaj, izčezne interakcije pri nizkih energijah:
– kaj se zgodi s tem ko gre $E_{sipanja} \to 0$?
Zastrta reakcija – ne bomo je opazili

nepomembno

$\mathcal{L}_{int} = \sum$ (produkti polj), v splošnem ∞ členov. Če zahtevamo renormaliz. s končnim št. členov $\Leftrightarrow$ $\dim(\mathcal{O}^i) \le 4$:

(da vse reakcije zapisano)

$\dim_{kanonična}(N) = \frac{3}{2}$, $\dim(\bar{N}N) = \frac{3}{2}+\frac{3}{2} = 3$, $\dim(\pi) = 1$, $\dim(\partial_\mu \pi) = 2$

skupaj 4 $\Rightarrow [g] = 1$, $[g'] = \frac{1}{[E]}$

ANTINUKLEONI V SU(2) IZOSPINSKI SIMETRIJI

300 MeV ali 770 MeV (nepert. skala QCD / prva ρ resonanca)

N .... Diracovo polje (prejšnji teden), dublet polj v $SU(2)_{izospin}$

C: $\psi^c \equiv C\psi C^{-1} \overset{\text{če razpišemo}}{=} i\gamma^2\gamma^0 \bar{\psi}^T \overset{\text{prejšnji teden}}{=} i\gamma^2 \psi^*$ (nabojna konjugacija)

$N^c_i = A_{ij}\, i\gamma^2 N^*_j$ (rotacija v izospinskem)

$SU(2)$: $\psi' = U\psi$

Zahtevamo, da se antinukleoni transformirajo enako kot nukleoni (konvencija)

$\psi^{c\prime} = U\psi^c$

$\Rightarrow iA\gamma_2 \,|\, \psi' = U\psi' \,/\, *$ (matrika v izospinskem)

$\psi'^c = iA\gamma U^* \psi^*$

$= A U^* \psi^c$

$UA = AU^*$

$U^* = A^{-1}UA \Rightarrow A = \varepsilon_{ij} = iI_2$ (popolnoma anti-simetričen Levi-Civita, včasih v literaturi)

$\Rightarrow N^c = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} p^c \\ n^c \end{bmatrix} = \begin{bmatrix} -n^c \\ p^c \end{bmatrix}$

$n^c = \bar{n}$ .... polje antinevtrona

$p^c = \bar{p}$ .... polje antiprotona

KIRALNA SIMETRIJA SU(3)

$N = \begin{bmatrix} p \\ n \end{bmatrix}$ v $SU(2)$; $m_p \approx m_n \neq 0$ in velike v primerjavi z tipičnimi jedrskimi reakcijami $\Rightarrow$ ne moremo zanemariti

$\Rightarrow$ imamo $SU(2)$ vektorsko, nimamo pa $SU(2)$ axialne. ∇ to smo prejšnji teden pokazali.

KVARKOVSKI OPIS NUKLEONOV

Zaenkrat samo ~~dinamična~~ števila, brez dinamike.

$q \equiv \begin{bmatrix} u \\ d \\ s \end{bmatrix}$ ..... u, d, s so najnižje ležeča kvarkovska stanja pod skalo močne int. (300 MeV). To bodo relevantne prostostne.

$m_u \neq m_d \neq m_s$, simulacije na mreži

miti približno, ampak $m_{u,d,s} \ll \Lambda_{QCD} \sim \left(\frac{m_p}{3}\right)$ (tipična skala močne int.)

v nasprotju z modelom nukleonov $m_{u,d,s} \ll \Lambda$ (absolutno, ne relat.) $\Rightarrow$ ne gledamo $SU(3)$ kjer so mase podobne, ampak tako kjer so 0

Predpostavimo: mase lahkih kvarkov so v močnih interakcijah zanemarljive. $\Rightarrow M_q = 0 \Rightarrow$ dobimo $SU(3)_A$ in $SU(3)_V$ (novo (tudi komutator zgine))

$\mathcal{L} = \bar{q}(i\not{\partial} - M_q) q$; $M_q = \begin{bmatrix} m_u & & \\ & m_d & \\ & & m_s \end{bmatrix}$ v masni bazi.

$q' = U q$; $U \in SU(3)_V$, $SU(3)_A$ „U pripada fund. upodobitvi $SU(3)$"

→ $SU(3)_V$: $U = e^{-i\varepsilon_i T^i}$; $T^i = \frac{\lambda^i}{2}$ in $[T^i, T^j] = i d_{ijk} T^k$

GML Metoda:

$q' \approx (1 - i\varepsilon_i(x) T^i) q$

$\bar{q}' \approx \bar{q}(1 + i\varepsilon_i(x) T^i)$

$\delta\mathcal{L} = \mathcal{L}' - \mathcal{L} = \bar{q}'(i\not{\partial} - M_q) q' - \bar{q}(i\not{\partial} - M) q$

$= \bar{q}(1 + i\varepsilon_i T^i)(i\not{\partial})(1 - i\varepsilon_i T^i) q - \mathcal{L}$

$= \bar{q}(1 + i\varepsilon_i T^i)\left[i\not{\partial} q + (\not{\partial}\varepsilon_i) T^i q + \varepsilon_i T^i \not{\partial} q \right] - \bar{q} M_q q + i\varepsilon_i \bar{q}[T^i, M] q - \mathcal{L}$

$= \bar{q} i\not{\partial} q + \bar{q}(\not{\partial}\varepsilon_i) T^i q - \bar{q} M_q q + i\varepsilon_i \bar{q}[T^i, M] q - \mathcal{L}$

$= \bar{q}\gamma^\mu T^i q (\partial_\mu \varepsilon_i) + \varepsilon_i i\bar{q}[T^i, M_q] q$

j^μ_i ... 8 tokov, ki bi bili ohranjeni (i: 8 generatorjev za vsak i en tok)

$\partial_\mu j^\mu_i$ ... 8 potencialno ohranjenih nabojev = pogoj, kdaj so tokovi ohranjeni $\partial_\mu j^\mu = 0 \Leftrightarrow [T, M] = 0$

če je $M = \mathbb{1}$ ali pa če $M = 0$.

(3. leto)

→ $SU(3)_A$: $U_A = e^{-i\varepsilon_i T^i \gamma_5}$

GML:

$q' = U_A q \approx$

$\bar{q}' = \bar{q} U_A \approx$

$\mathcal{L}' = \bar{q}'(i\not{\partial} - M_q) q'$

$-\bar{q}' M q' = -\bar{q}(1 - i\varepsilon_i T^i \gamma_5) M (1 - i\varepsilon_i T^i \gamma_5) q + \dots$

$\overset{=}{=} -\bar{q} M q +$

$\partial^\mu q' = (\partial^\mu U_A) q + U_A (\partial^\mu q) \approx$

$\delta\mathcal{L} = -i\partial^\mu \varepsilon_i \underbrace{\bar{q}\gamma_\mu T^i \gamma_5 q}_{j^\mu_A} + i\varepsilon_i \underbrace{\{T^i, M_q\}}_{\partial_\mu} \gamma_5 q$

OHRANJENI NABOJI SU(3) (v limiti $M_q \to 0$)

$$Q_i = \int d^3x\, \bar{q}\gamma_0 T^i q$$
$$Q_A^i = \int d^3x\, \bar{q}\gamma_0\gamma_5 T^i q$$

$$[Q_i^i, Q^j] = i f_{ijk} Q^k$$
$$[Q_A^i, Q_A^j] = i f_{ijk} Q^k$$
$$[Q_A^i, Q^j] = i f_{ijk} Q_A^k$$

To je zdej ena velika algebra dim 18. Radi bi, da razpade na 2 algebri dim 8: uvedeno novo bazo

$$Q_{L,R}^i = \tfrac{1}{2}(Q^i \mp Q_A^i)$$
$$[Q_{L,R}^i, Q_{L,R}^j] = i f_{ijk} Q_{L,R}^k$$
$$[Q_L^i, Q_R^j] = 0$$

$\Rightarrow$ kiralna simetrija $SU(3)_L \times SU(3)_R$

1) Ali velja tudi na kvantnem nivoju?

2) Ali je invariantno osnovno stanje (vakuum)?

Primer:

$$E_A = \langle A|\mathcal{H}|A\rangle$$
$$|B\rangle = U|A\rangle$$
$$E_B = \langle B|\mathcal{H}|B\rangle = \langle A|U^+\mathcal{H}U|A\rangle$$

$$\Rightarrow E_A = E_B \Longrightarrow U^+HU = H$$

$$|A\rangle = \phi_A|0\rangle$$
$$\langle B|H|B\rangle = \langle 0|\phi_B^+ H \phi_B|0\rangle = \langle 0|0|\phi_A^+ \underbrace{U^+ H U}_{H} \phi_A U^+|0\rangle$$

Zadnja enakost bo veljala če $\boxed{U|0\rangle = |0\rangle}$
$\Rightarrow$ vakuum mora bit invar.

Če $U|0\rangle \neq |0\rangle$ imamo prisoten spontani zlom simetrije.

Spontani zlom simetrije in teorem N-G

Naš $\mathcal{L}$ (in s tem $\mathcal{H}$) je invarianten na neke globalne zvezne simetrijske transformacije, vendar osnovno stanje ni invariantno.

Teorem Nambu-Goldstone: Spontani zlom simetrije zahteva obstoj brezmasnih skalarnih bozonov
(To velja za interne simetrije, s primera simetrij prostora-časa lahko dobimo vektorje)

Dokaz: predp. $\mathcal{L}' = \mathcal{L}$. Potem $\exists\, \partial_\mu j^\mu = 0$ in $\frac{dQ}{dt} = 0$.
$\Rightarrow$ Naboji zadoščajo algebri generatorjev.
Če je vakuum inv.: $e^{-i\varepsilon_i Q^i}|0\rangle = |0\rangle$
oz. $Q^i|0\rangle = 0$
V nasprotnem primeru $Q^i|0\rangle =$ neko stanje.
$\rightarrow$ lastnosti tega stanja:

$$\phi'(x) = e^{i\varepsilon_i Q_i}\phi(x)e^{-i\varepsilon_i Q_i} \simeq \phi(x) + i\varepsilon_i[Q^i, \phi(x)] + \dots$$

predpostavka, da tak komutator obstaja

$$0 = \int d^3x\, [\partial_\mu j_i^\mu(t,\vec{x}), \phi(0)]$$
$$= \partial_0 \int d^3x\, [j_i^0(t,\vec{x}), \phi(0)] + \int d\vec{S}\, [\vec{j}(t,\vec{x}), \phi(0)]$$

$\frac{d}{dt}[Q_i(t), \phi(0)] = 0$; to more bit tudi nic

$$\langle 0|[Q_i(t), \phi(0)]|0\rangle \equiv \eta \quad ; \quad \eta \neq f(t)$$
$$= \langle 0|\int d^3x\, (j_i^0(x)\phi(0) - \phi(0)j_i^0(x))|0\rangle$$

vrinemo $\sum_n |n\rangle\langle n| = \mathbb{1}$

translacijska invarianca:
$j_i^0(t,\vec{x}) = e^{iP\cdot x} j_i^0(0) e^{-iP\cdot x}$
$j_i^0(t) = e^{iHt} j_i^0(0) e^{-iHt}$

$$= \sum_n \langle 0|\int d^3x\, e^{iP\cdot x} j_i^0(0) e^{-iP\cdot x}|n\rangle\langle n|\phi(0)|0\rangle -$$
$$- \sum_n \langle 0|\phi(0)|n\rangle\langle n|\int d^3x\, e^{iPx} j_i^0(0) e^{-iPx}|0\rangle \quad ; \quad e^{-iPx}|n\rangle = e^{-iP_n x}|n\rangle,\ e^{-iPx}|0\rangle = |0\rangle$$
$$= \sum_n \Big(\langle 0|\int d^3x\, j_i^0(0) e^{-iP\cdot x}|n\rangle\langle n|\phi(0)|0\rangle -$$
$$- \langle 0|\phi(0)|n\rangle\langle n|\int d^3x\, e^{iP\cdot x} j_i^0(0)|0\rangle$$

$e^{-iP\cdot x} = e^{-i(P_n x_0 - \vec{P}_n\vec{x})}$
$\int d^3x\, e^{\pm iP_n x} = (2\pi)^3\delta^{(3)}(\vec{P}_n)$

$$= (2\pi)^3 \sum_n \delta^{(3)}(\vec{P}_n)\{e^{-iE_n t}\langle 0|j_i^0(0)|n\rangle\langle n|\phi(0)|0\rangle - \langle 0|\phi(0)|n\rangle\langle n|j_i^0(0)|0\rangle \cdot e^{iE_n t}\}$$

$\Rightarrow$ $E_\eta = 0$, $\vec{P}_n = 0$; $E_n = \sqrt{|\vec{P}_n|^2 + m_n^2}$ $\Big\}$ $m = 0$

Pa stanje z maso nič za nek ohranjen naboj ki je ohranjen

Primer:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\sigma\partial^\mu\sigma + \frac{1}{2}\partial_\mu\pi\partial^\mu\pi - \underbrace{V(\sigma^2+\pi^2)}$$

$$V(\sigma^2+\pi^2) = -\frac{\mu^2}{2}(\sigma^2+\pi^2) + \frac{\lambda}{4}(\sigma^2+\pi^2)^2$$

$$\mathcal{L}(\pi) = \frac{1}{2}\partial_\mu\pi\partial^\mu\pi + \frac{\mu^2}{2}\pi^2 + \dots \quad \text{če je } \mu^2 > 0$$

Interne simetrije modela: $\mathcal{L}' = \mathcal{L}$

$$\begin{bmatrix}\sigma'\\ \pi'\end{bmatrix} = \begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}\begin{bmatrix}\sigma\\ \pi\end{bmatrix}$$

Osnovno stanje:

$$\frac{\partial V}{\partial\sigma} = -\mu^2\sigma + \frac{\lambda}{4}2(\sigma^2+\pi^2)2\sigma = 0 \longrightarrow \sigma(-\mu^2+\lambda(\sigma^2+\pi^2)) = 0$$

$$\frac{\partial V}{\partial\pi} = -\mu^2\pi + \lambda(\sigma^2+\pi^2)\pi = 0 \longrightarrow \pi(-\mu^2+\lambda(\sigma^2+\pi^2)) = 0$$

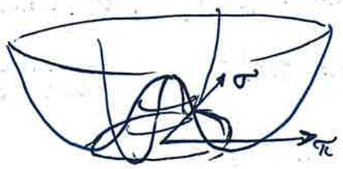
$$\Downarrow$$

$$\sigma^2+\pi^2 = \frac{\mu^2}{\lambda}$$

zahtevamo še da je

$\frac{\partial^2 V}{\partial\pi^2}, \frac{\partial^2 V}{\partial\sigma^2}, \frac{\partial^2 V}{\partial\pi\partial\sigma} \geq 0.$

$\Rightarrow$ minimum pri $v \equiv \sqrt{\frac{\mu^2}{\lambda}}$



To je blo klezično. Kvantno izberemo enega izmed minimumov (useeno kerga), recimo $(\vec{\pi},\pi) = (v,0)$.

$$\langle 0|\sigma|0\rangle = v$$

Majhne fluktuacije okoli vakuuma:

$$\sigma = v + \sigma' \quad \text{oz.} \quad \sigma' = \sigma - v$$

$$\Rightarrow \mathcal{L} = \frac{1}{2}\partial_\mu\sigma'\partial^\mu\sigma' + \frac{1}{2}\partial_\mu\pi\partial^\mu\pi - V(\sigma^2+\pi^2)$$

$$V = -\frac{\mu^2}{2}\left[v^2+\sigma'^2+2v\sigma'+\pi^2\right] + \frac{\lambda}{4}\left[(v^2+2v\sigma'+\sigma'^2)^2 + \pi^4 + 2\pi^2(v^2+2v\sigma'+\sigma'^2)\right]$$

$$= \sigma'^2\left[-\frac{\mu^2}{2}+\lambda v^2+\frac{v^2\lambda}{2}\right] + \pi^2\left[-\frac{\mu^2}{2}+\frac{\lambda v^2}{2}\right]$$

$$m_\pi = 0$$
$$m_{\sigma'} = \sqrt{2\mu^2}$$

Ohranjeni tokovi inu naboji:

$$\begin{bmatrix}\sigma\\ \pi\end{bmatrix} \longrightarrow \begin{bmatrix}\sigma'\\ \pi'\end{bmatrix} \overset{\text{inf.}}{=} \begin{bmatrix}1 & \alpha\\ -\alpha & 1\end{bmatrix}\begin{bmatrix}\sigma\\ \pi\end{bmatrix}$$

$\sigma' = \sigma + \alpha\pi$; $\delta\sigma = \sigma' - \sigma = \partial\pi$ $\qquad \frac{\delta\sigma}{\delta\alpha} = \pi$

$\pi' =$; $\delta\pi =$ $\qquad \frac{\delta\pi}{\delta\alpha} = -\sigma$

$$j_\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\pi)}\frac{\delta\pi}{\delta\alpha} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\sigma)}\frac{\delta\sigma}{\delta\alpha} = \partial_\mu\pi(-\sigma) + \partial_\mu\sigma\,\pi$$

$$Q = \int d^3x\, j_0(x) = \int d^3x\left(-(\partial_0\pi)\sigma + (\partial_0\sigma)\pi\right)$$

Algebra

$$\left[\partial_0\pi(t,x),\ \pi(t,x)\right] = -i\delta^{(3)}(\vec{x}-\vec{y})$$

$$\Rightarrow \left[Q,\pi(0)\right] = \int d^3x\left[(\partial_0\sigma(x))\pi - (\partial_0\pi(x))\sigma(x),\ \pi(0)\right]$$
$$= -\int d^3x\left[\partial_0\pi(x)\sigma(x),\ \pi(0)\right] = i\sigma(0)$$

$$\left[Q,\sigma(0)\right] = -i\pi(0)$$

$$\langle 0|\left[Q,\pi(0)\right]|0\rangle \overset{(*)}{=} i\langle 0|\sigma(0)|0\rangle \equiv v =$$

$$(*) = \sum_n (2\pi)^3\delta^3(\vec{p}_n)\left\{\langle 0|j_0(0)|0\rangle\langle n|\pi(0)|0\rangle e^{iE_n t} - \langle 0|\pi(0)|0\rangle\langle n|j_0(0)|0\rangle e^{-iE_n t}\right\}$$

$$\sum_n \longrightarrow \int\frac{d^3p_n}{2p_n^0(2\pi)^3} \quad ; \quad |n\rangle \longrightarrow |p_n\rangle \longrightarrow |\pi(p_n)\rangle$$

$$(*) = \int\frac{d^3p_n}{2p_n^0}\delta^{(3)}(p_n)\left\{\langle 0|j_0|\pi(p_n)\rangle\langle\pi(p_n)|\pi(0)|0\rangle e^{ip_n^0 t} - \langle 0|\pi(0)|\pi(\vec{p}_n)\rangle\langle\pi(\vec{p}_n)|j_0|0\rangle e^{-ip_n t}\right\}$$

$$\pi(x) = \int\frac{d^3p}{(2\pi)^3 2E_p}\left[a(p)e^{-ipx} + a^\dagger(p)e^{ipx}\right] \quad ; \quad \left[a(p),a^\dagger(p')\right] = (2\pi)^3 2p_0\delta^3(\vec{p}-\vec{p}')$$

$a^\dagger(p)|0\rangle = |\pi(p)\rangle$
$a(p)|0\rangle = 0$ $\Longrightarrow$ $\langle 0|\int\frac{d^3p}{(2\pi)^3 2p_0}\underbrace{a(p)a^\dagger(p)}_{[a(p),a^\dagger(p)]}|0\rangle = 1 = \langle 0|\pi(0)|\pi(p_n)\rangle$

$$(*) = \int\frac{d^3p_n}{2p_n^0}\delta^{(3)}(\vec{p}_n)\left[\langle 0|j_0(0)|\pi(p_n)\rangle - \underbrace{\langle\pi(\vec{p}_n)|j_0(0)|0\rangle}_{\langle 0|j_0(0)|\pi(\vec{p}_n)\rangle^\dagger}\right] = iv$$

$$\Rightarrow \langle 0|j_0(0)|\pi(\vec{p})\rangle = ivp^0$$

$$\boxed{\langle 0|j_\mu(0)|\pi(\vec{p})\rangle = iv\cdot p_\mu}$$

$\langle 0|\sigma(0)|0\rangle$

5. teden

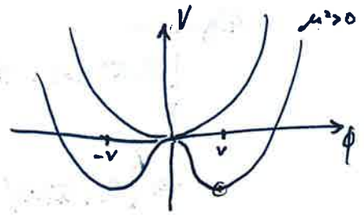
Spontani zlom diskretne

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}\mu^2\phi^2 - \frac{\lambda}{4}\phi^4$$

kjer je ϕ realno polje. Simetrija $\phi \to -\phi$ je posledica odsotnosti ϕ^3 člena.

Osnovno stanje:

$$\frac{\partial V}{\partial \phi} = -\mu^2\phi + \lambda\phi^3 = \phi(-\mu^2 + \lambda\phi^2)$$



1) $\mu^2 \neq 0$: $\phi_0 = 0$

2) $\mu^2 \lesssim 0$: $\phi_0 = \pm\sqrt{\frac{\mu^2}{\lambda}}$

V QFT : $\phi_0 = \langle 0|\phi|0\rangle = +v$

Razvijamo gibanje polja okoli minimuma $\phi = v + \phi'$

$$\mathcal{L}(\phi') = \frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}\mu^2(v+\phi')^2 - \frac{\lambda}{4}(v^2+\phi'^2+2v\phi')^2$$

$$= \frac{1}{2}(\partial_\mu\phi')^2 + \frac{1}{2}\mu^2v^2 + \mu^2v\phi' + \frac{1}{2}\mu^2\phi'^2 - \frac{\lambda}{4}(v^4+\phi'^4+4v^2\phi'^2+2v^2\phi'^2+4\ldots$$

$$= \ldots - \frac{\phi'^2}{2}\left[-\mu^2 + 2\lambda v^2 + \underbrace{\lambda v^2}_{\mu^2}\right] = -\frac{\phi'^2}{2}\,2\mu^2 + \mathcal{O}(\phi'^{k\neq 2})$$

nova masa^2

V diskretnem primeru ne dobimo ~~brezmasnih~~ prostostnih stopenj (ni brezmasnih bosonov).

MODEL σ SPONTANO ZLOMLJENE $SU(2)_L \times SU(2)_R$

→ Model pionov in nukleonov
→ polja $\vec{\pi} = (\pi^1\ \pi^2\ \pi^3)$, $N = (p, N)^T$, σ
→ prisotna je izospinska simetrija $SU(2)_V$

$$\begin{aligned}
\sigma &\longrightarrow \sigma' = \sigma \quad \text{(singlet)} ; && \delta\sigma = 0\\
\vec{\pi} &\longrightarrow \vec{\pi} + \vec{\alpha}\times\vec{\pi} ; && \delta\vec{\pi}^i = \varepsilon_{ijk}\alpha_j\pi^k\\
N &\longrightarrow N - i\vec{\alpha}_i\frac{\vec{\tau}}{2}N ; && \delta N_a = -i\vec{\alpha}\cdot\frac{\vec{\tau}}{2}_{ab}N_b
\end{aligned}$$

Invariantna Lagranova gostota

$$\mathcal{L} = \frac{1}{2}[(\partial_\mu\sigma)^2 + (\partial_\mu\vec{\pi})^2] + i\bar{N}i\not{\partial}N - g\bar{N}[\sigma + \vec{\tau}\cdot\vec{\pi}\gamma_5]N - V(\sigma^2+\vec{\pi}^2)$$

Invariianca na izospin ⟶ ohranjeni tokovi:

$$j^a_\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}\frac{\delta N}{\delta\alpha_a} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\pi^b)}\frac{\delta\pi^b}{\delta\alpha_a}$$

$$j^a_\mu = \bar{N}\gamma_\mu\frac{\tau^a}{2}N + \varepsilon_{abc}\pi^b\partial_\mu\pi^c$$

$$Q = \int j^a_0\,d^3x$$

$\mathcal{L}$ ima dodatno simetrijo

$$\begin{aligned}
\sigma &\longrightarrow \sigma + \vec{\beta}\cdot\vec{\pi} ; && \delta\sigma = \vec{\beta}\cdot\vec{\pi}\\
\vec{\pi} &\longrightarrow \vec{\pi} - \vec{\beta}\sigma ; && \delta\vec{\pi} = -\vec{\beta}\sigma\\
N &\longrightarrow N + i\vec{\beta}\cdot\frac{\vec{\tau}}{2}\gamma_5 N ; && \delta N = i\vec{\beta}\cdot\frac{\vec{\tau}}{2}\gamma_5 N
\end{aligned}$$

V matrični obliki za σ, π zapišemo po komponentah transf. lastnosti:

$$\begin{bmatrix}\sigma'\\ \pi^{1\prime}\\ \pi^{2\prime}\\ \pi^{3\prime}\end{bmatrix} = \begin{bmatrix}\sigma\\ \pi^1\\ \pi^2\\ \pi^3\end{bmatrix} + \beta_1\begin{bmatrix}0&1&&\\-1&0&&\\&&0&\\&&&0\end{bmatrix}\begin{bmatrix}\sigma\\ \pi^1\\ \pi^2\\ \pi^3\end{bmatrix} + \beta_2\begin{bmatrix}0&0&1&\\0&&&\\-1&&&\end{bmatrix}\begin{bmatrix}\sigma\\ \pi^1\\ \pi^2\\ \pi^3\end{bmatrix} + \ldots$$

Tvorijo algebro $SU(2)$

$$j^a_{A\mu} = \bar{N}\gamma_\mu\gamma_5\frac{\tau^a}{2}N + \underbrace{(\partial_\mu\sigma)\pi^a - (\partial_\mu\pi^a)\sigma}$$

$$Q^a_A = \int j^a_{A0}\,d^3x$$

Algebra nabojev:

$$\begin{aligned}
&[Q^a, Q^b] = i\varepsilon_{abc}Q^c\\
&[Q^a\ Q^b_A] = i\varepsilon_{abc}Q^c_A\\
&[Q^a_A, Q^b_A] = i\varepsilon_{abc}Q^c
\end{aligned}
\xrightarrow[\text{baza}]{\text{nova}}
\begin{aligned}
&Q^a_{L,R} = \frac{1}{2}(Q^a \mp Q^a_A) \qquad SU(2)_L \times SU(2)_R\\
&[Q^a_{L,R}, Q^b_{L,R}] = i\varepsilon_{abc}Q^c_{L,R}\\
&[Q_L, Q_R] = 0
\end{aligned}$$

Predpostavimo, da je pričakovana vrednost $\langle 0|\sigma|0\rangle = v \neq 0$, $\langle 0|\vec{\pi}|0\rangle = 0$.

$$[Q^a_A, \pi^b] = i\sigma\delta^{ab}$$

Nukleoni: $-g\bar{N}(\sigma' + i\vec{\pi}\cdot\vec{I}\gamma_5)N =$

$= -g\bar{N}(\sigma' + v + i\vec{\pi}\cdot\vec{I}\gamma_5)N =$

$= -gv\bar{N}N - g\bar{N}(\sigma' + i\vec{\pi}\cdot\vec{I}\gamma_5)N$

Spontani zlom $SU(2)_L \times SU(2)_R \longrightarrow SU$

Povezava med ostanim stanjem močne int. $\langle 0|\sigma|0\rangle$ in matričnimi elementi tokov (EM, šibki)

$$\langle 0|j_A^a|\pi^b(\vec{p})\rangle = i f_\pi p_\mu \delta^{ab} \Longrightarrow \frac{\Gamma(\pi^- \to \mu\nu)}{\Gamma(\mu \to \nu\nu)}$$

($\langle 0|\sigma|0\rangle$)

Translacijska inv. v Poinc. grupi (operator $\mathcal{O}$)

$\partial_\mu \mathcal{O} = -i[\mathcal{O}, P_\mu]$

$\langle 0|\partial_\mu j_A^{a\mu}|\pi^b(\vec{p})\rangle = \langle 0|-i[\int j_{A0}^a, p^\mu]|\pi(\vec{p})\rangle$

$= -i\langle 0|\int j_{A0}^a p^\mu - p^\mu \int j_{A\mu}^a|\pi^b(\vec{p})\rangle$

$= -ip^\mu \langle 0|\int j_{A\mu}^a|\pi^b(p)\rangle$

$f_\pi p^2 \delta^{ab} = f_\pi m_\pi^2 \delta^{ab}$

$P_\mu|\pi(\vec{p})\rangle = P_\mu|\pi(p)$

$\langle 0|P^\mu = 0$

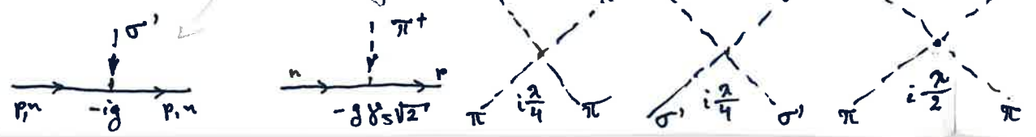
$\langle \partial_\mu j_A^{a\mu}\rangle = 0 \iff m_\pi^2 = 0$

„Delna ohranitev axialnega toka (PCAC)"

Interakcije med nukleoni in $\sigma, \vec{\pi}$

$$\mathcal{L}(N, \sigma', \vec{\pi}) = \tfrac{1}{2}[(\partial_\mu\sigma')^2 + (\partial_\mu\vec{\pi})^2] + \bar{N}i\partial\!\!\!/N - gv\bar{N}N - \mu^2\sigma'^2 - g\bar{N}(\sigma' + i\vec{\pi}\cdot\vec{I}\gamma_5)N + \sigma'^3 v\lambda + \tfrac{\lambda}{4}(\vec{\pi}^4 + \sigma'^4 + 2\sigma'^2\vec{\pi}^2)$$

Feynmanova pravila



Trik

$\vec{\pi}^2 = \pi^{+2} + \pi^{-2} + \pi^{3\,2} = 2\,Tr[\vec{I}\cdot\vec{\pi}] = \pi^{+0^2} + \pi^+\pi^- + \pi^-\pi^+$

$\hookrightarrow Tr[I^i\pi^i I^j\pi^j]$, $i\frac{\lambda}{2} = Tr[I^iI^j]\pi^i\pi^j$

V $\mathcal{L}$ dodamo člen $\mathcal{L}_1$; $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$:

$\mathcal{L}_1 = a\sigma \Longrightarrow \mathcal{L}_1' = a\sigma' \approx a(\sigma + \beta_i\pi^i) = \mathcal{L}_1 + a\beta_i\pi^i \neq \mathcal{L}_1$ (ta bo)

Spremenila se bo oblika axialnega toka

$$j_A^{a\mu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu N)}\frac{\delta N}{\delta\beta^i} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\pi^b)}\frac{\delta\pi^b}{\delta\beta^i} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\sigma)}\frac{\delta\sigma}{\delta\beta_i} = \bar{N}\gamma^\mu I^a\gamma_5 N - \partial^\mu\vec{\pi}\vec{\sigma} + \partial_\mu\sigma\vec{\pi}$$

Ta tok po metodi GML ni več ohranjen:

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\beta_i}\beta_i + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\beta_i)}\partial_\mu\beta_i$$

$\partial_\mu j_A^{i\mu}$; $j_A^{i\mu}$

Variacija $\mathcal{L}$: $N' \approx N + i\beta_i I^i\gamma_5 N$

$N'^\dagger \approx N^\dagger - i\beta_i I^i N^\dagger\gamma_5$

$\bar{N}' \approx \bar{N} + i\beta_i I^i\bar{N}\gamma_5$

$\Big\} \frac{\partial\mathcal{L}_0}{\partial\beta_i} = 0$

$\mathcal{L}_0' = \frac{1}{2}(\partial_\mu\sigma + \partial_\mu\beta_i\pi^i + \beta_i\partial_\mu\pi^i)^2$

$+ \frac{1}{2}(\partial_\mu\pi^i + \partial_\mu\beta_i\sigma - \beta_i\partial_\mu\sigma)^2 \Big\} (\partial_\mu\pi^i)^2$

$+ \bar{N}i\partial\!\!\!/N - \partial\!\!\!/\beta_i\bar{N}\gamma_\mu\gamma_5 IN - \beta_i\bar{N}\underbrace{\{\gamma_\mu\gamma_5\}}_{0} I^i\partial^\mu N$

$\delta\mathcal{L}_0 = \mathcal{L}_0' - \mathcal{L}_0 = a\beta_i\dot{\pi}_i$

$= \frac{\partial\mathcal{L}}{\partial\beta_i} = a\pi^i \longrightarrow \partial_\mu j_A^{i\mu} = a\pi^i$

$\langle 0|\partial_\mu j_A^{i\mu}|\pi^j(\vec{p})\rangle = m_\pi^2\delta^{ij}f_\pi$

$\Rightarrow a = m_\pi^2 f_\pi = m_\pi^2 v$, Pioni so zdaj psevdo~~bozoni~~ bozoni

$\mathcal{L}_1 \doteq m_\pi^2 v\sigma$

Razvoj perturbacij ne bo v $g \sim 10$, ampak v $\frac{m_\pi^2}{(4\pi v)^2}$, ali $\frac{|\vec{p}|^2}{(4\pi v)^2} \sim \frac{E^2}{(4\pi v)^2}$

Upodobitvena inv. modela σ

Poglejmo drugačno upodobitev:

Uvedemo polja $\Sigma \in$ ~~[illegible]~~ hermitska matrika: , N_L, N_R

$$\Sigma = \sigma \cdot I_{2\times 2} + 2i \underbrace{\vec{I}}_{\frac{\vec{\tau}}{2}} \cdot \vec{\pi} = \begin{bmatrix} \sigma + \pi_3 & \pi_1 - i\pi_2 \\ \pi_1 + i\pi_2 & \sigma - \pi_3 \end{bmatrix}$$

$$\psi_{L,R} = P_{L,R}\,\psi \quad ; \quad \bar{\psi}\gamma^\mu\psi = \bar{\psi}_L\gamma^\mu\psi_L + \bar{\psi}_R\gamma^\mu\psi_R$$
$$\bar{\psi}\psi = \bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L$$

$$\sigma^2 + \vec{\pi}^2 = \tfrac{1}{2}\,\mathrm{Trace}[\Sigma\Sigma^\dagger]$$
$$\sigma I = \tfrac{1}{4}(\Sigma^\dagger + \Sigma)$$
$$2i\vec{I}\cdot\vec{\pi} = \tfrac{1}{2}(\Sigma - \Sigma^\dagger)$$

$$\Rightarrow \mathcal{L} = \bar{N}_L i\partial\!\!\!/ N_L + \bar{N}_R i\partial\!\!\!/ N_R + \tfrac{1}{4}\mathrm{Trace}[\partial_\mu\Sigma\partial^\mu\Sigma^\dagger] + \tfrac{1}{4}\mu^2\mathrm{Trace}[\Sigma^\dagger\Sigma] - \tfrac{\lambda}{4}(\mathrm{Trace}[\Sigma^\dagger\Sigma])^2 - g(\bar{N}_L\Sigma N_R + \bar{N}_R\Sigma^\dagger N_L) = \mathcal{L}_0$$

$\mathcal{L}$ je manifestno inv. na $SU(2)_L \times SU(2)_R$

$$N_L' = U_L N_L \quad ; \quad U_L^\dagger U_L = 1\,,\ U_L = e^{i\vec{\alpha}_L\cdot\vec{I}P_L}$$
$$N_R' = U_R N_R \quad ; \quad U_R^\dagger U_R = 1\,,\ U_R = e^{i\vec{\alpha}_R\cdot\vec{I}P_R}$$

$$\Sigma' = U_L\Sigma U^\dagger, \quad \Sigma'^\dagger = U_L\Sigma^\dagger U_L^\dagger \quad \hookrightarrow \begin{matrix} P_L N_L = 0 & P_R N_R = N_R \\ P_L N_R = 0\,, & P_R N_L = 0 \end{matrix}$$

$$\mathcal{L}(N_{L,R}, \Sigma) \xrightarrow{U_L,U_R} \mathcal{L}(N', \Sigma') = \bar{N}_L' i\partial\!\!\!/ N_L' + \bar{N}' i\partial\!\!\!/ N' + \tfrac{1}{4}\mathrm{Trace}[\partial_\mu\Sigma'\partial^\mu\Sigma'^\dagger] + \tfrac{1}{4}\,{}^2\mathrm{Trace}[\Sigma'^\dagger\Sigma'] - \tfrac{\lambda}{4}(\mathrm{Trace}\,\Sigma'\Sigma'^\dagger)^2 - g(\bar{N}_L'\Sigma' N_R + \bar{N}_R'\Sigma'^\dagger N_L)$$
$$= \bar{N}_R N_L^\dagger i\partial\!\!\!/ N_L N_L + \bar{N}_R N_R^\dagger i\partial\!\!\!/ N_R N_R + \tfrac{1}{4}\mathrm{Trace}[U_L^\dagger\partial_\mu\Sigma U_R U_R^\dagger\partial^\mu\Sigma^\dagger U_L] + \tfrac{1}{4}\,{}^2\mathrm{Trace}[U_R\Sigma^\dagger U_L^\dagger U_L\Sigma U_R^\dagger] - \tfrac{\lambda}{4}(\mathrm{Trace}[U_L\Sigma U_R^\dagger U_R\Sigma^\dagger U_L])^2$$
$$- g(\bar{N}U_L^\dagger U_L\Sigma U_R^\dagger U_R N_R + \bar{N}_R N^\dagger U\,\Sigma^\dagger\gamma^0 UU) = \mathcal{L}(N_{L,R}, \Sigma)$$

$$\langle 0|\sigma|0\rangle = v = +\sqrt{\tfrac{\mu^2}{\lambda}} \quad \text{in} \quad \sigma = v + \sigma'$$
$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\sigma'\partial^\mu\sigma' - 2\mu^2\sigma'^2) + \tfrac{1}{2}\partial_\mu\vec{\pi}\partial^\mu\vec{\pi} - \lambda v\sigma'(\sigma'^2 + \vec{\pi}^2) - \tfrac{\lambda}{4}(\sigma'^2 + \vec{\pi}^2)^2$$
$$+ \bar{N}(i\partial\!\!\!/ - gv)N - g\bar{N}(\sigma' + i\vec{I}\cdot\vec{\pi}\gamma_5)N$$

Eksponentna upodobitev kiralne simetrije v zlomljeni fazi

Nelinearna

$$\Sigma = \sigma I + i2\vec{I}\cdot\vec{\pi} \longrightarrow (v+s)U \quad ; \quad U = e^{\frac{2i\vec{I}\cdot\vec{\pi}}{v}} \approx \mathbb{1} + \tfrac{2i}{v}\vec{I}\cdot\vec{\pi} + \dots$$

$$\Rightarrow \mathcal{L} = \tfrac{1}{2}((\partial_\mu s)^2 - 2\mu^2 s^2) + \tfrac{(v+s)^2}{4}\mathrm{Trace}[\partial_\mu U\partial^\mu U^\dagger] - \lambda v s^3 - \tfrac{\lambda}{4}s^4 + \bar{\psi}i\partial\!\!\!/\psi - g(v+s)[\bar{\psi}_L U\psi_R + \bar{\psi}_R U^\dagger\psi_L]$$

Fluktuacije skalarnega s nas zdej ne bodo zanimale. Poglejmo limito ko gre m_s oz $\mu^2 \to \infty$ ($v^2 = \frac{\mu^2}{\lambda} = f_\pi^2$) in s tem $\lambda \to \infty$ ker hočemo razpadno konst. = konč.:

$$\Rightarrow s(x) \longrightarrow 0$$
$$\Rightarrow \boxed{\mathcal{L} = \tfrac{f_\pi^2}{4}\mathrm{Tr}[\partial_\mu U\partial^\mu U^\dagger] + \bar{\psi}(i\partial\!\!\!/ - \not{v})\psi - vg(\bar{\psi}_L U\psi_R + \bar{\psi}_R U^\dagger\psi_L)}$$

($\not{v} = \psi_L\not{\partial}\psi_L + \bar{\psi}_R\not{\partial}\psi_R$; $\psi_L \to U_L\psi_L$, $U_L = e^{i\vec{\alpha}_L\cdot\vec{I}P_L}$; $\psi_R \to U_R\psi_R$)

Poglejmo ekvivalenco s tistim prvim

$$\Sigma = \sigma\cdot I + i2\vec{I}\cdot\vec{\pi} = \sigma\left[\mathbb{1} + 2i\frac{\vec{I}\cdot\vec{\pi}}{\sigma}\right] = (v+\sigma')\left[\mathbb{1} + 2i\frac{\vec{I}\cdot\vec{\pi}}{v+\sigma'}\right] \approx v\left[\mathbb{1} + 2i\frac{\vec{I}\cdot\vec{\pi}}{v}\right] + \mathcal{O}(\sigma')$$

Eksplicitni zlom $SU(2)$

$$\mathcal{L}_1 = \frac{f_\pi^2}{4}m_\pi^2\,\mathrm{Trace}[U^\dagger + U]$$

Uvedemo matriko χ:

$$\frac{f_\pi^2}{2}\mathrm{Trace}[\chi U^\dagger + U\chi^\dagger]$$

$$\chi = B_0\begin{bmatrix} m_u & \\ & m_d \end{bmatrix} \Rightarrow m_\pi \propto \mathrm{Trace}[\chi]$$

$$\mathcal{L}_0 = \tfrac{1}{4}[\dots + 2\partial_\mu\pi^+\partial^\mu\pi^- + 2\partial_\mu\pi^-\partial^\mu\pi^+] = \dots + \partial_\mu\pi^+\partial^\mu\pi^-$$

$$\mathcal{L}_1 = -\frac{B_0}{2}f_\pi^2\,\mathrm{Trace}\left[\begin{bmatrix} m_u & \\ & m_d \end{bmatrix}\frac{2}{f_\pi^2}\begin{bmatrix} \frac{\pi^{0\,2}}{4} + \frac{\pi^+\pi^-}{2} & \\ & \frac{\pi^{0\,2}}{4} + \frac{\pi^+\pi^-}{2} \end{bmatrix}\right] + \frac{2}{f_\pi^2}\begin{bmatrix} \frac{\pi^{0\,2}}{4} + \frac{\pi^+\pi^-}{2} & \\ & \frac{\pi^{0\,2}}{4} + \frac{\pi^+\pi^-}{2} \end{bmatrix}\begin{bmatrix} m_u & \\ & m_d \end{bmatrix}$$
$$= -B_0(m_u + m_d)\pi^+\pi^-$$

$$m_{\pi^\pm}^2 = B_0(m_u + m_d) \quad \Rightarrow \quad m_{\pi^\pm}^2 > 0$$
$$m_{\pi^0}^2 = B_0(m_u + m_d) \qquad\qquad m_{\pi^\pm}^2 = m_{\pi^0}^2$$

Razsplošimo na večje prostotnih stopenj : kaoni

$SU(3)_L \times SU(3)_R \xrightarrow{\text{spont. zlom}} SU(3)_V$ + eksplicitni zlom

$T^a = \frac{\lambda_a}{2}$

$$U = e^{\frac{\sqrt{2}\,i}{f}\frac{\lambda_a}{\sqrt{2}}\phi^a}$$

$$\pi = \begin{bmatrix} \frac{\pi^0}{\sqrt{2}} + \frac{\eta_8}{\sqrt{6}} & \pi^+ & K^+ \\ \pi^- & -\frac{\pi^0}{\sqrt{2}} + \frac{\eta_8}{\sqrt{6}} & K^0 \\ K^- & \bar{K}^0 & -\frac{2\eta_8}{\sqrt{6}} \end{bmatrix}$$

$$U \approx \mathbb{1} + \frac{\sqrt{2}\,i}{f}\pi - \frac{1}{2}\pi^2 + \dots$$

$$\pi^2 = \begin{bmatrix} \left(\frac{\pi^0}{\sqrt{2}} + \frac{\eta_8}{\sqrt{6}}\right)^2 + \pi^+\pi^- + K^+K^- & & \\ & \pi^-\pi^+ + \left(-\frac{\pi^0}{\sqrt{2}} + \frac{\eta_8}{\sqrt{6}}\right)^2 + \bar{K}^0K^0 & \\ & & K^-K^+ \end{bmatrix}$$

$$\mathcal{L} = \frac{f^2}{4}\,\mathrm{Trace}\left[\partial_\mu U \partial^\mu U^\dagger\right] + \frac{f^2}{2}B_0\,\mathrm{Trace}\left[\chi^\dagger U + U^\dagger \chi\right]$$

$\chi = \mathrm{diag}(m_u, m_d, m_s)$

Mase pseudo-skalarnih bozonov

$m_{\pi^\pm}^2 = B_0(m_u + m_d)$

$m_{K^0}^2 = B_0(m_s + m_d)$

$m_{K^\pm}^2 = B_0(m_s + m_u)$

$m_{\eta_8}^2 = \frac{1}{3}B_0(4m_s + m_u + m_d)$

V izospinskem limiti $m_u = m_d$:

$$\left.\begin{aligned} m_K^2 &= B_0(m_u + m_s) \\ m_\pi^2 &= B_0(2m_u) \end{aligned}\right\} \quad m_s B_0 = m_K^2 - \frac{1}{2}m_\pi^2$$

$$\Rightarrow \quad m_{\eta_8}^2 = \frac{1}{3}(4m_K^2 - m_\pi^2) \qquad \text{Gell-Mann–Okubo}$$

Eksperimentalno : $m_\pi \approx 0{,}139\ \mathrm{GeV}$, $m_K \approx 0{,}495\ \mathrm{GeV}$ $\Big\}$ $m_{\eta_8} = 0{,}565\ \mathrm{GeV}$

$m_{\eta'} = 0{,}958\ \mathrm{GeV}$

$m_\eta = 0{,}58\ \mathrm{GeV}$

Posledica $U(1)_A$ axialne anomalije

Povezava s teorijo lahkih kvarkov in kromodin.

$$\begin{aligned} \mathcal{L}_{QCD} &= \sum_{i=u,d,s} \bar{q}_i(i\not{\partial} - m_i)q_i + \dots \\ &= \sum_i \left(\bar{q}_{iL}\, i\not{\partial}\, q_{iL} + \bar{q}_{iR}\, i\not{\partial}\, q_{iR} - \bar{q}_{iL}\, m_i\, q_{iR} - \bar{q}_{iR}\, m_i\, q_{iL}\right) + \dots \\ &= \bar{Q}_L\, i\not{\partial}\, Q_L + \bar{Q}_R\, i\not{\partial} Q_R - \bar{Q}_L M_q Q_R - \bar{Q}_R M_q^\dagger Q_L \end{aligned}$$

$\mathcal{L}$ ima približno ($M_q \to 0$) $SU(3)_L \times SU(3)_R$ simetrijo in M_q predstavlja eksplicitni zlom.

$$\left.\begin{aligned} m_K^2 &\approx \frac{m_s}{2B_0} \\ m_\pi^2 &\approx \frac{1}{2B_0}(2m_u^2 - m_\pi^2) \end{aligned}\right\} \quad \frac{m_u}{m_s} = \frac{1}{26}$$

Teorija lahkih kvarkov ima to isto $SU(3) \times SU(3) \longrightarrow SU(3)_V$

$\Rightarrow$ ujetje kvarkov in barve

$\Rightarrow$ $\sigma \sim \frac{1}{2}\bar{q}_L q_R \frac{1}{2}$ in ker $\langle\sigma\rangle \neq 0$ sledi $\langle \bar{q}_L q_R\rangle \neq 0$ „kvarkovski kondenzat"

$\Rightarrow$ $\chi = M_q \implies m_{\pi,K}^2 \propto m_q$

med tem ko $m_{\eta,\rho} \propto \langle \bar{q}_i q_i\rangle \equiv \Lambda_{\chi=V}$

Trdimo, da je to efektivna teorija QCD pri nizkih energijah.
Poglejmo si torej perturbat. razvoj v kir. teoriji.

$$\mathcal{L}_0 = \frac{f_\pi^2}{4}\,\mathrm{Trace}\left[\partial_\mu U \partial^\mu U^\dagger\right] \approx \frac{1}{2}\partial_\mu \vec{\pi}\,\partial^\mu \vec{\pi} + \frac{1}{6}\frac{1}{f_\pi^2}\left[(\vec{\pi}\,\partial_\mu \vec{\pi})^2 + \dots\right]$$

$(\vec{\pi}\,\partial_\mu\vec{\pi})^2 \propto \frac{p^2}{f_\pi^2}$ nerenormal. že 1. int. v razvoju

Sipanje $\pi\pi \longrightarrow \pi\pi$:

$$\mathcal{M} \sim \frac{s}{f_\pi^2} + \dots + \dots + \dots$$

$s = (p_1 + p_2)^2$ potencialno krči unitarnost

$\pi(p_1)$, p_3, $\pi(p_2)$, p_4

$\sim \frac{s^2}{16\pi^2 f_\pi^4}\log\left(\frac{\Lambda^2}{s}\right)$ še hujša kršitev unitarnosti

Pri nizkih energijah je $\sigma \propto s$ po naporedi

Pri $2E_\pi \sim 0{,}77\,\mathrm{GeV}$: resonanca ρ

$$\Lambda \leq m_\rho \leq 4\pi f_\pi$$

$\frac{E_\pi}{4\pi f_\pi}$ konvergira za $E_\pi < \frac{m_\rho}{2}$

Kontra členi, ki poravnajo divergence

$$\mathcal{L}_{c.t.} = l_1 \cdot \mathrm{Trace}\left[(\partial_\mu U \partial^\mu U^\dagger)^2\right] + l_2 \cdot \mathrm{Trace}\left[\partial_\mu U \partial_\nu U^\dagger \partial^\mu U \partial^\nu U^\dagger\right] + \text{ostalih}$$

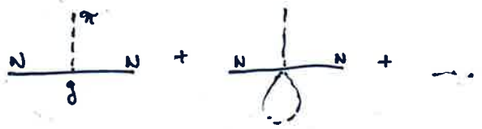
$$\Rightarrow \mathcal{M}(\pi\pi \to \pi\pi) \sim \frac{s}{f_\pi^2}\left[1 + \frac{s}{(4\pi f_\pi)^2}\left[\frac{1}{2}\log\frac{\mu^2}{s} + \frac{1}{3}\bar{l}_1(\mu) + \dots\right]\right.$$

Nizkoenergijski razvoj $\left(\frac{E}{\Lambda}\right)^2 \sim \left(\frac{s}{(4\pi f_\pi)^2}\right) \iff \mathcal{L}$ po $(\partial_\mu U)^p$

$$\mathcal{L} = \underset{\uparrow 2}{\mathcal{L}^{(p=2)}} + \underset{\uparrow 10}{\mathcal{L}^{(4)}} + \underset{\uparrow}{\mathcal{L}^{(6)}} + \dots$$

$$\mathcal{M} = \underbrace{\mathcal{M}^{(2)}_{tree}}_{\text{končno } \sim s^1} + \underbrace{\left[\mathcal{M}^{(2)}_{1\,loop} + \mathcal{M}^{(4)}_{tree}\right]}_{\text{končno } \sim s^2} + \underbrace{\left[\mathcal{M}^{(2)}_{2\,loop} + \mathcal{M}^{(4)}_{1\,loop} + \mathcal{M}^{(6)}_{tree}\right]}_{\text{končno } \sim s^0} + \dots$$

Perturbativni razvoj $\frac{E_\pi}{\Lambda}$ namesto $g \sim 10$



Ob eksplicitnem členu : dvojni razvoj

$$\frac{s}{(4\pi f_\pi)^2}, \left(\frac{m_\pi}{4\pi f}\right)^2$$

f^2 ; χ

Nerelativistična kvantna mehan

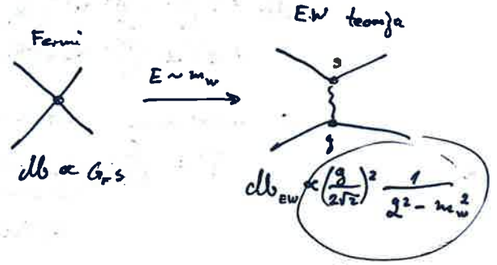
$\left(\frac{\vec{p}_\pi}{m_\pi}\right)$, $4\pi f \gg m_\pi$

EFEKTIVNA TEORIJA POLJA

Vsebuje vse relevantne (v spektru) prostostne stopnje in pri tem ne rabimo skrbet za renormaliz.

Primer : Fermijeva teorija šibke

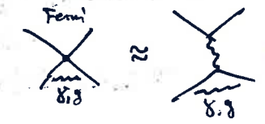
$$\mathcal{L}_{Fermi} = \frac{G_F}{\sqrt{2}} j_\mu j^{\mu\dagger}$$



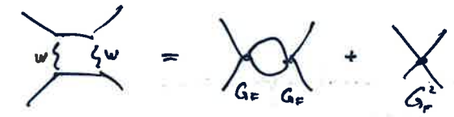
Lahko sistematično razvijemo do višjih redov (zanke + kontračleni)

$$\mathcal{L}_{Fermi} \longrightarrow \mathcal{L}^{d=4}_{QED+QCD} + \mathcal{L}^{d=6}_{Fermi} + \mathcal{L}^{d=8}_{c.t.} + \dots$$

$$\mathcal{M}_{weak} = \mathcal{M}^{d=6}_{tree} + \left[\mathcal{M}^{d=6}_{1\,loop} + \mathcal{M}^{d=8}_{tree}\right] + \dots$$

1) 

$$\longrightarrow G_F = G_F^{(0)}\left[1 + \frac{\alpha}{4\pi}\log\frac{\mu^2}{m_W^2}\right]$$

2) 

popravljamo nove kontrt. člene dim 8

Vaja : Razpadna širina $\Gamma(\pi^+ \to \mu^+\nu)$

$$\mathcal{L}_{Fermi} = 4\frac{G_F}{\sqrt{2}} j_\mu j^\mu_\dagger \quad ; \quad j^\mu = \underbrace{\bar{\psi}^{(\nu)}\gamma^\mu P_L \psi^{(\nu)}}_{j^\mu_L = \frac{1}{2}j_V - \frac{1}{2}j^\mu_A} + V_{ud}\,\bar{\psi}^{(d)}\gamma^\mu P_L \psi^{(u)}$$

$$-\frac{1}{2}\left[(\partial^\mu\sigma)\pi^+ - \sigma(\partial^\mu\pi^+)\right] = \left[\text{s=0} \downarrow + \dots\right] = \frac{f_\pi}{2}\partial^\mu\pi^+ + \dots$$

$$(2\pi)^4\delta^{(4)}(\Sigma p_i)\, i\mathcal{M}(\pi^+ \to \mu^+\nu) = 4\frac{G_F}{\sqrt{2}}\int d^4x\, \langle \mu\nu|\, \underbrace{\mathcal{L}_{int}}_{\mathcal{L}_{Fermi}}\, |\pi^+\rangle$$

$$\to S = \langle e^{-i\int \mathcal{H}_{int} d^4x}\rangle$$

$$\mathcal{M} = 4\cdot\frac{G_F}{\sqrt{2}}\bar{u}^{(\nu)}\gamma_\mu P_L v^{(\nu)}\underbrace{\langle 0|j^\mu|\pi(p_\pi)\rangle}_{\frac{if_\pi}{2}p_\pi V_{ud}}$$



$p = k_1 + k_2$

$(\not{k}_1 + m_\mu)\, v^{(\mu)}(k_1) = 0$

$\bar{u}^{(\nu)} \not{k}_2 = 0$

$$d\Gamma = 4 \frac{G_F}{\sqrt{2}} \frac{g_\pi}{2} \bar{u}^{(\nu)} (\not{k}_1 + \not{k}_2) P_L v^{(\mu)} =$$

$$= 4 V_{ud} \frac{G_F}{\sqrt{2}} m_\mu \frac{g_\pi}{2} \bar{u}^{(\nu)} P_R v^{(\mu)}$$

V mirovnem sistemu π :

$$\Gamma = \frac{1}{8\pi m_\pi^2} \sum_s d\Gamma_s d\Gamma_s^+ = \frac{G_F^2 |V_{ud}|^2}{8\pi} g_\pi^2 m_\mu^2 m_\pi \left(1 - \frac{m_\mu^2}{m_\pi^2}\right)^2 \left[1 + \mathcal{O}\left(\frac{m_e^2}{m_\mu^2}\right) + \mathcal{O}\left(\frac{m_\pi^2}{(4\pi f_\pi)^2}\right)\right]$$

$$2 |V_{ud}|^2 G_F^2 m_\mu^2 g_\pi^2 \sum_{s,s'} \bar{u}_{e'} P_R v_e \bar{v}_e P_L u_{e'} =$$

$$= 2 |V_{ud}|^2 G_F^2 m_\mu^2 g_\pi^2 \, \mathrm{Trace}\left[\not{k}_2 P_R (\not{k}_1 - m_\mu) P_L\right]$$

$(k_1 + k_2)^2 = p^2 = m_\pi^2$

$k_1^2 + 2k_1 k_2 + k_2^2 = m_\pi^2$

$$\mathrm{Trace}\left[\not{k}_2 P_L \not{k}_1\right] = \frac{1}{2} \mathrm{Trace}\left[\not{k}_2 \not{k}_1\right] = 2 k_2 \cdot k_1$$

$$\Rightarrow k_1 \cdot k_2 = \frac{1}{2}(m_\pi^2 - m_\mu^2)$$

Razpad β :

$\pi^- \to p\, e^- \bar{\nu}$

Matrični element axialnega toka

$\langle p(k') | j_A^\mu | n(k) \rangle$

$j_A = \frac{1}{2} j_V^\mu - \frac{1}{2} j_A^\mu$

$\langle p | j_V | n \rangle = \langle p | j_V | p \rangle$ (izosp.) $\uparrow$ EM

$$\langle p(k') | j_A^{(+)\mu} | n(k) \rangle = \bar{u}^{(p)}(k') \left[\gamma^\mu \gamma_5 g(q^2) + q^\mu \gamma_5 \quad \right] u^{(n)}(k)$$

PCAC : $\partial_\mu j_A^{i\mu} = m_\pi^2 f_\pi \pi^i$

$P_\mu j^\mu = [j^\mu, P_\mu]$

1) $\langle p(k') | j_A^\mu | n(k) \rangle = i \bar{u}(k') \left[2 m_N g_A(q^2) + q^2 A_A(q^2) \right] \gamma_5 u(k)$

$(\not{k} - m_N) u(k) = 0$, $\bar{u}(k')(\not{k}' - m_N) = 0$

$\overset{q^2 \to 0}{=} 2 i m_N g_A \bar{u} \gamma_5 u$

2) $\langle p(k') | \partial_\mu j_A^{i\mu}(0) | n(k) \rangle = \sqrt{2}\, m_\pi^2 f_\pi \langle p(k') | \pi^+(0) | n(k) \rangle$

$\delta_A^{ij} \; T^i \propto \pi^i$

$T^{A_+} : T^2 \propto \pi^+$

$|n\rangle\langle n| e^{-i \int H_{int} d^4x}$

$H_{int} = g \bar{N} \vec{\tau} \cdot \vec{\pi} g N$

$$\overset{(-i)}{=} i 2 m_\pi^2 f_\pi g \int d^4x \, \langle p(k') | \bar{\psi}^{(p)}(x) \pi^-(x) \gamma_5 \psi^{(n)}(x) \pi^+(0) | n(k) \rangle$$

$\bar{u}(k') e^{ik'x}$; $\int \frac{d^4q}{(2\pi)^4} \frac{i}{q^2 - m_\pi^2} e^{-iqx}$; $u(k) e^{-ikx}$

$q^2 \to 0$

$$= i 2 f_\pi g \, \bar{u}^{(p)}(k') \gamma_5 u^{(n)}(k)$$

(1) = (2) : $g_\pi g \overset{\text{Goldberger-T}}{=} m_N g_A \quad (q^2 \to 0)$ relacija

Experimentalno razpad β : $g_A(q^2 \to 0) \approx 1{,}26$

AKSIALNI TOKOVI IN ANOMALIJE

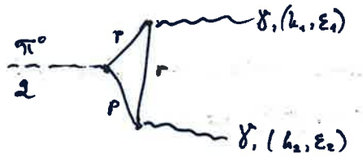
Axialni tokovi in anomalije

Ali klasične simetrije veljajo tudi na kvantnem nivoju ? Ne.

Primer : Razpad nevtralnega piona in PCAC.

$\Gamma(\pi^0 \to 2\gamma) \approx 7\,eV$

V σ modelu kiralne teorije poteka na nivoju 1 zanke



$$d\Gamma = \lim_{q^2 \to m_\pi^2} \varepsilon_1^\mu \varepsilon_2^\nu \, T_{\mu\nu}(q^2)$$

prispevajo vse nabite prostostne stopnje v zanki, ki se sklapljajo s pionom

PCAC: $\partial_\mu j_A^\mu = m_\pi^2 f_\pi \pi(0)$

$$T_{\mu\nu}(q^2) = e^2 \int d^4x \, d^4x' \langle 0 | T\left[j_{em\,\mu}(x) \, j_{em\,\nu}(x') \, \pi(0) \right] | 0 \rangle$$

$\pi(0)$ mirno

$$= e^2 \frac{m_\pi^2 - q^2}{f_\pi^2 m_\pi^2} \, i^5 \underbrace{\int d^4x \, d^4x' \, e^{i(k_1 x + k_2 x')} (q^2 - m_\pi^2) \langle 0 | T\left[j_{em\,\mu}(x) \, j_{em\,\nu}(x') \, j_{A_5} \right] | 0 \rangle}_{T_{\mu\nu 5}(q^2)} \quad \text{LSZ}$$

(7. teden)

* Realni fotoni imajo transv. polarizacije; zanje veljaju Wardove id. za QED:

$$k_1^\mu T_{\mu\nu\varsigma} = k_2^\nu T_{\mu\nu\varsigma} = 0$$

* Simetrije na zamenjavo:

$$T_{\mu\nu\varsigma}(k_1,k_2) = T_{\nu\mu\varsigma}(k_2,k_1)$$

* Brezmasnost fotonov $k_1^2 = k_2^2 = 0$

Te 3 lastnosti nam dajo

$$T_{\mu\nu\varsigma}(k_1,k_2) = \varepsilon_{\mu\nu\sigma\tau} k_1^\sigma k_2^\tau q_\varsigma T_1(q^2) + (\varepsilon\ \ k_{2\nu} - \varepsilon\ \ k_1)k_1^2 k_2^2\ T_2(q^2)$$
$$+\left[(\varepsilon_{\mu\varsigma\sigma\tau}k_{1\nu} - \varepsilon\ \ k_{2\mu})k_1^2 k_2^\tau - \varepsilon\ \ (k_1^2 - k_2^2)k_{1\cdot}k_2\right]\cdot T_3(q^2)$$

$$q^\varsigma T_{\mu\nu\varsigma} = \varepsilon_{\mu\nu\sigma\tau} k_1^\sigma k_2^\tau q^2\cdot\left[T_1(q^2) + T_3(q^2)\right]$$

$$db(\pi^0 \to \gamma\gamma^*) = \lim_{q^2\to m_\pi^2} \varepsilon_1\varepsilon_2 e^2 \frac{m_\pi^2 - q^2}{g_\pi m_\pi^2} q^\varsigma T_{\mu\nu\varsigma}(q^2)$$

Zgornja enačba $q^\varsigma T_{\mu\nu\varsigma} = \ldots$ nima ničle, ne pola pri $q^2 = m_\pi^2$.
Od tod sledi da je $db(\pi^0 \to \gamma\gamma^*) = 0$

Explicitен model v σ modelu pa da, daje amplitudo

$$db = \frac{e^2 g}{4\pi^2 m_p}\varepsilon_{\mu\nu\sigma\varsigma}\varepsilon_1^\mu\varepsilon_2^\nu k_1^\sigma k_2^\varsigma \neq 0$$

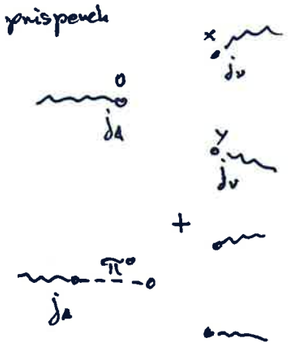
Zdej sumimo, da je nekej narobe s PCAC. Poglejmo

V splošnem bo imel $T_{\mu\nu\varsigma}$ dva prispevka:

$$T_{\mu\nu\varsigma} = \frac{g_{\pi\gamma\gamma}\, g_{\pi\varsigma}}{m_\pi^2 - q^2} T_{\mu\nu} + \hat{T}_{\mu\nu\varsigma}$$

$T_{\mu\nu}$: $db = \lim_{q^2\to m_\pi^2} \varepsilon_1^\mu\varepsilon_2^\nu T_{\mu\nu}(q^2)$

$\hat{T}_{\mu\nu\varsigma}$: intrinzični prispevek



$$T_{\mu\nu\varsigma} = T^{(a)}_{\mu\nu\varsigma}(k_1,k_2) + T^{(b)}_{\mu\nu\varsigma}(k_2,k_1)$$

$$\hookrightarrow = g e^2 \int \frac{d^4p}{(2\pi)^4} \frac{\mathrm{Tr}\left[(\not q - \not p - \not k_1 + m)\gamma_\mu(\not p + m)\gamma_5(\not p + \not k_2 + m)\gamma_\nu\right]}{\left[(q-p-k_1)^2 - m^2\right]\left[(q-p)^2 - m^2\right](p^2-m^2)}$$

$$= 4 g e^2 m \varepsilon_{\mu\nu\sigma\varsigma} k_1^\sigma k_2^\varsigma \frac{1}{\pi^2}\int_0^1 dx \int_0^{1-x} dy \frac{1}{m^2 - xy\, q^2}$$

$$\tau(q^2) = \frac{e^2 g m}{2\pi^2}\varepsilon_{\mu\nu\sigma\varsigma}\varepsilon_1^\mu\varepsilon_2^\nu k_1^\sigma k_2^\varsigma \int_0^1 dx \int_0^{1-x} dy \frac{1}{\ldots}, \qquad \tau(0) \neq 0$$

$$\Gamma = \frac{1}{2m_\pi}\sum_{\varepsilon_1,\varepsilon_2}\int \frac{d^3k_1\, d^3k_2}{4\pi\,(2\pi)^6 E_1 E_2}|\tau|^2 (2\pi)^4 \delta^{(4)}(q - k_1 - k_2)$$

$$= \frac{q^2}{8\pi\, m_\pi^3 m_p^2 g}\left(\frac{E_\pi}{2}\right)^5 \approx 7{,}8\,\mathrm{eV} \longleftrightarrow 10^{-16}\,\mathrm{s}$$



$$\hat{T}_{\mu\nu\varsigma}(k_1,k_2) = ie^2\int\frac{d^4p}{(2\pi)^4}\,\mathrm{Trace}\left[\frac{i}{\not p - m}\gamma_\mu\frac{i}{\not p + \not k_1 - m_p}\gamma_\nu\frac{i}{\not p - \not k_2 - m_p}\gamma_5\right]$$

poenostavimo

$$q^\varsigma\hat{T}_{\mu\nu\varsigma} = e^2\int\frac{d^4p}{(2\pi)^4}\frac{\mathrm{Trace}\left[(\not p + m_p)\gamma_\mu(\not p + \not k_1 + m_p)(\not k_1 + \not k_2)\gamma_5(\not p - \not k_2 + m_p)\gamma_\nu\right]}{(p^2 - m_p^2)\left[(p+k_1)^2 - m_p^2\right]\left[(p-k_2)^2 - m_p^2\right]}$$

$$\to = (\not p + \not k_1 + m_p)(\not k_1 + \not p - m_p + \not k_2 - \not p + m_p)\gamma_5(\not p - \not k_2 + m_p)$$
$$= 2m_p(\not p + \not k_1 + m_p)\gamma_5(\not p - \not k_2 + m_p) + \left[(p+k_1)^2 - m_p^2\right]\gamma_5\cdot$$
$$\cdot(\not p - \not k_2 + m_p) +$$
$$+ (\not p + \not k_1 + m_p)\gamma_5\left[(p-k_2)^2 - m_p^2\right]$$

$\partial_\mu j_\Delta^\mu \propto 2m\,\bar\psi\gamma_5\psi$

Ta del je prišteloven za masivni proton

$$\Rightarrow = \frac{m_p}{g}db(\pi^0 \to 2\gamma)$$

$$q^\varsigma\hat{T}_{\mu\nu\varsigma} = \frac{m_p}{g}db + e^2\int\frac{d^4p}{(2\pi)^4}\,\mathrm{Trace}\left[\frac{(\not p + m)\gamma_5\gamma_5(\not p - \not k_2 + m_p)\gamma_\nu}{(p^2 - m^2)\left[(p+k_2)^2 - m_p^2\right]} - \right.$$
$$\left. \frac{(\not p + \not k_1 + m)\gamma_\nu\gamma_5(\not p + m)\gamma_\mu}{\left[(p+k_1)^2 - m^2\right](p^2 - m^2)}\right]$$

$g(k_2)$ $\qquad$ $g(k_2)$

Če sta ta dva integrala končna, potem je to nič. To ni res, vsak zase linearno divergira $\Rightarrow$ Razlika?

Uporabimo Pauli-Villas regularizacijo:

$$\frac{1}{p^2 - m^2} \xrightarrow{\text{P-V}} \left(\frac{1}{p^2 - m^2} - \frac{1}{p^2 - M^2}\right)\Big|_{M\to\infty}$$

$$q^\varsigma\hat{T}_{\mu\nu\varsigma} = \left[\frac{m_p}{g}db(m_p) - \frac{M}{g}db(M)\right]_{M\to\infty}$$

Ta trik se izide zaradi predfaktorja m_p. Drugi člen je končen:

$$\frac{M}{g}db(M) = \frac{e^2}{4\pi^2}\varepsilon_{\mu\nu\sigma\varsigma}\varepsilon_1^\mu\varepsilon_2^\nu k_1^\sigma k_2^\varsigma = \text{končno.}$$

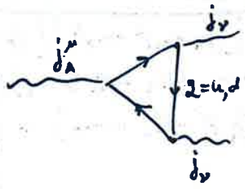
Od tod dobimo dodaten prispevek

$$\boxed{\partial_\mu j_\Delta^{3\mu} = g_{\pi\gamma\gamma} m_\pi^2\pi^0 - \frac{\alpha}{8\pi}\varepsilon_{\mu\nu\sigma\varsigma}F^{\mu\nu}F^{\sigma\varsigma}}$$
$$(+2m_N\bar N\gamma_5\tau^3 N)$$

Anomalna kršitev aksialnega toka

Posledica

i) Prispevek k $U(1)_A$ anomaliji ob prisotnosti QCD na kvarkovskem nivoju



$$\partial_\mu j^{3\mu}_A = -\frac{2}{8\pi}\sum_q Q_q^2 \varepsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma} = \frac{1}{3} N_{color}$$

$Q_u = \frac{1}{3}e_0$,
$Q_d = -\frac{1}{3}e_0$

dobimo isti rezultat le, če obstajajo 3 barve

Povezava med EM naboji kvarkov in številom barvnih nabojev.
⇒ Odsotnost umeritvenih anomalij

ii) Zakaj klasično ninamo $\frac{2}{8\pi}\varepsilon_{\mu\nu\rho\sigma} F\,F$?
Lahko jo zapišemo

$$\mathcal{L} = \mathcal{L}_{QED} + \mathcal{L}_0 - \theta \frac{2}{8\pi} \tilde{F}_{\mu\nu} F^{\mu\nu}$$

$\mathcal{L}_\theta$

$\tilde{F}_{\mu\nu} = \frac{1}{2}\varepsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$

nov prost parameter.

$$T\mathcal{L}_\theta T^{-1} = -\mathcal{L}_\theta$$
$$P\mathcal{L}_\theta P^{-1} = -\mathcal{L}_\theta$$

krši parnost, obrat časa in CP

Tega pa ne opazimo her lahko zapišemo kot divergenco toka

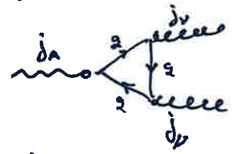
$$\mathcal{L}_\theta = \partial_\mu k^\mu$$

$$k^\mu = \frac{\theta d}{4\pi}\varepsilon^{\mu\nu\rho\sigma}\left(A_\nu \partial_\rho A_\sigma + \frac{2}{3} A_\nu A_\rho A_\sigma\right)$$

Akcijo $S_\theta = \int d^4x\, \mathcal{L}_\theta$

⇒ ne vodi do Feynmanovih pravil.

iii) Prispevek zaradi gluonskih interakcij



Bistveno: Močna int. ne razlikuje med okusi kvarkov, EM loči med naboji.

Nevtralni axialni tokovi v okusni SU(3):

$$j^3_A(\pi^0) \sim +\bar{u}\gamma^\mu\gamma_5 u - \bar{d}\gamma^\mu\gamma_5 d$$
$$j^8_A(\eta_8) \sim +\bar{u}\gamma^\mu\gamma_5 u + \bar{d}\gamma\gamma_5 d - 2\bar{s}\gamma^\mu\gamma_5 s$$

pokrajša, se pokrajša

$$SU_A(3) \times U_A(1) = U_A(3)$$
$$\underbrace{8}_{e^{-i T_i \omega_i}} + \underbrace{1}_{e^{i\omega}} = 9$$

in

$$j^{a\mu}_A = \bar{u}\gamma^\mu\gamma_5 u + \bar{d}\gamma^\mu\gamma_5 d + \bar{s}\gamma\gamma_5 s = \bar{Q}\gamma^\mu\gamma_5 Q \quad \left[\begin{smallmatrix} u \\ d \\ c \end{smallmatrix}\right]$$

$$U = e^{i\frac{\sqrt{3}}{8\pi}\pi}$$
$$\pi \longrightarrow \pi + I\eta_1$$
$$U \longrightarrow e^{i\frac{\pi}{f_\pi}\left[\underbrace{\frac{\lambda_a}{\sqrt{2}}\phi_a}_{\pi} + \underbrace{\phi_0}_{\eta_1}\right]}$$ ← to je exponat

$$\partial_\mu j^{4\mu}_A = -\frac{2}{8\pi}\varepsilon_{\mu\nu\rho\sigma} G^{\mu\nu} G^{\rho\sigma} = m^2_{\eta_1} f_\eta \eta_1$$

gluonski tenzor

Posledica: $m_{\eta'} >> m_\eta$
Imamo nov prispevek

QCD: $U(3)_A \longrightarrow SU(3)_A$

V zoreh zloma okusne simetrije lahkih kvarkov (hadronov)

Barionsko število $\longrightarrow$ B+L

$$U(3)_L \times U(3)_R \cong \underbrace{SU(3)_L \times SU(3)_R}_{\text{spontano zlomi}} \times \underbrace{U(1)_A}_{\text{kvantni popravki QCD, anomalije}} \times U(1)_V$$

$SU_V(3)$
+ pioni in kaoni so goldstonovi

približna ~~stanja~~ ($m_d \neq m_u$ in QED)
→ masa kvarkov
→ QED razlikuje med naboji

QCD

Primer: QED

$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi \longleftrightarrow$ Diračova enačba

* $\exists$ globalna $U(1)$ simetrija: $\psi(x) \longrightarrow e^{-iq\omega}\psi(x)$, ki ohranja $\mathcal{L}$.

* lokalna $U(1)$ simetrija: $\psi(x) \longrightarrow e^{-iq\omega(x)}\psi(x)$
  Kaj se zgodi z $\mathcal{L}$ v tem primeru?

$$\begin{aligned}\mathcal{L}' &= \bar{\psi}e^{iq\omega(x)}(i\not{\partial} - m)e^{-iq\omega(x)}\psi \\ &= \bar{\psi}e^{iq\omega(x)}i\gamma^\mu e^{-iq\omega(x)}(\partial_\mu\psi) + \bar{\psi}\ldots \\ &= \bar{\psi}i\not{\partial}\psi + q\bar{\psi}\gamma^\mu\psi(\partial_\mu\omega) - m\bar{\psi}\psi \\ &= \mathcal{L} + \underbrace{q\bar{\psi}\gamma^\mu\psi}_{j^\mu}\partial_\mu\omega\end{aligned}$$

Dodamo člen oblike $-j^\mu A_\mu$ kjer velja:

$$A_\mu \longrightarrow A + \partial_\mu\omega$$

$$\begin{aligned}\mathcal{L}' &= \bar{\psi}(i\not{\partial} - m)\psi - q\bar{\psi}\gamma^\mu\psi A_\mu \\ &= \bar{\psi}(i\not{D} - m)\psi\end{aligned}$$

$$D_\mu\psi \longrightarrow (\partial_\mu + iqA')\psi'$$
$$= (\partial_\mu + iq(A_\mu + \partial_\mu\omega))\psi\cdot e^{-iq\omega}$$
$$= e^{-iq\omega}(-iq)(\partial_\mu\omega)\psi + e^{iq\omega}(\partial_\mu\psi) + iq(\partial_\mu\omega)e^{-iq\omega}\psi + iqA_\mu e^{-iq\omega}\psi$$
$$= e^{-iq\omega}D_\mu\psi$$

$\Rightarrow D_\mu\psi$ in ψ se transformirata enako.

A_μ je nekaj dinamično polje $\Rightarrow$ kinetični člen

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$
$$F_{\mu\nu} \longrightarrow \partial_\mu A'_\nu - \partial_\nu A'_\mu = \partial_\mu(A_\nu + \partial_\nu\omega) - \partial_\nu(A_\mu + \partial_\mu\omega)$$
$$= \partial_\mu A_\nu + \partial_\mu\partial_\nu\omega - \partial_\nu A_\mu - \partial_\nu\partial_\mu\omega$$
$$= F_{\mu\nu}$$

$\Rightarrow \mathcal{L}_A = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$

Obstaja še priročna zveza

$$[D_\mu, D_\nu]\psi = iqF_{\mu\nu}\psi$$

Neabelske ali nekomutativne Umeritvene grupe $U(1)$

$$\psi \longrightarrow U\psi \;;\quad U = e^{-ig\alpha_i T_i\omega_i},$$

T_i so generatorji algebre, tvorijo algebro $[T_i, T_j] = if_{ijk}T_k$,
So brezsledni in hermitski $T = T^\dagger$

Če je U globalna transf.

$$\mathcal{L}_0 = \bar{\psi}(i\not{\partial} - m)\psi = \mathcal{L}'_0$$

Invarianca $\mathcal{L}_0$ vodi do ohranjenih tokov:

$$j^i_\mu = \bar{\psi}_a T^i_{ab}\gamma_\mu\psi_b$$

Če je U lokalna transf.

$$\mathcal{L}'_0 = \bar{\psi}'(i\not{\partial} - m)\psi = \bar{\psi}U^\dagger(i\not{\partial})U\psi - m\bar{\psi}\underbrace{U^\dagger U}_{\mathbb{1}}\psi$$
$$\vdots$$
$$= \mathcal{L}_0 + i\bar{\psi}U^\dagger(\not{\partial}U)\psi$$

Podobno kot v QED

$$\mathcal{L}_1 = \mathcal{L}_0 - g\bar{\psi}\gamma^\mu A_\mu\psi$$
$$\mathcal{L}'_1 = \mathcal{L}'_0 - g\bar{\psi}'\gamma^\mu A'_\mu\psi' = \mathcal{L}_0 + \underbrace{i\bar{\psi}U^\dagger\not{\partial}U\psi - g\bar{\psi}U^\dagger\not{A}U\psi}_{\overset{!}{=} -g\bar{\psi}\not{A}\psi}$$

$$-gA_\mu = -gU^\dagger A'_\mu U + iU^\dagger\partial_\mu U \qquad /U^\dagger, U/$$
$$-gUA_\mu U^\dagger = -gA'_\mu + i(\partial_\mu U)U^\dagger$$
$$\Rightarrow \boxed{A'_\mu = UA_\mu U^\dagger + \tfrac{i}{g}(\partial_\mu U)U^\dagger}$$

Ponovitev prejšnjega tedna:

(8. teden 3. u.)

$$U = e^{-ig\alpha_i T_i\omega_i}$$
$$U \overset{inf.}{\simeq} \mathbb{1} - ig\omega_i T^i$$
$$\psi' = U\psi \simeq \psi + \delta\psi = (\mathbb{1} - ig\omega_i T^i)\psi$$
$$\delta\psi = \psi' - \psi = -ig\omega_i T^i\psi$$
$$\delta\psi_a = -ig\,\omega_i T^i_{ab}\psi_b$$

$$A_\mu' \approx (1 - ig\omega_i T^i) A_\mu (1 + ig\omega_j T^j) + \underbrace{\frac{i}{g}(-ig\partial_\mu \omega_k T^k)(1 + ig\omega_l T^l)}_{(\partial_\mu \omega_i) T^i + \mathcal{O}(\omega^2)}$$

$$= A_\mu + ig\omega_j [A_\mu, T^j] + (\partial_\mu \omega_i) T^i$$

$$\Rightarrow A_\mu = A_\mu^i T^i$$

T_r, T_k

$$A_k^{i'} T^i \approx A_\mu^i T^i + ig\omega_j A_\mu^i \underbrace{[T^i, T^j]}_{if_{ijk}T^k} + (\partial_\mu \omega_i) T^i$$

$$A_\mu^{i'} \tfrac{1}{2} \delta_{ki} \approx A_k^i \tfrac{1}{2} \delta_{ki} - g\omega_j A_\mu^i f_{ijl} \tfrac{1}{2} \delta_{kl} + (\partial_\mu \omega_i) \tfrac{1}{2} \delta_{ki}$$

$$A_\mu^{k'} \approx A_\mu^k - \underbrace{g\omega_j A_\mu^i f_{ijk}}_{\text{nov člen ki ne obstaja v}} + \partial_\mu \omega_k$$

Neabelski kovariantni odvod

$$D_\mu \equiv \partial_\mu + igA_\mu \quad ; \qquad D_\mu \Psi = (\partial_\mu + igA_\mu)\Psi$$

$$(D_\mu \Psi)' = (\partial_\mu + igA_\mu')\Psi' = (\partial_\mu + igUA_\mu U^\dagger + ig\tfrac{1}{g}(\partial_\mu U)U^\dagger) U\Psi$$

$$= (\partial_\mu U)\Psi + U(\partial_\mu \Psi) + igUA_\mu \Psi - (\partial_\mu U)\Psi$$

$$= U(\partial_\mu + igA_\mu)\Psi$$

$$= U D_\mu \Psi$$

$$[D_\mu, D_\nu]\Psi \overset{?}{=} ig F_{\mu\nu}\Psi :$$

$$\ldots = [(\partial_\mu + igA_\mu)(\partial_\nu$$

$$= \partial_\mu\partial_\nu \Psi + igA_\mu \partial_\nu \Psi + ig\partial_\mu(A_\nu \Psi) - \partial_\mu\partial_\nu \Psi - igA_\nu(\partial_\mu \Psi)$$

$$- ig\partial_\mu(A_\nu \Psi) - g^2 A_\mu A_\nu \Psi + g^2 A_\nu A_\mu \Psi$$

$$= igA_\mu(\partial_\nu \Psi) + ig$$

$$- ig(\partial_\nu A_\mu)\Psi - igA_\mu(\partial_\nu \Psi) - g^2[A_\mu, A_\nu]\Psi$$

$$= ig(\partial_\mu \partial_\nu - \partial_\nu \partial_\mu + ig[A_\mu, A_\nu])\Psi$$

$F_{\mu\nu}$

$$[D_\mu', D_\nu']\Psi' = ig F_{\mu\nu}' \Psi' = ig F_{\mu\nu}' U\Psi$$

$$U[D_\mu, D_\nu]\Psi = igF_{\mu\nu} U\Psi \qquad /U^\dagger$$

$igF_{\mu\nu}\Psi$

$$F_{\mu\nu}' = U F_{\mu\nu} U^\dagger$$

$$F_{\mu\nu}^i T^i = \partial_\mu T^i A_\nu^i - \partial_\nu T^i A_\mu^i + ig\underbrace{[T^i, T^j]}_{if_{ijk}T^k} A_\mu^i A_\nu^j \qquad \text{Trace, } T^l$$

$$\tfrac{1}{2} F_{\mu\nu}^{l\prime} = \tfrac{1}{2}\partial_\mu A_\nu^l - \tfrac{1}{2}\partial_\nu A_\mu^l - g\tfrac{1}{2} f_{ijl} A_\mu^i A_\nu^j$$

$$F_{\mu\nu}^l = (\partial_\mu A_\nu^l - \partial_\nu A_\mu^l) - g f_{ijl} A_\mu^i A_\nu^j$$

$$\mathcal{L} = \bar\Psi(i\not{\partial} - m)\Psi + \mathcal{L}_g$$

$\partial_\mu + igA_\mu$; $\mathcal{L}_g$ kinetični členi za umeritv

$$\mathcal{L}_g = -\tfrac{1}{2}\mathrm{Tr}[F_{\mu\nu}F^{\mu\nu}] = -\tfrac{1}{2}\underbrace{\mathrm{Tr}[T^i, T^j]}_{\frac{1}{2}\delta_{ij}} F_{\mu\nu}^i F^{\mu\nu j}$$

$$= \tfrac{1}{4} F_{\mu\nu}^i F^{\mu\nu i} = -\tfrac{1}{4}(\partial_\mu A_\nu^i - \partial_\nu A_\mu^i - g f_{ijl} A_\mu^j A_\nu^l)(\partial^\mu A^{\nu i} - \partial^\nu A^{\mu i} - g f_{ijk} A^{\mu j} A^{\nu k})$$

$$= -\tfrac{1}{4}[(\partial_\mu A_\nu^i - \partial_\nu A_\mu^i)(\partial^\mu A^{\nu i} - \partial^\nu A^{\mu i}) - 2g\cdot f_{ijl} A_\mu^j A_\nu (\partial^\mu A^{\nu i} - \partial^\nu A^{\mu i}) +$$

$$+ g^2 f_{ijl} f_{ikl} A_\mu^j A_\nu^k A^{\mu k} A^{\nu \kappa})$$

Feynmanova pravila v QCD

$$A(p) = \int d^4x\, A(x) e^{ipx}$$

$A_\mu^i \longrightarrow G_\mu^i$, $i = 1, \ldots 8$ gluonska polja $(T_{ab}^i = \frac{\lambda_{ab}^i}{2})$

$\Psi_a \longrightarrow q_a$, $a = 1, 2, 3$ kvarkovska polja

$g \longrightarrow g_s$, mogoča skalirna „barve"

Kvarkovski propagator

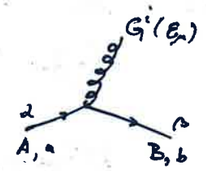
$$i S_F(p)_{\lambda\beta}\, \delta_{ab}\, \delta_{AB} = \quad a\,\lambda \circ\!\!\longrightarrow\!\!\circ b, B \quad \alpha \qquad \beta \quad = \left(\frac{i}{\not{p} - m_a + i\varepsilon}\right)_{\lambda\beta} \delta_{AB}\, \delta_{ab}$$

spin, barva, okus

Interakcije med kvarki

$$\mathcal{L}_{int} = -g_s \bar q_a \gamma_\mu \frac{\lambda_{ab}^i}{2} q_b G^{i\mu}$$

$$\Rightarrow -ig_s(\gamma^\mu)_{\lambda\beta}\frac{\lambda_{ab}^i}{2}\delta_{AB} =$$



gluonski propagator (ostvar ni trivialna)

$$\mathcal{L}_{Gauge} = -\frac{1}{4}(\partial_\mu G_{i\nu} - \partial_\nu G_{i\mu})(\partial^\mu G^{\nu i} - \partial^\nu G^{\mu i}) = -\frac{1}{4}(\partial_\mu G_{i\nu}\partial^\mu G_i^\nu - \partial_\nu G_{i\mu}\partial^\mu G_i^\nu - \dots$$
$$= -\frac{1}{2}\partial_\mu G_{i\nu}(\partial^\mu G_i^\nu - \partial^\nu G_i^\mu) + \dots$$

$$G_i^\mu(x) = \int \frac{d^4p}{(2\pi)^4} e^{-ipx} G_i^\mu(p)$$

$$\int d^4x\, \mathcal{L}_{Gauge} \approx -\frac{1}{2}\underbrace{\int d^4x \frac{1}{(2\pi)^8}}_{\int e^{-i(p+p')x}d^4x = (2\pi)^4\delta^4(p+p')} \int d^4p' G_{i\nu}(p') e^{-ip'x}(-ip'_\mu)\int d^4p\, e^{-ipx}\left[G_i^\nu(p)(-ip^\mu) - (-ip^\nu)G_i^\mu(p)\right]$$
$$= -\frac{1}{2}\int \frac{d^4p\, d^4p'}{(2\pi)^4}\delta(p+p')\left[G_{i\nu}(p')G_i^\nu(p)(-p'_\mu p^\mu) + p'_\mu p^\nu G_{i\nu}(p')G_i^\mu(p)\right]$$
$$= -\frac{1}{2}\int \frac{d^4p}{(2\pi)^4} G_i^\mu(-p)\, p^2 \underbrace{\left[g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right]}_{P^T} G_i^\nu(p)$$

$P^T P^T \overset{?}{=}$ inverz propagatorja $D^{-1}_{\mu\nu}(p)$

$$P^T_{\mu\nu} = g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2};$$
$$P^T_{\mu\nu} + P^L_{\mu\nu} = g_{\mu\nu}$$
$$P^T_{\mu\nu}P^{T\nu}{}_\rho = \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right)\left(\delta^\nu_\rho - \frac{p^\nu p_\rho}{p^2}\right)$$
$$= g_{\mu\rho} - \frac{p_\mu p_\rho}{p^2} - \frac{p_\mu p_\rho}{p^2} + \frac{p^2 p_\mu p_\rho}{p^4}$$
$$= g_{\mu\nu} - \frac{p_\mu p_\rho}{p^2} = P^T_{\mu\rho}$$

$$\boxed{\begin{aligned} P^L_{\mu\nu} &= \frac{p_\mu p_\nu}{p^2} \\ P^L_{\mu\nu}P^{L\nu}{}_\rho &= P^L_{\mu\rho} \\ P^T_{\mu\nu}P^{L\nu}{}_\rho &= 0 \end{aligned}}$$

Propagator not Greenova, klasična primer: QED

$$\mathcal{L}_{QED} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad ; \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$\Rightarrow$ Enačbe gibanja: $\partial_\mu F^{\mu\nu} = 0$
$$\partial_\mu(\partial A) - \partial\partial A = 0$$
$$\Box A - \partial(\partial A) = 0$$

$\Rightarrow$ Propagator not Greenova

$$(\Box g_{\mu\nu} - \partial_\mu\partial_\nu)D^{\mu\rho}(x-y) = \delta^\rho_\nu \delta(x-y)$$

$$D^{\mu\rho}(x-y) = \frac{1}{(2\pi)^4}\int e^{-ip(x-y)} D^{\mu\rho}(p) d^4p$$
$$(\Box g_{\mu\nu} - \partial\partial)\frac{1}{(2\pi)^4}\int d^4p\, D^{\mu\rho}(p) e^{-ip(x-y)} = \delta^\rho_\nu \frac{1}{(2\pi)^4}\int d^4p\, e^{-ip(x-y)}$$
$$\Rightarrow (-p^2 g_{\mu\nu} + p_\mu p_\nu)D^{\mu\rho}(p) = \delta^\rho_\nu$$

Splošni nastavek:
$$(-p^2 g_{\mu\nu} + p_\mu p_\nu)(A(p) g^{\mu\rho} + B(p) p^\mu p^\rho) = \delta^\rho_\nu$$
$$\underbrace{-p^2 A \delta_\nu^\rho + p^\rho p_\nu A}_{\partial} \underbrace{- p^2 B p_\nu p^\rho + B p^2 p_\nu p^\rho}_{\partial} = \delta_\nu^\rho$$

$\Rightarrow$ B ni določen

$$G_\mu \longrightarrow U G_\mu U^\dagger + \frac{i}{g_s}(\partial_\mu U)U^\dagger$$

Rešitev: Fiksiramo umeritev

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}\partial_\mu G_i^\mu \partial_\nu G_i^\nu \qquad \begin{cases} \xi = 1 & \dots \text{ Feynmanova} \\ \xi = 0 & \dots \text{ Landauova} \\ \vdots & \\ \xi \to \infty & \dots \text{ Unitarna} \end{cases}$$

popravi: $\mathcal{L}_{Gauge} \downarrow \mathcal{L}_{GF}$?

Akcija: $S \longrightarrow \int d^4x (\mathcal{L}_{Gauge} + \mathcal{L}_{GF})$, po istem postopku kot prej ...

$$S = \frac{1}{2}\int \frac{d^4p}{(2\pi)^4} G_i^\mu(p)\underbrace{\left[-g_{\mu\nu}p^2 + \left(1 - \frac{1}{\xi}\right)p_\mu p_\nu\right]}_{D^{-1}_{\mu\nu}(p) = -g_{\mu\nu}p^2 + (1-\xi^{-1})p_\mu p_\nu} G_i^\nu(p)$$

Splošni nastavek: $D^{\mu\rho} = A g^{\mu\rho} + B p^\mu p^\rho$

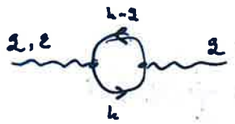
$$\left[-g_{\mu\nu}p^2 + \left(1-\tfrac{1}{\xi}\right)p_\mu p_\nu\right]\left[A g^{\rho\sigma} + B p^\nu p^\sigma\right] = \delta^\sigma_\nu$$
$$-\delta_\nu^\sigma A p^2 + A\left(1-\tfrac{1}{\xi}\right)p^\sigma p_\nu - B p^2 p_\nu p^\sigma + \left(1-\tfrac{1}{\xi}\right)B p^2 p_\nu p^\sigma = \delta^\sigma_\nu$$
$$-3Ap^2 + \left(1-\tfrac{1}{\xi}\right)A p^\sigma p_\nu - \frac{1}{\xi}B p^2 p_\nu p^\sigma = \delta^\sigma_\nu \qquad / \delta^\nu_\sigma$$
$$-4Ap^2 + A\left(1-\tfrac{1}{\xi}\right)p^2 - \frac{Bp^4}{\xi} = 4$$
$$\| \quad -3+1$$

$$\Rightarrow A = -\frac{1}{p^2} \quad ; \quad \frac{1}{\xi}\left[1 - Bp^2\right] = 1 \quad \text{poljubno}$$
$$\Downarrow$$
$$B = \frac{1-\xi}{p^4}$$

$$\boxed{D^{\mu\rho}(p) = \frac{\delta^{ij}}{p^2 + i\varepsilon}\left(-g^{\mu\rho} + \frac{1-\xi}{p^2}p^\mu p^\rho\right) = \overset{i}{\mu}\overset{p}{\circ\!\!\!\!\!\!\!\!\circ}\overset{j}{\rho}}$$

Asimptotska svoboda in koncept drseče sklopitve

Primer: DRSEČA SKLOPITEV v QED



$i\mathcal{M} = \varepsilon_\mu \varepsilon'_\nu \, \Pi^{\mu\nu}(\ell)$

$i\Pi^{\mu\nu}(\ell) = -1 \int \frac{d^4k}{(2\pi)^4} \mathrm{Trace}\left[ie\gamma^\mu \frac{i}{\not k - m} ie\gamma^\nu \frac{i}{\not k - \not\ell - m} \right]$

zanemari zaključ fermionske zanke

$= ie^2 \int \frac{d^4k}{(2\pi)^4} \mathrm{Trace}\left[\frac{\gamma^\mu(\not k + m)\gamma^\nu(\not k - \not\ell + m)}{(k^2-m^2)[(k-\ell)^2-m^2]} \right]$

→ Feynmanov trik: $\frac{1}{ab} = \int_0^1 \frac{dx}{[ax + b(1-x)]^2}$

→ Dimenzijska regulariz.: $\int \frac{d^4k}{(2\pi)^4} \longrightarrow M^{4-n} \int \frac{d^nk}{(2\pi)^n}$

$$\Pi^{\mu\nu}(\ell) = \frac{4ie^2}{(2\pi)^n} \int_0^1 dx \int d^nk \left\{ \frac{2k^\mu k^\nu - k^\mu \ell^\nu - k^\nu \ell^\mu - g^{\mu\nu}(k^2 - k\ell - m^2)}{k^2 - 2k\ell + \ell^2 x - m^2} \right\}$$

$D(k,\ell)$

→ P integrali: $C \equiv i \frac{\pi^{\frac{n}{2}}}{2} \Gamma(2-\tfrac{n}{2}) \left[m^2 - \ell^2 x(1-x) \right]^{\frac{n}{2}-2}$

$\int d^nk \frac{2k^\mu k^\nu}{D} = C\left\{ 2\ell^\mu \ell^\nu x^2 - g^{\mu\nu}\left[m^2 - \ell^2 x(x-1) \right] \frac{\Gamma(1-\,)}{\Gamma(1-\,)} \right\}$

$\int d^nk \frac{-k^2 g^{\mu\nu} - k^\mu \ell^\nu}{D} = C \cdot 2\ell^\mu \ell^\nu \cdot x$

$\int d^nk \frac{-g^{\mu\nu}k^2}{D} = C\, g^{\mu\nu}\left[\ell^2 x^2 - \frac{n}{2}\left(m^2 - \ell^2 x(1-x) \right) \frac{\Gamma(1-\,)}{\Gamma(2-\,)} \right]$

$\int d^nk \frac{g^{\mu\nu}(k\cdot\ell + m^2)}{D} = \cdots \cdot g^{\mu\nu}(\ell^2 x + m^2)$

$\Rightarrow \Pi^{\mu\nu}(\cdot) = (\ell^2 g^{\mu\nu} - \ell^\mu \ell^\nu)\, \Pi(\ell)$

zahteva Wardovih

$$\Pi(\ell) = \mu^{4-n} \frac{4e^2}{(4\pi)^{n/2}} \underbrace{\Gamma(2-\tfrac{n}{2})}_{\varepsilon = 4-n} \int_0^1 dx \, \frac{2x(1-x)}{\left[m^2 - \ell^2 x(1-x) \right]^{2-\frac{n}{2}}}$$

$\Gamma(\frac{\varepsilon}{2}) = \frac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$

Eulerjeva konst. 0,5772...

$\rightarrow \Delta^\varepsilon = 1 + \varepsilon \log \Delta + \mathcal{O}(\varepsilon^2)$

$$= -\frac{e^2}{12\pi^2}\left\{ \frac{2}{\varepsilon} - \gamma_E + \log(4\pi) - \log\frac{m^2}{\mu^2} - 6\int_0^1 dx\, x(1-x) \log\left[1 - \frac{\ell^2}{m^2} x(1-x) \right] + \mathcal{O}(\varepsilon) \right\}$$

Renormalizacija fotonskega

$\sim\!\!\sim + \sim\!\!\bigcirc\!\!\sim + \sim\!\!\bigcirc\!\!\!+\!\!\bigcirc\!\!\sim + \cdots$ (17.2)

$D_{\mu\nu}(\ell^2) = -\frac{ig_{\mu\nu}}{\ell^2} - i\frac{g_{\mu\nu}}{\ell^2} i\Pi^{\mathrm{tot}} \frac{ig_{\alpha\nu}}{\ell^2} + \cdots$

$D_{\mu\nu}(\ell^2) = \frac{-ig_{\mu\nu}}{\ell^2\left[1 - \Pi(\ell^2)\right]}$

$D_{\mu\nu}(\ell^2 \to 0)$: $-i\frac{g_{\mu\nu}}{\ell^2} \longrightarrow -i\frac{g_{\mu\nu}}{\ell^2\left[1-\Pi(0)\right]} \equiv -iZ_3 \frac{g_{\mu\nu}}{\ell^2}$

$\Pi(0) = 1 - \frac{1}{Z_3}$

$Z_3 = \frac{1}{1-\Pi(0)} \simeq 1 + \Pi(0) + \cdots$

$= 1 - \frac{e^2}{12\pi^2}\left[\frac{2}{\varepsilon} - \gamma_E + \log\frac{4\pi\mu^2}{m^2} \right] + \mathcal{O}(\varepsilon, e^4)$

V Pauli-Villars

$Z_3 = 1 - \frac{e^2}{12\pi^2} \log\frac{\Lambda^2}{m^2} + \mathcal{O}(e^4, \Lambda^{-2})$

Lagrangian z golimi polji:

$\mathcal{L} = -\frac{1}{4}(\partial_\mu A_\nu^0 - \partial_\nu A_\mu^0)^2 + \bar\psi^0(i\not\partial - m_0)\psi^0 + e_0 \bar\psi^0 \gamma_\mu \psi^0 A^{0\mu}$

$m = m_0 - \delta m_0$

Renormalizacija: $A_\mu^0 = \sqrt{Z_3} A_\mu$

$\psi^0 = \sqrt{Z_2}\,\psi$

$e = e_0 \frac{\sqrt{Z_2} Z_0}{Z_1}$

$\mathcal{L}_{\mathrm{ren.}} = -\frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + \bar\psi(i\not\partial - m)\psi + e\bar\psi\gamma_\mu\psi A^\mu + \mathcal{L}_{\mathrm{c.t.}}$

$\mathcal{L}_{\mathrm{c.t.}} = -\frac{1}{4}\delta_3(\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + \bar\psi(i\delta_2\not\partial - \delta m) + e\delta_1\bar\psi\gamma^\mu\psi A_\mu$

$\delta_i = Z_i - 1$; $\delta_m = m\delta_2 - Z_2\delta m_0$

δ_3: $\sim\!\!\bigcirc\!\!\sim + \sim\!\!\otimes^{\delta_3}\!\!\sim = i(-g^{\mu\nu}\ell^2 + \ell^\mu\ell^\nu)\delta_3$, $\Pi_{\mathrm{REN}} = \Pi(\ell^2) - \delta_3$

δ_2:  $= \Sigma_{\mathrm{REN}} = \Sigma(p) - \delta_2(\not p - m)$

δ_1:  $= i\gamma^\mu\delta_1 e$, $\Gamma^\mu_{\mathrm{REN}} = \Gamma^\mu(p', p) + \delta_1\gamma^\mu e$

4. vrsta

Postopek:

1) Izračunamo popravke k golemu Lagrangianu.
2) Divergence izoliramo z izbrano regulariz. shemo
3) def. renormalizac. pogoje za končne dele amplitud

⇒ Relacije med merljivimi količinami morajo bit neodvisni od renormalizacijske skale

⇒ Enačbe renormalizacijske grupe

Drseča sklopitvena konstanta $e(e_0)$:

$$e = e_0 \frac{Z_2\sqrt{Z_3}}{Z_1} = e_0\sqrt{Z_3}$$

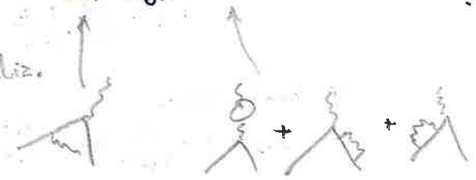
↑ v QED: $Z_1 = Z_2$

V splošnem: v vsaki renormalizabilni teoriji lahko definiramo renorm. sklopitev

$$g \equiv g(\mu) = \frac{g_0}{\tilde{Z}_1} \prod_{noge} \sqrt{Z_{noge}}$$



skala renormaliz.



Lahko posplošimo na poljubno k-točkovno funkcijo

$$G^k(x_1, \dots x_k) \equiv \langle 0|T[\phi(x_1) \dots \phi(x_k)|0\rangle$$

→ 2-točkovna f. = propag
→ 3-točkovna f. = vertex
⋮

gola k-točkovna funkc.

Enačbe renormalizacijske grupe:

$$G^k_{REN.}(p_i, \mu, g) \equiv [Z(\mu, g_0)]^{-\frac{k}{2}} G^k_0(p_i, g_0)$$

v k-prostoru ↑ renorm. skala; ↑ sklopitve; to vsebuje kvantne popravke

mi odvisna od ren. sheme ⇒ mi odv. od ren. skale

$$\frac{d}{d\mu} G^k_0(p_i, g_0) = 0$$

$$\Rightarrow \frac{d}{d\mu}\left[G^k_{REN} \cdot Z^{k/2}\right] = 0$$

$$Z^{k/2}\left[\frac{d}{d\mu} + \frac{k}{2}\frac{d\ln Z}{d\mu}\right] G^k_{REN.} = 0$$

↑ $\frac{d}{d\mu} = \frac{\partial}{\partial\mu} + \frac{\partial}{\partial g}\frac{\partial g(\mu)}{\partial\mu}$

$$\Rightarrow \left[\mu\frac{\partial}{\partial\mu} + \beta\frac{\partial}{\partial g} + \gamma\frac{k}{2}\right] G^k_{REN} = 0$$

↑ $\beta = \mu\frac{\partial g(\mu)}{\partial\mu}$ ↑ $\gamma = \mu\frac{\partial \ln Z}{\partial\mu}$ Anomalna dim.

$$\left[\mu\frac{\partial}{\partial\mu} + \beta(e)\frac{\partial}{\partial e} + \frac{1}{2}N_e\underbrace{\gamma_2(e)}_{Z_2} + N_\gamma\underbrace{\gamma_3(e)}_{Z_3}\right] G^k_{REN}(p_i, \mu, e) = 0$$

↑ renorm G^k; ↑ renorm. e

$K = N_e + N_\gamma$

↑ št. elektron; ↑ št. fotonov

$$\beta(e) = e\mu\frac{\partial}{\partial\mu}\left[-\delta_1 + \frac{1}{2}(2\delta_2 + \delta_3)\right]$$

$$\left.\begin{aligned}\delta_1 - \delta_2 &= -\frac{e^2}{6\pi^2}\log\frac{\Lambda^2}{m^2}\\ \delta_3 &= -\frac{e^2}{12\pi^2}\log\frac{\Lambda^2}{m^2}\end{aligned}\right\}$$ Pauli-Villars

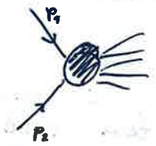
$$\beta(e) = e\mu\frac{\partial(\frac{1}{2}\delta_3)}{\partial\mu} = e\mu\frac{\partial}{\partial\mu}\left[-\frac{e^2}{12\pi^2}\left[\log\frac{\mu^2}{m^2} + \dots\right]\right]$$ enačba ren. grupe.

↓ Rešitev: $\beta(e) = \frac{de}{dt}$; $t = \log\mu$

$$\frac{1}{e^2(t)} - \frac{1}{e^2(t_0)} = -\frac{t - t_0}{6\pi^2}$$

$$\rightarrow \alpha_{EM} = \frac{e^2}{4\pi}$$

$$\Rightarrow \boxed{\alpha_{EM}(\mu^2) = \frac{\alpha_{EM}(\mu_0^2)}{1 - \frac{\alpha_{EM}(\mu_0^2)}{3\pi}\log\frac{\mu^2}{\mu_0^2}}}$$



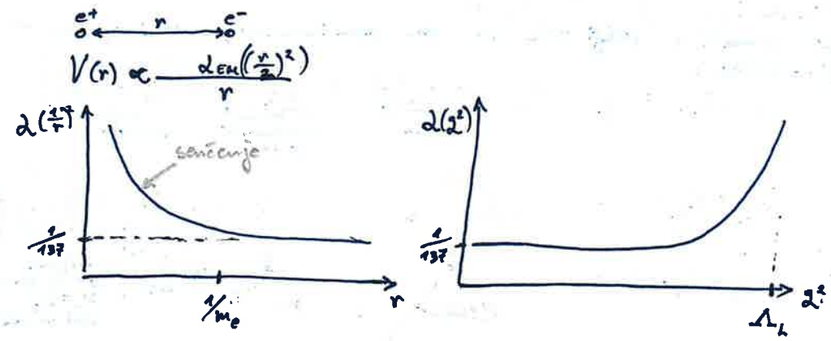
$q^2 = (p_1 - p_2)^2$

Za $q^2 < m_e^2$:

$q^2 \to 0$: $\alpha(q^2 \to 0) = \text{konst.} = \frac{1}{137,\dots} = \alpha(m_e^2) \equiv \alpha_{EM}$

↑ μ^2

$$\Rightarrow \boxed{\alpha_{ef.}(q^2) = \frac{\alpha_{EM}}{1 - \frac{\alpha_{EM}}{3\pi}\log\left(\frac{q^2}{m_e^2}\right)}}$$



9. teden
16.4.

Bistvena razlika med QED in QCD: SU(3) je neabelska
⇒ samo interakcije gluonov ⇒ dodatni popravki k polariz. vakuuma



isto kot v QED

Vzrok: člen v $\mathcal{L}_0$ bi nam dal šibkejši propag. nima inverza.

Pri neabelskih umeritvenih teorijah postopek ki smo ga naredili pri QED ne zadošča (tam smo fiksirali umeritev ⇔ znebili nefizikalnih prostostnih stopenj do dovoljšnje mere ...)

V primeru neabelskih umeritvenih teorij je fiksiranje umeritve bolj kompleksno:

METODA FADEEV - POPOV:

Dodamo nefizikalna polja skalarjev (s=0), ki upoštevajo Fermi-Diračovo statistiko (antikomutirajo). S tem spet zadostimo Wardovi identiteti.

$$\mathcal{L}_{FP} = -\partial_\mu \bar{\phi}_a D^\mu \phi^a \quad ; \quad \phi^a; \ a = 1, \dots 8 \text{ adj. upod.}$$

to je ϕ^*, z bar notacijo poudarimo antikomutacijske.

$$D_\mu \phi^a = \partial_\mu \phi^a - g_s f_{abc} \phi^b G^c_\mu$$

D_μ se obnaša enako kot D_μ za adj. upod.

Formalno konsistentnost zagotavlja BRST = zakaj točno taka oblika. Ne bomo...



⇐ iz tega sledi dodaten prispevek k renormalizaciji gluonske valovne funkc., pokrajša tist 3-gluonski prispevek (in nefizikalne prostostne stopnje), oba dva skupaj zadoščata Wardovi.

Funkcija β v QCD:

Ko uporabimo gornji trik in vse poračunamo, dobimo β to je enak v QED.

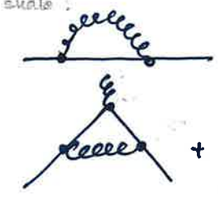
Formalno:

$$\beta_{QCD}(g_s) = g_s \mu \frac{\partial}{\partial \mu}\left(-\delta_1(\mu) + \delta_2(\mu) + \frac{1}{2}\delta_3(\mu) \right)$$

odvod po logaritmu renorm. skale ?

vertex; kvarkovsko polje; gluonsko

$\delta_2 = Z_2 - 1$:



$$\delta_2 = -\frac{g_s^2}{16\pi^2} \cdot \frac{N_c^2 - 1}{2N_c} \xi \cdot \frac{\Gamma(2-\frac{n}{2})}{\mu^{4-n}}$$

v QED: $\frac{e^2}{16\pi^2}$, seštevek po 8 gluonih $N_c = 3$, sled po Gell-Mannovih; R_ξ umeritev; dimenzijska regularizacija: n .... # dimenzij, μ .... poljubna fizikalna konst. = ren. skala

komentar: dahovi imajo samo kin. člen ⇕ se ne sklapljajo s kvarkie ne ravni 1 zanke

$\delta_1 = Z_1 - 1$:

$$\delta_1 = -\frac{g_s^2}{16\pi^2}\left[\frac{N_c - 1}{2N_c}\xi + \left(1 - \frac{1-\xi}{4}\right)N_c\right]\frac{\Gamma(2-\frac{n}{2})}{\mu^{4-n}}$$

$\delta_3 = Z_3 - 1$:



$$\delta_3 = \frac{g_s^2}{16\pi^2}\frac{\Gamma(2-\frac{n}{2})}{\mu^{4-n}}\left[\left(\frac{5}{3} + \frac{1-\xi}{2}\right)N_c - \frac{2}{3}N_2\right]$$

št. kvarkov

Vstavimo nazaj v β funkcijo: vsa odvisnost od ξ se pokrajša med ločenimi prispevki

$$\beta(g_s) = -\frac{g_s^3}{16\pi^2}\left(\frac{11\,N_c}{3} - \frac{2\,N_2}{3}\right) \qquad b_0$$

$$\Rightarrow \boxed{\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + b_0 \frac{\alpha_s(\mu^2)}{4\pi}\log\left(\frac{Q^2}{\mu^2}\right)}}$$

Primerjava QED in QCD:

	NEABELSKA: QCD: $\left[\frac{g_s^2}{16\pi^2}\frac{\Gamma(2-\frac{n}{2})}{\mu^{4-n}}\right]$	ABELSKA: QED: $\left[\frac{e^2}{16\pi^2}\frac{\Gamma(2-\frac{n}{2})}{\mu^{4-n}}\right]$
δ_1 (vertex)	$-\frac{N_c^2-1}{2N_c}\xi - \left(1-\frac{1-\xi}{4}\right)N_c$	$-\xi$
δ_2 (fermion) $_{n=3}$	$-\frac{N_c-1}{2N_c}\xi$	$-\xi$
δ_3 (boson) $_{n=3}$	$\left(\frac{5}{3} + \frac{1-\xi}{2}\right)N_c - \frac{2}{3}N_2$	$-\frac{2}{3}\left(1+0\right)\sum_f Q_f^2$

št. Diracovih fermionov v fund. upodob. ⇒ kvantizacija naboja barve: možne upodobitve določajo koliko ima dele kvantnega naboja oz. upodobitelj fiksira naboje

teorija ne pove koliko ima kater delc naboja = poljubno (v obsegu QED je $e(e^+) = +1$ vs $e(e^-) = -1$, čisti mistezij)

Če nam enačbe renormalizacijske grupe povežejo vrednost te efektivne sklopitve na različnih skalah, še vedno jo moramo izmeriti nekako iz meritev in lahko jo izključimo pri kateri koli skali. Da se prevrit da so procesi na različnih skalah kompletno opisani z zgornjo enačbo.

→ $\Gamma(Z \to \text{hadroni}) \simeq \Gamma(Z \to \text{kvarki} + \text{gluoni})$: Tu je najbolj natančno.

da se pokazat da so pri 90 GeV = m_Z taj popravki ki opisujejo razvoj kvarkov v hadrone zelo zastrti

$\Rightarrow \alpha_s(m_Z^2) = 0{,}117 \pm 0{,}005$ .... α_s je „majhen“, perturbativen razvoj je smiseln.

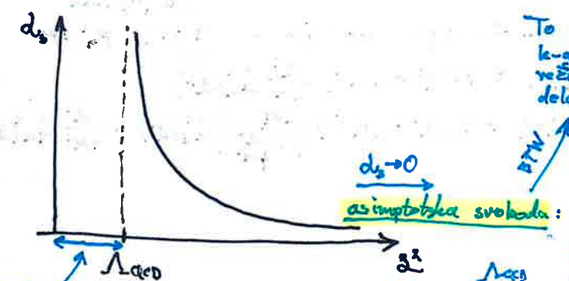
→ $\Gamma(\tau \to \nu X_{\text{hadroni}})$: m_τ

$\alpha_s(m_\tau^2) \approx 0{,}37$ ... To je meja perturbativnosti (1,77 GeV)

lahko $\tau \to \mu\nu\bar{\nu}$, $\tau \to e\nu\bar{\nu}$ ali $\tau \to X\nu$. Znamo razvejitveno razmerje $\frac{\Gamma(\tau\to\mu\nu)}{\Gamma(\tau\to e\nu)}$ in razpadnično, kar ostane je $\tau \to X\nu$. ⇒ α_s

Skala ujetja barve: $\alpha_s \to \infty$

$$1 = b_0 \frac{\alpha_s(m_\tau^2)}{4\pi}\log\frac{m_\tau^2}{\Lambda_{QCD}^2} \Rightarrow \Lambda_{QCD} \simeq 220\,MeV$$



To ni kvark-gluon plasma: k-g plasma je v perturbativnem režimu, dimpak je zasedenost delcev zelo velika

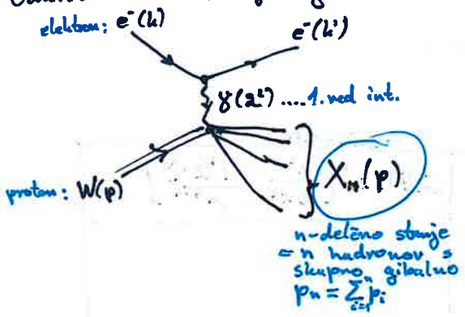
pri ∞ energijah/izmenjanih q so kvarki prosti (in gluoni). Delci se ne sklapljajo več. To je lastnost edino neabelskih teorij.

ujetje barv: to je povezano s spontanim zlomom približne kiralne simetrije (majhne mase kvarkov). Dobimo teorijo pionov ⇔ dinamika dolgega dosega, relevantne prostostne so

GLOBOKO NEELASTIČNO SIPANJE V QCD:

* želimo videti asimp. svobodo za opis procesov, kjer so na začetku hadroni
⇒ dinamika dolgega dosega $\xrightarrow{\text{trčimo vedno}}$ perturbativen opis.

Odkrito v Kaliforniji SLAC (1968) $m_e = 0$, $E \approx |\vec{k}|$, $E' \approx |\vec{k}'|$



n-delčno stanje = n hadronov s skupno gibalno $p_n = \sum_{i=1}^{n} p_i$

$$q^2 = (k - k')^2 \simeq -2k\cdot k' = -2(EE' - \vec{k}\cdot\vec{k}') = -2EE'(1 - \cos\theta)$$

Laboratorijski sistem: $p = (M, \vec{0})$ (nukleon na začetku miruje)

$k = (E, \vec{k})$, $k' = (E', \vec{k}')$ $\Big\}$ $\nu \equiv E - E'$

Also:

$$\cos\theta = \frac{\vec{k}\cdot\vec{k}'}{|\vec{k}||\vec{k}'|} \Rightarrow q^2 = -4EE'\sin^2\frac{\theta}{2}$$

Also:

$q^2 < 0 \Rightarrow Q^2 \equiv -q^2 > 0$ (to se da pokazat)

Sipalna amplituda:

$$\mathcal{M}_0 = \underbrace{e^2\,\bar{u}_e(k')\gamma^\mu u_e(k)}_{\text{matrični el. leptonskega toka}}\frac{1}{q^2}\,\langle X_n(p_n)|\,\underbrace{j^\mu_{em}(0)}_{\text{EM tok v točki 0=x, da se znebimo šibkega faznega faktorja ki nastopa v } q}\,|N(p)\rangle$$

v splošnem ne znamo ⇒ „grobo ji"

Sipalni presek:

$$d\sigma = \frac{1}{|\vec{v}|}\,\frac{1}{2M}\,\frac{1}{2E}\,\frac{d^3\vec{k}'}{(2\pi)^3 2E'}\prod_{i=1}^{n}\left[\frac{d^3\vec{p}_i}{(2\pi)^3 2p_i}\right]\frac{1}{4}\sum|\mathcal{M}_0|^2(2\pi)^4\,\delta^{(4)}(p+k-k'-p_n)$$

$p_n = \sum_i p_i$

$|\vec{v}|$: flux e^-; $2M$: flux p^+; $\frac{1}{2E}\frac{d^3\vec{k}'}{(2\pi)^3 2E'}$: fazni prost. e^-; $\prod_{i=1}^{n}\left[\frac{d^3\vec{p}_i}{(2\pi)^3 2p_i}\right]$: fazni prostor hadronov; $\frac{1}{4}$: normalic. začet. stanja; $\sum$: spini začetnih in končnih stanj

$$\frac{d\sigma}{dE'd\Omega} = \frac{\alpha^2}{q^4}\left(\frac{E'}{E}\right)L^{\mu\nu}W_{\mu\nu}$$

($L^{\mu\nu}$: leptonski, $W_{\mu\nu}$: hadronski)

$$\rightarrow L_{\mu\nu} = \frac{1}{2}\,\mathrm{Tr}\left[k'\gamma_\mu k\gamma_\nu\right] = 2\left(k_\mu k'_\nu + k_\mu k_\nu + q^2\cdot g_{\mu\nu}\right)$$

$$\rightarrow W_{\mu\nu} = \frac{1}{4M}\sum_{spini}\sum_{n}\int\prod_{i=1}^{n}\frac{d^3\vec{p}_i}{(2\pi)^3 2p_i}\cdot\langle N(p)|\,j^{em}_\mu(0)\,|X_n(p_n)\rangle\langle X_n(p_n)|\,j^{em}_\nu(0)\,|N(p)\rangle\cdot$$

$\sum_n |n\rangle\langle n| = \mathbb{1}$ tudi če so notri stanja ki se sklapljajo z j_{em}, ni važno, matrični element bo postrbel da bodo 0.

: koračno na lahko 1 hadron, 2 hadr, 3 ...

$\int d^4x\, e^{-i(p_n - p - q)x}$ .... tok izrecnotno v točki x namesto 0

$\cdot(2\pi)^4\,\delta(p_n - p - q)$

$$= \frac{1}{4M}\sum_{spini}\int d^4x\, e^{iqx}\,\langle N(p)|\,j^{em}_\mu(0)\,j^{em}_\nu(x)\,|N(p')\rangle$$

Ta trik se pogosto uporablja pri poizkih hadron → karkoli, karakteriziramo s pomočjo matričnega elementa nukleona med gluonskimi stanji: nekega produkta tokov. Podobno smo naredili pri axialni anomaliji

Vidimo da:
* v splošnem je odvisen samo od q in od p
* simetričen na zamenjavo Lor. indeksov

$$\Rightarrow W_{\mu\nu} = g_{\mu\nu}Y_1 + p_\mu p_\nu Y_2 + q_\mu q_\nu Y_3 + Y_4(p_\mu p_\nu + q_\mu q_\nu)$$

v splošnem $Y = f(p, q)$

Also: Wardove identitete: $q^\mu W_{\mu\nu} = 0$ (mora veljat ker:)

$$= \left[-W_1\left(g_{\mu\nu} - \frac{q_\mu q_\nu}{q^2}\right) + \frac{W_2}{M^2}\left(p_\mu - \frac{p\cdot q}{q^2}q_\mu\right)\left(p_\nu - \frac{p\cdot q}{q^2}q_\nu\right)\right]$$

Limita elastičnega sipanja:

$n = 1$ (en hadron na koncu = proton), $p_n^2 = M^2$

$$W_1^{el}(q^2, \nu) = \delta(q^2 + 2M\nu)\,\frac{q^2}{2M}\,G_m^2(q^2)$$

$$W_2^{el}(q^2, \nu) = \delta(q^2 + 2M\nu)\,\frac{2M}{1 - \frac{q^2}{4M^2}}\left[G_E^2(q^2) - \frac{q^2}{4M^2}G_m^2(q^2)\right]$$

Parametrizacija matričnega elementa EM med stanji nukleonov:

$$\langle N(p')|\,j^{em}_\mu(0)\,|N(p)\rangle = \bar{u}_N(p')\left[\gamma_\mu F_1(q^2) + \frac{i\sigma_{\mu\nu}q^\nu}{2M}F_2(q^2)\right]u_N(q^2)$$

to je antisim. na zamenjavo ind. ⇒ delokalizacija z magnetnim in nevarnim dipol.

bispinorske invariante, ki zadoščajo Diracovi en.

ohranitev j_μ nam da: $F_1^p(0) = 1$, $F_2^p(0) = 0$ (ker je naboj); $F_1^n(0) = 0$ (ker je nevtron brez naboja)

to bi bilo za točkasta p^+ in n^0

$F_1^p(0) = 2{,}79$, $F_1^n(0) = -1{,}91$ (anomalna mag. momenta)

to je za razsežna proton in nevtron

Povezava F_i in G_i: Wardovi id.! $\partial^\mu j_\mu = 0$

$$G_E(q^2) = F_1(q^2) + \frac{q^2}{4M^2}F_2 \qquad G_M(q^2) = F_1(q^2) + F_2(q^2)$$

$\mathcal{M}\,|j_\mu|N\rangle = 0$

$\Rightarrow G_E^p(0) = F_1^p(0)$

$W_{1,2}(q^2, \nu) \xrightarrow{Q^2 \gg M^2} W_{1,2}\left(\frac{q^2}{2M\nu}\right)$ približno

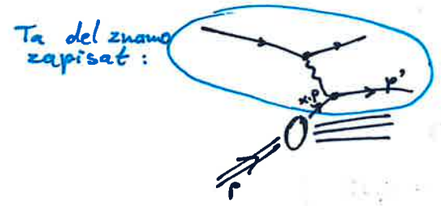
.... Povod za partonski model, predhodnik QCD.

(3. ura)

PARTONSKI MODEL NUKLEONOV

Predpostavka: proton je skupek brezmasnih prostostnih stopenj oka partonov in elektron se siplje na enem izmed teh partonov.

Ta del znamo zapisat:



$$x \equiv \frac{Q^2}{2p\cdot q}$$ ..... delež gibalne količine, ki dejansko sodeluje pri sipanju

Gibalna količina partona $x \cdot p$

⇒ „masa" $x \cdot M$ se spremenja s kinematiko (v CMS je to masa nukleona)

masa je odvisna od tega koliksen delež $|\vec{p}|$ nosi.... nesmisel.

to ima skebe lastnosti Lor. transformacij

To je približek za $Q^2 \gg M^2$

$$W_1(Q^2, \nu) = \frac{e_q^2 Q^2}{4x M^2 \nu}\,\delta\left(x - \frac{Q^2}{2M\nu}\right)$$

$$W_2(Q^2, \nu) = \frac{e_q^2 x}{\nu}\,\delta\left(x - \frac{Q^2}{2M\nu}\right)$$

$$F_1(x, Q^2) \equiv M W_1(Q^2, \nu) = \frac{1}{2}e_q^2\,\delta\left(x - \frac{Q^2}{2M\nu}\right)$$

$$F_2(x, Q^2) \equiv \nu W_2(Q^2, \nu) = e_q^2 x\,\delta\left(x - \frac{Q^2}{2M\nu}\right)$$

„Bjorken scaling"

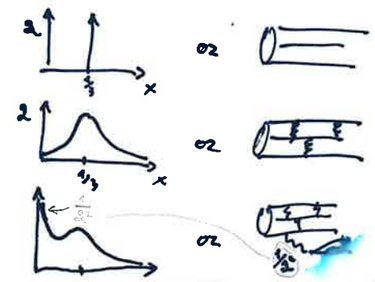
Partonske distribucijske funkcije: verjetnost, da znotraj protona sipljemo na partonu z deležem gibalne količine x.

$q_i(x)$ .... parametrizacija

$$\Rightarrow F_2(x, Q^2) = 2x\,F_1(x, Q^2) = \sum_i \int_0^1 dx\, q_i(x) e_q^2 x^2\,\delta\left(x - \frac{Q^2}{2}\right)$$

VSOTNA PRAVILA

* en prost kvarek: $q(x) = \delta(x - 1)$
* trije prosti kvarki: $q(x) = \delta(x - \frac{1}{3})$
* interakcije med kvarki (gluoni)
* tvorba parov



$$j^{em}_\mu = \tfrac{2}{3}\bar u \gamma_\mu u - \tfrac{1}{3}\bar d\gamma_\mu d - \tfrac{1}{3}\bar s\gamma_\mu s$$

* izospinska simetrija za PDF : $p \leftrightarrow n$

$u_p(x) = d_n(x) \equiv u(x)$ $\qquad \bar u$

$d_p(x) = u_n(x) \equiv d(x)$ $\qquad$ (u isto) za $\bar d$

$s_p(x) = s_n(x) \equiv s(x)$ $\qquad \bar s$

$$F_2^{ep}(x) \propto \tfrac{4}{9}\left(u(x)+\bar u(x)\right) + \tfrac{1}{9}\left(d(x)+\bar d(x)\right) + \tfrac{1}{9}\left(s(x)+\bar s(x)\right)$$

$$\frac{F_2^{ep}(x)}{F_2^{en}(x)} = \frac{4(u+\bar u)+(d+\bar d)+(s+\bar s)}{(u+\bar u)+4(d+\bar d)+(s+\bar s)} \quad\Longrightarrow\quad \frac{1}{4} \le \frac{F_2^{ep}(x)}{F_2^{en}(x)} \le 4$$

lahko tudi $q(x) = q_v(x) + q_{sea}(x)$

* za gibalno količino

$$\int dx\, x\left[u+\bar u+d+\bar d+g+\bar s\right] = 1 - \varepsilon_g$$

$$\left.\begin{aligned}\int dx\, x F_2^{ep}(x) &\simeq \tfrac{4}{9}\varepsilon_u + \varepsilon_d + \ldots = 0{,}18 \\ \int dx\, x F_2^{en}(x) &\simeq \tfrac{1}{9}\varepsilon_u + \tfrac{4}{9}\varepsilon_d + \ldots = 0{,}12\end{aligned}\right\} \quad \varepsilon_g = 1-\varepsilon_u-\varepsilon_d-\ldots \simeq 0{,}46$$

($\varepsilon_q = \int dx\, x(q+\bar q)$)

$\Rightarrow$ skoraj 50 % gibalne količine nosijo gluoni

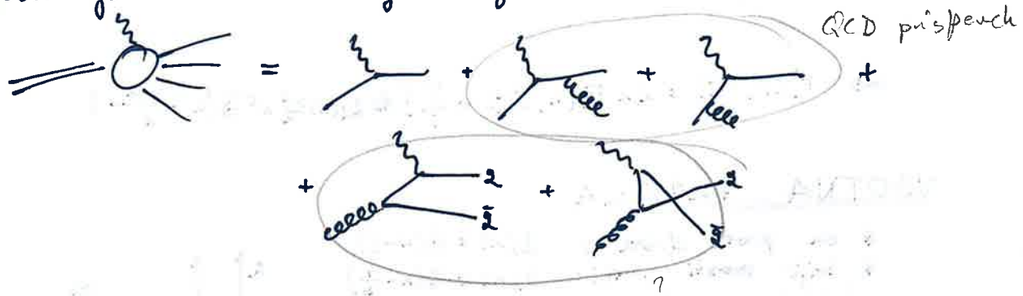
QCD v globokem neelastičnem

→ kvarki imajo barvni & EM naboj

→ barvne interakcije posredujejo 8 gluonov

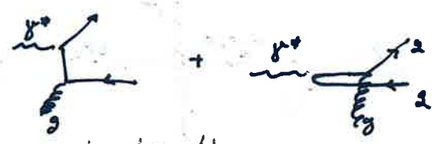
→ na majhnih razdaljah je močna sklopitev perturb

Upoštevanje barvnih interakcij v globokem neelastičnem:



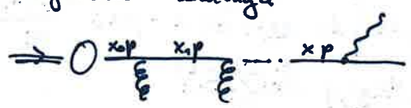
→ kinematika





$\Rightarrow$ kvarki dobijo p_T

→ krušen Bjorken scaling



$\Rightarrow$ oba pojava lahko kvalitativno pojasnimo zvotraj QCD.

$\sigma(\gamma^* p \to X)$, γ^* je virtualen:

$\sigma = \sigma_L + \sigma_T \longrightarrow 2F_1 = \frac{\sigma_T}{\sigma_0}$ $\qquad \frac{F_2}{x} = \frac{\sigma_T+\sigma_L}{\sigma_0}$

(σ_L: prispevek long, σ_T: prispevek transv. γ^*)

$\sigma_0' = \frac{4\pi\alpha}{S}$ $\qquad S = (q+p)^2 = Q^2 + 2Mv + M^2$

$\sigma(\gamma p \to X)$, γ je realen ($q^2 = 0$)

$\mathcal{L}_{int} = e A_\mu j^\mu_{em}$

$-i\mathcal{M} = -ie\,\varepsilon_\mu \langle X | j^\mu_{em} | p\rangle$

$$\sigma(\gamma p \to X) = \frac{4\pi^2\alpha}{K}\,\varepsilon^{\mu\nu}\varepsilon^* W_{\mu\nu}$$

$$W_{\mu\nu} = \frac{1}{4\pi M}\sum \tfrac{1}{2}\sum_{spin}\int \sum_i \frac{d^3\vec p_i}{(2\pi)^3 2E_i}\langle p,s|j_\mu^{em}|X\rangle\langle X|j_\nu^{em}|p,s\rangle (2\pi)^4\delta^{(4)}(\ldots - \Sigma p_i)$$

→ za $q^2 = 0$ je flux $4\pi K$

$\Rightarrow$ za virtualen foton je flux arbitraren

$$K = \frac{S-M^2}{2M} = \nu + \frac{q^2}{2M}$$

→ za realne fotone prispevata dva transv.

$\lambda = \pm 1$: $\varepsilon_\pm = -\frac{1}{\sqrt 2}(0, 1, \pm i, 0)$

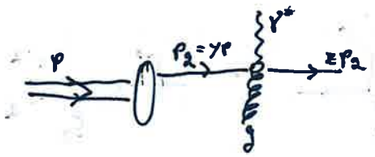
$\Rightarrow$ za virtualne fotone prispevata tudi long.

$\lambda = L$: $\varepsilon_L = \frac{1}{\sqrt{-q^2}}(\sqrt{\nu^2 - q^2}, 0, 0, \nu)$

$$\sigma_\lambda = \frac{4\pi^2\alpha}{K}\,\varepsilon^{\mu *}_\lambda \varepsilon^\nu_\lambda W_{\mu\nu}$$

$$\sigma_T = \tfrac{1}{2}(\sigma_+ + \sigma_-) = \frac{4\pi^2\alpha}{K} W_1(\nu, q^2)$$

$$\sigma_L = \frac{4\pi^2\alpha}{K}\left(\left(1-\frac{\nu^2}{q^2}\right)W_2(\nu,q^2) - W_1(\nu,q^2)\right)$$



; $z = \frac{Q^2}{2p_2\cdot q} = \frac{x}{y}$; $X \equiv \frac{Q^2}{2pq}$

partonski sip.

$$\left(\frac{\sigma_T(x,Q^2)}{\sigma_0}\right)_{\gamma^* p} = \sum_q \int_0^1 dz \int_0^1 dy\; q(y)\,\delta(x - yz)\,\frac{\hat\sigma_T(x,Q^2)}{\sigma_0}$$
$$= \sum_q \int_0^1 \frac{dy}{y} q(y)\,\frac{\hat\sigma_T(\frac{x}{y},Q^2)}{\sigma_0}$$

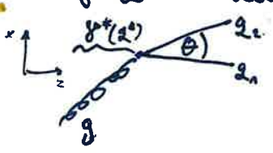
Predno poračunamo ta popravek si poglejmo kako to izgleda v partonskem modelu; v limiti brezmasnih prostih partonov

$\gamma^* q \to q$: $(q+p_2)^2 = 0 = q^2 + 2p_2\cdot q$

$\Rightarrow z = \frac{Q^2}{2p_2\cdot q} = 1$,

$\frac{\hat\sigma(t,Q^2)}{\sigma_0} \approx \delta(1-z)$

Vaja: $\gamma^* q \to qg$



$q = (q_0, 0, 0, k)$

$q_1 = (k, 0, 0, -k)$

$q_2 = (k', k'\sin\theta, 0, k'\cos\theta)$

$g = (k', -k'\sin\theta, 0, -k'\cos\theta)$

$q + q_1 = q_2 + g$

$q_0 + k = 2k' \longrightarrow q_0 = kk' - k$

Partonske Mandel Stamove

$\hat s = (q+q_1)^2 = (q_2+g)^2 \iff \hat s = kk'^2$

$\hat t = (q-q_2)^2 = (g-q_1)^2 \iff \hat t = q^2 - 2q\cdot q_2 + q_2^2 = -Q^2 - 2(k'q_0 - kk'\cos\theta)$

$\hat u = (q-g)^2 = (q_2-q_1)^2 \iff \hat u = -2q_1\cdot q_2 = -2kk'(1+\cos\theta)$

$\rightarrow p_T = k'\sin\theta$

$\hat t\cdot\hat u = (2kk')^2(1-\cos\theta)(1+\cos\theta)$
$= 4(kk')^2\sin^2\theta$

$\hat t\hat u\hat s = 4k'^2\, 4(k\cdot k')^2\sin^2\theta$
$= \dots$
$= (-Q^2 - \hat s)^2 p_T^2$

$$\Rightarrow \boxed{p_T^2 = \frac{\hat s\hat t\hat u}{(Q^2+\hat s)^2}}$$

V limiti $-\hat t \ll \hat s$: $-\hat u \approx \hat s + Q^2$

$$p_T^2 \approx -\frac{\hat s\hat t(\hat s+Q^2)}{(\hat s+Q^2)^2} = -\frac{\hat s\hat t}{\hat s+Q^2}$$

Za majhne sipalne kote θ : $\cos\theta \approx 1$

$dp_T^2 = k'^2\, d\sin^2\theta = 2k'\sin\theta\cos\theta\, d\theta$

$\Rightarrow d\Omega = \sin\theta\, d\theta\, d\varphi$
$= 2\pi\sin\theta\, d\theta$

$$\boxed{d\Omega = \frac{4\pi}{\hat s}\,dp_T^2}$$

Virtualno komptonovo sipanje:

$\gamma^* e \to \gamma e$



$s = (k_1+p_1)^2$
$t = (k_1-k_2)^2$
$u = (k_1-p_2)^2$

$$\overline{|\mathcal{M}|^2} = 32\pi^2\alpha^2\left(-\frac{u}{s} + \frac{s}{u} + \frac{2+Q^2}{su}\right)$$

Realno sipanje

$$\sum_{polar} \varepsilon_\mu^* \varepsilon_\nu = -g_{\mu\nu}$$

Posplošimo na

$$\overline{|\mathcal{M}|^2} = 32\pi^2(e_q^2\cdot 2\cdot 2_3\,\tfrac{4}{3})\left(-\frac{\hat t}{\hat s} - \frac{\hat s}{\hat t} + \frac{2\hat u Q^2}{\hat s\hat t}\right)$$

ξ_q

$\frac{1}{2}\overline{T_{ij}^a T_{ji}^a} = \frac{1}{N_c}\sum_{a,i,j} T_{ij}^a T_{ij}^a = \frac{1}{N_c}\sum_a \underbrace{\mathrm{Tr}\,[T^a T^a]}_{\frac{1}{2}} = \frac{4}{3}$

$$\hat\sigma = \frac{1}{64\pi^2 s}\int \overline{|\mathcal{M}|^2}\, d\hat\Omega \qquad d\hat\Omega = \frac{dp_T^2}{\dots}$$

$$\frac{d\hat\sigma}{dp_T^2} = \frac{8\pi e_q^2 \alpha\alpha_s}{3\hat s^2}\left(-\frac{1}{\hat t}\right)\left[\hat s + \frac{2(\hat s+Q^2)Q^2}{\hat s}\right], \quad za\ \hat t \ll \hat s$$

V partonskih koordinatah:

$$z = \frac{Q^2}{2p_2\cdot q} = \frac{Q^2}{(p_2+q)^2 - q^2} = \frac{Q^2}{\hat s + Q^2} \qquad \Longrightarrow \quad \hat s = \frac{1-z}{z}Q^2$$

to je pomembno

$$\frac{d\hat\sigma}{dp_T^2} \simeq e_q^2\,\sigma_0\,\frac{1}{p_T^2}\,\frac{\alpha_s}{2\pi}\,P_{qq}(z)$$

kaj je σ_0, $\hat\sigma$?

„Cepitvena funkcija": $P_{qq}(z) = \frac{4}{3}\frac{1+z^2}{1-z}$

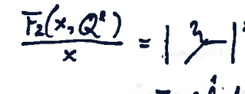
KRŠITEV BJORKEN SCALINGA

10. t 23. 4.

$$\hat\sigma(\gamma^* q \to qg) = \int dp_T^2 \frac{d\sigma}{dp_T^2} = e_q^2\hat\sigma_0 \int_{p_{T0}^2}^{\hat s/4} \frac{dp_T^2}{p_T^2}\frac{d\hat s}{2\pi} P_{qq}(\,) =$$

* $\frac{d\hat\sigma}{dp_T^2} = e_q^2$
* $P_{qq}(z) = \frac{4}{3}\frac{1+z^2}{1-z^2}$
* $p_T^2 = k'^2\sin^2\theta \Longrightarrow (p_T^2)_{max} = k'^2 = \frac{\hat s}{4}$

$$\frac{F_2(x,Q^2)}{x} = \left|\ \right|^2 + \left|\ \right|^2 + \left|\ \right|^2$$



$$= \sum_q e_q^2 \int_x^1 \frac{dy}{y} q(y)\left[\delta(1-\frac{x}{y}) + \frac{\alpha_s}{2\pi} P_{qq}\left(\frac{x}{y}\right)\log\left(\frac{Q^2}{\mu^2}\right)\right]$$

$$= \sum_q e_q^2\left[\underbrace{q_0(x)}_{q(x,Q^2)} + \Delta q(x,Q^2)\right]$$

μ^2: $\Delta q(x,Q^2) = \frac{\alpha_s}{2\pi}\log\frac{Q^2}{\mu^2}\int_x^1 \frac{dy}{y} q(y) P_{qq}\left(\frac{x}{y}\right)$

$$\frac{dg_2(x,Q^2)}{d\log Q^2} = \frac{\alpha_s}{2\pi}\int_x^1 \frac{dy}{y}\, g_2(x,Q^2)\, P_{gg}\left(\frac{x}{y}\right)$$

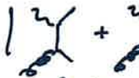
„lowest order";
danes: next-to-next-to-low. ord. + QED

Enačbe DGLAP



$\gamma^* q \to g q$ $\quad \gamma^* q \to g q$ $\quad \gamma^* g \to q\bar{q}$ $\quad \gamma^* g \to q\bar{q}$ $\quad \gamma^* \bar{q} \to g\bar{q}$ ...

Dodajmo najprej $\gamma^* g \to q\bar{q}$

$$\frac{F_2(x,Q^2)}{x}\Big|_{\gamma^* g \to q\bar{q}} = \Big|$$  $$\Big|^2$$

$$= \sum_q e_q^2 \int_x^1 \frac{dy}{y}\, g(y)\, \frac{\alpha_s}{2\pi}\, P_{qg}\left(\frac{x}{y}\right) \log\frac{Q^2}{\mu^2}$$

$$\frac{dg_2(x,Q^2)}{d\log Q^2} = \frac{\alpha_s}{2\pi}\int_x^1 \frac{dy}{y}\left[q(y,Q^2) P_{qq}\left(\frac{x}{y}\right) + g(y,Q^2) P_{qg}\left(\frac{x}{y}\right)\right]$$

cepitev ↑

$P_{qg}(z) = P_{gq}(1-z)$

$P_{gq}(z) = \frac{1}{2}\left[z^2 + (1-z)^2\right]$

Pa dodajmo še

$$\frac{dg_2(x,Q^2)}{d\log Q^2} = \frac{\alpha_s}{2\pi}\int_x^1 \frac{dy}{y}\left[q(y,Q^2) P_{gq}\left(\frac{x}{y}\right) + g(y,Q^2) P_{gg}\left(\frac{x}{y}\right)\right]$$



$P_{gq}(z) = P_{qg}(1-z) = \frac{4}{3}\frac{1+(1-z)^2}{z} + P_{gg}\left(\frac{x}{y}\right) g(y,Q^2)$



$P_{gg}(z) = P_{gg}(1-z) = 6\left[\frac{1-z}{z} + \frac{z}{1-z} + z\cdot(1-z)\right]$

Prispevki šibkega sipanja:

$eN \to \nu X$, $\nu N \to eX$ } nabiti šibki bozoni
$eN \to eX$, $\nu N \to \nu X$ } nevtralni šibki bozoni

$$j_\mu^{em} = \frac{2}{3}\bar{u}\gamma_\mu u - \frac{1}{3}\bar{d}\gamma_\mu d - \frac{1}{3}\bar{s}\gamma_\mu s + \dots$$

$$j_\mu^{W} = \bar{u}\gamma_\mu(1-\gamma_5)\left[\cos\theta_c\, d + \sin\theta_c\, s\right] + \text{ostali kvarki} \dots$$

Cabibov kot, $\approx 0{,}23 \doteq 13°$

$$j_\mu^{Z} = \sum_q \bar{q}\gamma_\mu(v_q + a_q\gamma_5) q \quad \to F_3(x,Q^2)$$

v_q: popravek EM toka, $F_2 = F_2^{em} + \frac{Q^2}{Q^2+m_Z^2} F_2^{\gamma Z} + \left[\frac{Q^2}{Q^2+m_Z^2}\right]^2 * F_2^{Z}$

γ-Z interferenca; prispevek samo Z kvarka; ta dva sta pomembna pri velikih Q^2

Pomemben komentar:

Zgodovina: Glashow-Weinberg-Salam (1964) – model $SU(2)_L \times U(1)_Y$, NP 79'

→ Hoft, Weltman (1971): renormalizacija spontano zlomljenih neabelskih teorij (umeritvenih), NP 99'

Eksperimentalno
* nevtralno-šibki bozoni 1973, CERN
* odkritje W, Z bozonov 1983, CERN

Razvoj teorije šibkih interakcij cega v 30-a leta: Fermijev model 1933, zavrnjen v Nature
* Fermijev model

$$\mathcal{L}_F = -\frac{G_F}{\sqrt{2}}\left[\bar{p}(x)\gamma_\mu n(x)\right]\left[\bar{e}(x)\gamma^\mu \nu(x)\right] + \text{h.c.}$$

$G_F \approx 10^{-5} m_p^{-4}$

* Opažanje kršitve parnosti v β

* Feynman-Gell-Mann (1958)

$$\mathcal{L}_{eff} = -\frac{G_F}{\sqrt{2}} J_\mu(x) J^\mu(x) + \text{h.c.} \quad ; \quad J_\mu(x) = j_\mu^h(x) + j_\mu^l(x)$$

$$j_\mu^l = \bar{\nu}_e\gamma_\mu(1-\gamma_5)e + \bar{\nu}_\mu\gamma_\mu(1-\gamma_5)\mu + \dots$$
$$j_\mu^h = \bar{u}\gamma_\mu(1-\gamma_5)d' \quad (d' = \cos\theta_c\, d + \sin\theta_c\, s)$$

} kiralni tokovi

v nabitih šibkih interakcijah nastopajo le levo vrteči

$$\psi_{L,R} \equiv \frac{1 \mp \gamma_5}{2}\psi, \quad \psi_L + \psi_R = \psi$$

OMEJITVE EFEKTIVNEGA PRISTOPA

a) Tak opis deluje le pri nizkih energijah. Pri visokih energijah $E \geq G_F^{-\frac{1}{2}}$ kvantne amplitude kršijo (perturbativno) unitarnost

b) Višji redi perturbacije dajo rezultate, ki so vse bolj divergentni. Teh divergenc ne moremo absorbo. v končno število golih parrnm. (npr G_F, θ_c

a) Unitarnost matrike S:

$SS^\dagger = 1 \longleftrightarrow \sum_c S_{ac} S^*_{bc} = S_{ab}$

$\rightarrow S_{ab} = \delta_{ab} + i (2\pi)^4 \delta^{(4)}(p_a - p_b) T_{ab}$; $\overline{|\mathcal{M}|^2}$

Če je $\overline{|\mathcal{M}|^2}$ omejen navzgor, potem

$\Rightarrow \sigma < \frac{\pi}{2k^2}$

Medtem ko elastično sipanje $e\nu_e \longrightarrow e\nu$

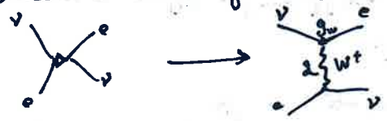
$\frac{d\sigma}{d\Omega} = G_F^2 \frac{k^2}{\pi^2}$ $\quad k \leq 300$

b) Renormalizabilnost:



$\sim \int \frac{d^4k}{k^2}$

Poskušamo z imitacijo QED in predp. masivne vektorske bozone W



$\frac{d\sigma}{d\Omega} = \frac{2 g_W^4 k^2}{\pi^2 (q^2 - m_W^2)^2}$; $q^2 = -2k^2(1-\cos\theta)$

Za $q^2 \longrightarrow 0$: $\frac{g_W}{m_W^2} \sim \frac{G_F}{\sqrt{2}}$

Za $k^2 \longrightarrow \infty$: $\sigma = \frac{4 G_F^2 k^2}{\pi} \frac{1}{1 - \frac{c}{m_W^2}} \longrightarrow$ konst.

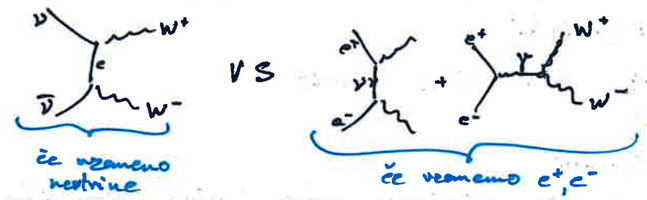
$\mathcal{L}_{int} = \frac{G_F}{\sqrt{2}} j^\mu J_\mu^\dagger \Rightarrow \mathcal{L}_{int} = \ldots j_\mu W^\mu + \ldots - \frac{1}{4} W^{\mu\nu} W_{\mu\nu} - m_W^2 W_\mu W^{\mu\dagger}$

Sipanje $\bar{\nu}_e \nu_e \longrightarrow W^+ W^-$:

$\varepsilon_T(k) = \varepsilon_T(0) \longrightarrow \sigma(\nu\bar{\nu} \longrightarrow W_T^+ W_T^-) \overset{k\to\infty}{=} \text{konst.}$

$\varepsilon_L(k) = \left(\frac{|\vec{k}|}{m_W}, 0, 0, \frac{k^0}{m_W}\right) \overset{k\to\infty}{\simeq} \frac{k^\mu}{m_W} + \mathcal{O}\left(\frac{m_W}{k_W}\right)$

$\Rightarrow \sigma(\nu\bar{\nu} \longrightarrow W_L^+ W_L^-) \propto \frac{k^2}{m_W^4}$



Renormalizabilnost:

Propagator masivnega vektorja: Proca polje

$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} m_A^2 A_\mu A^\mu$; $F_{\mu\nu} = \partial A - \partial A$

Enačbo gibanja

$\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial^\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu A_\mu} = 0$

$m_A^2 A^\mu + \partial^\nu F_{\nu\mu} = 0$

$\vdots$

$(\partial\partial + m^2) A_\nu - \partial \partial A_\nu = 0$

Propagator: Rešitev greenove enačbe

$[\Box + m^2) - \partial\partial\,] \Delta$

Rešitev dobimo preko Fourierove

$\mathcal{J}^{(4)}(x-y) = \frac{1}{(2\pi)^4} \int e^{-iq(x-y)} d^4q$

$\Delta_{\nu\rho}(x-y) = \frac{1}{2\pi} \int \Delta_{\nu\rho}$

$\Rightarrow [(-q^2 + m^2) g_{\mu\nu} + q_\mu q_\nu] \Delta^{\nu\rho}(q^4) = \mathcal{J}^\rho{}_\mu$

Nastavek: $\Delta^{\nu\rho}(q^2) = A g^{\nu\rho} + B q^\nu q^\rho$

$\Rightarrow [(-q^2 + m^2) g + q_\mu q_\nu](A g^{\nu\rho} + B q^\nu q^\rho) = \mathcal{J}^\rho{}_\mu$

$(q^2 + m^2) \cdot A\,\mathcal{J} + q_\mu \rho^\rho A + (-q^2 + m^2) B q_\mu q^\rho + B q^2 q_\mu q^\rho = \mathcal{J}_\mu{}^\rho$ / $\cdot \mathcal{J}^\mu_\rho$, $\cdot q^\mu$

a) $(-q^2 + m_A^2) 4A + q^2 A + (-q^2 + m^2) B q^2 + B q^4 = 4$

b) $(-q^2 + m_A^2) A q^\rho + q^2 A q^\rho + (-q^2 + m^2) B q q^\rho + B q^4 q^\rho = q^\rho$

$\Rightarrow A = \frac{1}{-q^2 + m_A^2}$

$B = \frac{1}{m_A^2} \frac{-1}{-q^2 + m_A^2}$

$i\Delta_{\mu\nu}(q^2) = i \frac{g_{\mu\nu} - \frac{q_\mu q_\nu}{m_A^2}}{-q^2 + m_A^2} \xrightarrow{q^2 \to \infty} \text{konst}$



Rešitev: HIGGSOV MEHANIZEM (tukaj semiklasično)

Predp. kompleksno skalarno polje $\phi = (\phi_1 + i\phi_2)\frac{1}{\sqrt{2}}$

$$\mathcal{L} = D_\mu\phi (D^\mu\phi)^\dagger - \mu^2\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2 - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

$\uparrow D_\mu = \partial_\mu - igA_\mu$

Tak $\mathcal{L}$ je inv. pod $U(1)$ umerit.

$$\phi' = e^{-i\alpha(x)}\phi$$
$$A'_\mu = A_\mu - \frac{1}{g}\partial_\mu\alpha(x)$$

Osnovno stanje teorije poiščemo preko min. potenciala

$$V(\phi) = -\mu^2\phi^\dagger\phi + \lambda(\phi^\dagger\phi)^2$$

Nas bo zanimal primer $\mu^2 > 0$:



$$v^2 = \frac{\mu^2}{\lambda}$$

V QFT velja da $\langle 0|\phi|0\rangle = \frac{v}{\sqrt{2}}$. Izberemo eno stanje, npr. $\langle 0|\phi_1|0\rangle = v$ in $\langle 0|\phi_2|0\rangle = 0$.

Fluktuacije

$$\left.\begin{aligned}\phi'_1 &= \phi_1 - v\\ \phi'_2 &= \phi_2\end{aligned}\right\}$$

$\Rightarrow$ Iz Goldstonovega: ϕ'_1 je masivno, ϕ'_2 pa ne.

Umeritveno kovar. kinetični člen:

$$\begin{aligned}(D_\mu\phi)(D^\mu\phi)^\dagger &= \tfrac{1}{2}[(\partial_\mu - igA_\mu)(\phi_1 + i\phi_2)][(\partial + igA)(\phi_1 - i\phi_2)]\\ &= \tfrac{1}{2}(\\ \to &= \tfrac{1}{2}(\partial_\mu\phi'_1 + i\partial_\mu\phi'_2 - igA_\mu v - igA_\mu\phi'_1 + gA_\mu\phi'_2)\cdot\\ &\quad\cdot(\partial^\mu\phi'_1 - i\partial^\mu\phi'_2 + igA^\mu v + igA^\mu\phi'_1 + gA^\mu\phi'_2)\\ &= \tfrac{1}{2}[(\partial_\mu\phi'_1 + gA_\mu\phi'_2)(\partial^\mu\phi'_1 + gA^\mu\phi'_2) +\\ &\quad + (\partial_\mu\phi'_2 - gA_\mu\phi'_1 - gA_\mu v)(\partial^\mu\phi'_2 - gA^\mu\phi'_1 - gA^\mu v)]\\ &= \tfrac{1}{2}[(\partial_\mu\phi'_1 + gA_\mu\phi'_2)^2 + (\partial_\mu\phi'_2 - gA_\mu\phi'_1)^2 +\\ &\quad + \underbrace{v^2g^2A_\mu A^\mu}_{\text{masni člen } m_A^2 = v^2g^2} - \underbrace{2gvA_\mu(\partial^\mu\phi'_2 - g\phi'_1A^\mu)}_{\text{kinetično mešanje}}\end{aligned}$$

Rešitev:

Rešitev: Primerno fiksiranje umeritve (R_ξ)

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2, \quad \text{brez Hio}$$

Brez Higgsovega mehanizma modificiramo

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu + \xi m_A\phi')^2$$

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2 - \underbrace{\frac{1}{2\xi}2\xi m_A\phi'_2\partial_\mu A^\mu}_{-m_A\phi'_2\partial_\mu A^\mu} - \underbrace{\frac{\xi^2}{2\xi}m_A^2\phi'^2_2}_{\text{masni člen za } \phi'_2}$$

masni člen za ϕ'_2 (odvisen od umeritve, nefizikalna polja $\equiv$ Would-be-Goldstone)

$$\begin{aligned}\int d^4x\,\mathcal{L}_{GF} &= \ldots - \int dx^4\, m_A\phi'_2\partial_\mu A^\mu\\ &= \ldots - \int d^4x\,[\underbrace{-m_A\partial_\mu(\phi'_2A^\mu)}_{\text{totčao pokrajšà mešàni člen}} + m_A(\partial_\mu\phi'_2)A^\mu]\end{aligned}$$

$\Rightarrow$ primerno fiksiranje umeritve nam pokrajša mešanje ϕ'_2-A_μ, a nam da nefizikalno maso ϕ'_2.

Propagator za vektorsko polje A_μ je v R_ξ umeritvi

$$\mathcal{L}(A) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \ldots$$

$$\mathcal{L}(A) = -\frac{1}{4}F_{\mu\nu}F + \frac{1}{2}m_A^2A_\mu A - \frac{1}{2\xi}(\partial_\mu A^\mu)^2$$

$$[(\Box + m_A^2)g_{\rho\beta} + (1-\tfrac{1}{\xi})\partial_\beta\partial_\rho]\,\Delta^{\nu\beta}(x-y) = \delta^\nu_\alpha\,\delta^{(4)}(x-y)$$

$$[(-q^2+m^2)g_{\rho\beta} + (1-\tfrac{1}{\xi})q_\rho q_\beta](Ag^{\beta\delta} + Bq^\beta q^\delta) = \delta^\delta_\rho$$

$$\begin{aligned}(-q^2+m^2)A + (1-\tfrac{1}{\xi})q^2A + (-q^2+m^2)Bq^2 + (1-\tfrac{1}{\xi})Bq^2 &= 4\\ (-q^2+m^2)A + (1-\tfrac{1}{\xi})q^2A + (-q^2+m^2)Bq^2 + (1-\tfrac{1}{\xi})Bq^4 &= 1\end{aligned}$$

Iz teh dveh enačb dobimo

$$A = \frac{1}{-q^2+m^2}$$

$$B = (1-\tfrac{1}{\xi})\frac{1}{-q^2-m^2} = (\xi-1)\cdot\frac{1}{-q^2+m^2}\cdot\frac{1}{\xi m^2 - q^2}$$

Propagator ima torej obliko

$$\Delta_{\mu\nu}(q^2) = -\frac{i}{q^2-m^2}\left(g_{\mu\nu} - \frac{q_\mu q_\nu(1-\xi)}{q^2-\xi m^2}\right.$$

ξ se mora pokrajšat na koncu izračunov sipalne matrike!

$\xi = 1$: Feynmanova umeritev

$\xi = 0$: Landauova ; vektorski propagator ima poseben obliko, bozoni so brezmasni

$\Delta_{\mu\nu}(q^2) \xrightarrow{q^2 \to 0} \frac{1}{q^2}$

Skalarne prostostne

ϕ_1': $m_{\phi_1'} = \sqrt{2\mu^2}$; $i\Delta(q) = \frac{i}{q^2 - 2\mu^2}$, to je Higgsov

ϕ_2': $m_{\phi_2'} = \sqrt{\xi} m_A$; $i\Delta(q^2) = \frac{i}{q^2 - \xi m_A^2}$

Higgsov mehanizem v unitarni umeritvi

Malo drugače parametriziramo ϕ:

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x)) e^{i\frac{\xi(x)}{v}} \simeq \frac{1}{\sqrt{2}}(v + \eta(x) + i\xi(x) + \dots)$$

majhne gibalne.

Ključni korak : uporabimo umeritveno invarianco $U(1)$ in fiksiramo umeritev

$$\phi'(x) = e^{-i\alpha(x)}\phi(x)$$
$$A_\mu'(x) = A_\mu(x) - \frac{1}{g}\partial_\mu \alpha(x)$$

in

$$\alpha(x) = \frac{\xi(x)}{v} .$$ To je unitarna umeritev.

$$\phi^u(x) \equiv \frac{1}{\sqrt{2}}(v + \eta(x)) = e^{-i\frac{\xi(x)}{v}}\phi(x)$$
$$A_\mu^u(x) \equiv B_\mu(x) = A_\mu(x) + \frac{1}{gv}\partial_\mu \xi(x)$$

$$D_\mu \phi = \left(\partial_\mu - igB_\mu + \frac{i}{v}\partial_\mu \xi(x)\right) e^{i\frac{\xi(x)}{v}} \cdot \frac{1}{\sqrt{2}}(v + \eta(x))$$
$$= i\frac{\partial_\mu \xi}{v} e^{i\frac{\xi}{v}} \frac{v+\eta}{\sqrt{2}} + e^{i\frac{\xi}{v}} \frac{1}{\sqrt{2}}\partial_\mu \eta - igB_\mu e^{i\frac{\xi}{v}} \phi^u - i\partial_\mu \frac{\xi}{v} \phi^u e^{i\frac{\xi}{v}}$$
$$= e^{i\frac{\xi}{v}} \frac{1}{\sqrt{2}}(\partial_\mu \eta - ig(v+\eta)B_\mu) =$$
$$= e^{i\frac{\xi}{v}} D_\mu \phi^u$$

$$(D_\mu \phi)(D^\mu \phi)^\dagger = \frac{1}{2}|\partial_\mu \eta(x) - igB_\mu(v + \eta(x))|^2$$

$$\mathcal{L} = \frac{1}{2}|\partial_\mu \eta - igB_\mu(v+\eta)|^2 + \frac{A^2}{2}(v+\eta)^2 - \frac{\lambda}{4}(v+\eta)^4 - \frac{1}{4}(\partial_\mu B_\nu - \partial_\nu B_\mu)(\partial^\mu B^\nu - \partial^\nu B^\mu)$$
$$= \mathcal{L}_0 + \mathcal{L}_1$$

kinetični členi, int.

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \eta)^2 - \frac{1}{2}\mu^2 \eta^2 - \frac{1}{4}(\partial_\mu B_\nu - \partial_\nu B_\mu)^2 + \frac{1}{2}(gv)^2 B_\mu B^\mu$$

m_η

To ustreza teoriji masivnega skalarja in masivnega vektorja brez umeritvene invariance.

Feynmanova pravila v R_ξ umeritvi

2. vaja

$$\mathcal{L} = \frac{1}{2}\left[(\partial^\mu \phi_1')^2 - 2\mu^2 \phi_1'^2\right] + \frac{1}{2}\left[(\partial^\mu \phi_2')^2 - \xi m_A^2 \phi_2'^2\right]$$
$$- \frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + \frac{1}{2}m_A^2 A_\mu A^\mu - \frac{1}{2\xi}(\partial^\mu A_\mu)^2$$

ϕ_1' : $\frac{i}{k^2 - 2\mu^2 + i\varepsilon}$ $\quad \circ\!\!-\!\!-\!\!\circ$

ϕ_2' : $\frac{i}{k^2 - \xi m_A^2 + i\varepsilon}$ $\quad \circ\!\!-\!\!-\!\!\circ$

A_μ : $\frac{-i}{k^2 - m_A^2 + i\varepsilon}\left(g_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_A^2}\right)$ $\quad A_\mu \circ\!\!-\!\!-\!\!\circ A_\nu$

$\mathcal{L}_1 = gA_\mu(\partial$

Od tod preberemo Feyn. pravila za int.

$A_\mu \phi_1' \phi_2'$:  $g(p_1 + p_2)_\mu$

$A_\mu A^\mu \phi_1'$:  $2ig^2 v g_{\mu\nu}$

$A_\mu A^\mu \phi\phi$:  $2ig^2 g_{\mu\nu}$

$\phi\phi\phi$:  $-2i\lambda v$ ali  $-6i\lambda v$

$\phi\phi\phi\phi$:  $-2i\lambda_q$ ali $-6i\lambda$

Nobelski Higgsov mehanizem

Spontani zlom SU(2)

Vzemimo najmanjšo netrivialno upodobitev, dublet skalarjev

$$\phi = \begin{bmatrix}\phi_0\\ \phi_2\end{bmatrix} \;;\quad \phi_{1,2} = \frac{1}{\sqrt{2}}(\mathrm{Re}\,\phi_{0,2} + \mathrm{Im}\,\phi_{0,2})$$

$$\mathcal{L} = (D_\mu\phi)^\dagger (D^\mu\phi) - V(\phi) - \frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}$$

$D_\mu = (\partial_\mu - ig\frac{\vec\sigma}{2}\cdot A_\mu)$; $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - g\varepsilon^{abc}A^b_\mu A^c_\nu$

$$V(\phi) = -\mu^2\phi^\dagger\phi + \lambda(\phi^\dagger\phi)^2$$

Spontani zlom SU(2) za $\mu^2 > 0$

$$\frac{\partial V}{\partial\phi} = 0 \quad \text{in} \quad \frac{\partial^2 V}{\partial\phi\partial\phi} > 0.$$

Najdemo minimum na 4-sferi $\langle 0|\phi^\dagger\phi|0\rangle = \frac{v^2}{2}$; $v^2 = \frac{\mu^2}{\lambda}$

Izberemo eno stanje npr.

$$\langle\phi\rangle_0 = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\ v\end{bmatrix}$$

$$(D_\mu\phi)^\dagger(D^\mu\phi) = \left[\left(\partial_\mu - ig\frac{\vec\sigma}{2}\vec A_\mu\right)(\phi' + \langle\phi\rangle_0)\right]^\dagger\left[\ \dots\right.$$
$$= \left[(D_\mu\phi')^\dagger + \langle\phi\rangle_0 ig\frac{\vec\sigma}{2}\cdot\vec A_\mu\right]\cdot\left[(D^\mu\phi') - ig\frac{\vec\sigma\cdot\vec A^\mu}{2}\langle\phi\rangle_0\right]$$

Poglejmo samo produkt teh dveh členov. Samo pri

$$= +\frac{g^2}{4}\langle\phi\rangle_0^\dagger\underbrace{(\vec\sigma\cdot\vec A_\mu)(\vec\sigma\cdot\vec A^\mu)}\langle\phi\rangle_0.$$

$$\sigma^a A^a_\mu \sigma^b A^{b\mu} = \sigma^a\sigma^b A^a_\mu A^{b\mu}$$
$$\to [\sigma^a,\sigma^b] = 2i\varepsilon^{abc}\sigma^c$$
$$\to \{\sigma^a,\sigma^b\} = \sigma^a\sigma^b - \sigma^b\sigma^a = 2\delta^{ab}$$
$$\Rightarrow 2\sigma^a\sigma^b = 2\delta^{ab} + 2i\varepsilon^{abc}\sigma_c \qquad \varepsilon_{abc}A^a_\mu A^{b\mu} = 0$$

$$= \frac{g^2}{4}(0,\tfrac{v}{\sqrt2})A^a_\mu A^{a\mu}\begin{bmatrix}0\\ \frac{v}{\sqrt2}\end{bmatrix} =$$
$$= \frac{g^2v^2}{8}A^a_\mu A^{a\mu}$$

$\frac{m_A^2}{2} \Rightarrow \boxed{m_A = \frac{gv}{2}}$ masni člen vektorskega

$$\phi^\dagger\phi = \phi'^\dagger\phi' + \langle\phi\rangle_0^\dagger\phi' + \phi'^\dagger\langle\phi\rangle_0 + \langle\phi\rangle_0^\dagger\langle\phi\rangle_0$$
$$= (\phi_0'^*, \phi_2'^*)\begin{bmatrix}\phi_0'\\ \phi_2'\end{bmatrix} + \frac{1}{\sqrt2}(0,v)\begin{bmatrix}\phi_0'\\ \phi_2'\end{bmatrix} + \begin{bmatrix}\phi_0'^*, \phi_2'^*\end{bmatrix}\frac{1}{\sqrt2}\begin{bmatrix}0\\ v\end{bmatrix} + \frac{v^2}{2}$$
$$=$$

$$(\phi^\dagger\phi)^2 = \dots$$
$$+ \dots$$

$$V(\phi) = -\mu^2\phi^\dagger\phi + \lambda(\phi^\dagger\phi)^2$$
$$= -\mu^2(\phi_0'^*\phi_0' + \phi_2'^*\phi_2' + \frac{v}{\sqrt2}(\phi_2' + \phi_2'^*) + \frac{v^2}{2})$$
$$+ \lambda\Big[(\phi_0'^*\phi_0')^2 + (\phi_2'^*\phi_2')^2 + \frac{v^2}{2}(\phi_2' + \phi_2'^*)^2 + \frac{v^2}{4}$$
$$+ 2\phi_0'^*\phi_0'\phi_2'^*\phi_2' + \frac{2v}{\sqrt2}(\phi_2' + \phi_2'^*)($$
$$+ v^2(\phi_0'^*\phi_0' + \phi_2'^*\phi_2') + \frac{v^2}{\sqrt2}(\phi_2'^* + \phi_2')\Big]$$
$$\uparrow\ v^2 = \frac{\mu^2}{\lambda}$$

$$\frac{\lambda v^2}{2}(\phi_2' + \phi_2'^*)^2 = \frac{\mu^2}{2}(\phi_2' + \phi_2'^*)^2 \quad\Longrightarrow\quad (\phi_2' + \phi_2'^*)\frac{1}{\sqrt2} \text{ je masivno}$$

1) $\phi = e^{i\frac{\vec\sigma\cdot\vec\xi(x)}{v}}\begin{bmatrix}0\\ \frac{v+\eta(x)}{\sqrt2}\end{bmatrix}$; 4 realna skalarna polja

2) Izbira umeritve:

$$\vec\xi(x) = -\frac{2\vec\xi(x)}{v}$$
$$U(x) = e^{-i\frac{\vec\sigma\cdot\vec\xi(x)}{v}}$$
$$\phi^U(x) \equiv U(x)\,\phi(x)$$

$$\to\quad D_\mu\phi^U(x) = U(x)D_\mu\phi = \left(\partial_\mu - ig\frac{\vec\sigma}{2}\vec B_\mu\right)\phi^U$$

$$\frac{\vec\sigma}{2}\vec B_\mu(x) \equiv U(x)\frac{\vec\sigma}{2}\cdot\vec A_\mu(x)\,U^{-1}(x) - \frac{1}{ig}[\partial_\mu U(x)]^{-1}U(x)$$

Celoten $\mathcal{L}$:

$$\phi^{U\dagger}\phi^U = \frac{1}{2}[0, v+\eta(x)]\begin{bmatrix}0\\ v+\eta(x)\end{bmatrix} = \frac{(v+\eta(x))^2}{2}$$

$$\mathcal{L} = \underbrace{(D_\mu\phi^U)^\dagger(D^\mu\phi^U)} + \frac{\mu^2}{2}(v+\eta)^2 + \frac{\lambda}{4}(v+\eta)^4 - \frac{1}{4}G^c_{\mu\nu}G^{c\mu\nu}$$

$$\frac{1}{2}(\partial_\mu\eta)^2 + \underbrace{\frac{g^2}{8}[0,v](\vec\sigma\cdot\vec B_\mu)\begin{bmatrix}0\\ v\end{bmatrix}} + \dots$$

$\frac{1}{2}(\frac{gv}{2})^2\vec B_\mu\cdot\vec B^\mu$ 3 masivni vektorji

„Would-be Goldstonovi"

Število Would-Be-Goldstonovih

Goldstonov teorem: $\mathcal{L}$ inv. na neko zvezno simetrijo, njen vakuum pa ni, potem teorija vsebuje Goldstonove

Predpostavimo: $\phi_i'(x) = e^{i\varepsilon_a T^a_{ij}}\phi_j(x)$, ϕ_i realna polja

$$\partial_{ij} + i\varepsilon_a T^a_{ij} \longrightarrow \delta\phi_i = \phi_i' - \phi_i = i\varepsilon_a T^a_{ij}\phi_j$$

$V(\phi)$ je invarianten pod takimi transf.

$$\delta V = \frac{\partial V}{\partial \phi_i}\delta\phi_i = i\frac{\partial V}{\partial \phi_i}\varepsilon_a T^a_{ij}\phi_j = 0 \quad \text{za } \forall \varepsilon_a$$

$$\Rightarrow \underline{\frac{\partial V}{\partial \phi_i} T^a_{ij}\phi_j = 0}$$

Izberemo minimum v $\phi_i = v \Longrightarrow \left.\frac{\partial V}{\partial \phi_i}\right|_{\phi_i = v_i} = 0$

$$\frac{\partial^2 V}{\partial\phi_i\partial\phi_k} T^a_{ij}\phi_j + \cancel{\frac{\partial V}{\partial\phi_i} T^a_{ij}\delta_{jk}} = 0$$

$$\left.\frac{\partial^2 V}{\partial\phi_i\partial\phi_k}\right|_{\phi_i = v_i} T^a_{ij} v_j = 0$$

$$V(\phi_i) = V(v_i) + \frac{1}{2}\left.\frac{\partial^2 V}{\partial\phi_i\partial\phi_k}\right|_{v_i}(\phi_i - v_i)(\phi_i - v_i) \qquad M^2_{ik}$$

$$= V(v_i) + \frac{1}{2}M^2_{ik}(\phi - v)(\phi - v)$$

a) $T^a_{ij}v_j = 0$

b) $T^a_{ij}v_j \neq 0$

Lahko gornjo enačbo prepišemo v

$$M^2_{ik}T^a_{ij}v_j = 0$$

a) $T^a_{ij}v_j = 0 \Longrightarrow M^2_{ik} \neq 0$

b) $T^a_{ij}v_j \neq 0 \Longrightarrow M^2_{ik} = 0$

↖ nekatere lastne vrednosti bodo enake nič

$SU(2)_L \times U(1)_Y$ UPODOBITVE

Mamo nabiti bilki tok

$$j^w_\mu \equiv \frac{1}{2}\bar{\nu}_e\gamma_\mu(1-\gamma_5)e + \frac{1}{2}\frac{1}{2}\bar{u}\gamma_\mu(1-\gamma_5)d$$

$$j^w_\mu = \left[\bar{\nu}_e, \bar{e}\right]\gamma_\mu\frac{1}{2}(1-\gamma_5)\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}\nu_e\\ e\end{bmatrix} + \dots$$

ustreza I^+ generatorju $SU(2)$

Pripadajoča(a) ohranjajoča(a) naboja(e)

$$I^+_w \equiv \int d^3x\,(\nu^\dagger_{eL}e_L + u^\dagger_L d_L)$$

$$I^-_w \equiv (I^+_w)^\dagger$$

Če želimo izpolniti $SU(2)$ algebro $[I^i_w, I^j_w] = i\varepsilon^{ijk}I^k_w$, mora obstajati še 3. tok

$$I^3_w = \frac{1}{2}\int d^3x\,(\nu^\dagger_{eL}\nu_{eL} - e^\dagger_L e_L + u^\dagger_L u_L - d^\dagger_L d_L)$$

↖ To ni EM tok

Levo ročne fermione tretiramo kot dublete pod $SU(2)$

$$l_L = \begin{bmatrix}\nu_e\\ e\end{bmatrix} \text{ in } q_L = \begin{bmatrix}u_L\\ d_L\end{bmatrix}, \qquad \psi_{L,R} = \frac{1}{2}(1\mp\gamma_5)\psi$$

Desno ročni fermioni (predpost. da) so singleti pod $SU(2)$

Potem zapišemo EM tok

$$j^{em}_\mu \equiv -\bar{e}\gamma_\mu e + \frac{2}{3}\bar{u}_i\gamma_\mu u - \frac{1}{3}\bar{d}\gamma_\mu d =$$

$$= -\bar{e}_L\gamma e_L - e_R\gamma e_R + \frac{2}{3}\bar{u}_L\gamma u_L + \frac{2}{3}u_R\gamma u_R -$$

In EM ohranjen naboj

$$Q \equiv \int d^3x\left(-e^+e^- + \frac{2}{3}u^+u - \frac{1}{3}d^+d\right)$$

$$Q - I^3_w = \int d^3x\left[-\frac{1}{2}(\nu^\dagger_{eL}\nu_{eL} + e^\dagger_L e_L) + \frac{1}{6}(u^\dagger_L u_L + d^\dagger_L d_L) - e^\dagger_R e_R + \frac{1}{3}u^\dagger_R u_R - \frac{1}{3}d^\dagger_R d_R\right]$$

$$\Rightarrow \left[Q - I^3_w, I^\pm_w\right] = 0$$

Definiramo hipernaboj $Y \equiv 2(Q - I^3_w)$

KVANTNA ŠTEVILA

Za kiralne

$$Y(e_L) = -1, \qquad Y(e_R) = -2$$
$$Y(q_L) = \frac{1}{3} \qquad Y(u_R) = \frac{4}{3}, \qquad Y(d_R) = -\frac{2}{3}$$

Skupna umeritvena simetrija bo torej $SU(2)_L \times U(1)_Y$.
Ta simetrija deluje za kiralne fermione

Težave:

i) V masivni fermijoni imajo maso

ii) Možna prisotnost axialnih anomalij

iii) V naravi je foton brezmasen, W masiven.

„Rešitev": Šibko interakcijo moramo opisati z masivnimi bosoni, zatorej v našo teorijo vpeljemo

ODSOTNOST KVANTNIH KIRALNIH ANOMALIJ

(7.5.)
12? teden

Splošni pogoj: $\mathrm{Trace}\left[\{T_a(R), T_b(R)\} T_c(R)\right] = 0.$

To je splošni pogoj za pokvarjenje anomalij kiralnih fermionov upodobitvi R_g z matriko $T(R)$.

Poglejmo si ta pogoj za $SU(2)_L \times U(1)_Y$:

1) $\mathrm{Trace}\left[\{I_i, I_j\} I_k\right] = 2\delta_{ij} \cdot \mathrm{Trace}[I_k] = 0 \Rightarrow$ Ni anomalij,
($\{I_i, I_j\} = 2\delta_{ij}$; I_k je traceless)

2) $\mathrm{Trace}[I^i Y Y] \propto \mathrm{Trace}[I^i] = 0 \Rightarrow$ isto,

3) $\mathrm{Trace}[\{I^i, I^j\} Y] = 2\delta_{ij} \cdot \mathrm{Trace}[Y] = 0 \Rightarrow$ isto.

$$\mathrm{Trace}\, Y = \sum_{\text{leptoni}} Y_L + \sum_{\text{kvarki}} Y_L$$
$$= \underbrace{2\cdot(-1)}_{e_L} - \underbrace{2}_{e_R} + \underbrace{3}_{\text{barve}} \cdot \Big(\underbrace{2\cdot\tfrac{1}{3}}_{\text{dublet}} + \underbrace{\tfrac{4}{3}}_{u_R} - \underbrace{\tfrac{2}{3}}_{d_R}\Big)$$
$$= -4 + 4$$
$$= 0$$

4) $\mathrm{Trace}[Y Y Y] = 6 \cdot \mathrm{Trace}[Q^3 - 3Q^2 I_3 + 3Q I_3^2 - I_3^3]$
($Y = 2(Q - I_3)$; Q^3: to je vektorski tok - ni kiralne kompon. $= 0$; $3Q I_3^2$: $\mathrm{Trace}[Q^2 I_3 - Q I_3^2] = \sum (Q^2 I_3 - Q I_3^2)$; I_3^3: $= 0$, $SU(2)_L$ tok, $\mathrm{Trace}[I_3^3] \propto \mathrm{Trace}[I_3] = 0$.)

$$= \sum_{\text{leptoni}} (Q^2 I_3 - Q I_3^2) + \sum_{\text{kvarki}} (Q^2 I_3 - Q I_3^2)$$
(e_L: $(-1)^2(-\tfrac{1}{2}) - (-1)(-\tfrac{1}{2})^2 = -\tfrac{1}{4}$; u_L, d_L: $(\tfrac{4}{9}\tfrac{1}{2} - \tfrac{2}{3}\tfrac{1}{4})\cdot 3 + (\tfrac{1}{9}(-\tfrac{1}{2}) + \tfrac{1}{3}\cdot\tfrac{1}{4})\cdot 3 = +\tfrac{1}{4}$)
$$= 0$$

5) $\mathrm{Trace}[Y] = 2 \cdot \mathrm{Trace}[Q - \underbrace{I_3}_{\mathrm{Tr}=0}] = 2 \cdot \mathrm{Trace}[Q]$

Vsem pogojem za (ne)obstoj anomalije je zadoščeno, če velja $\mathrm{Trace}[Y] = 2\,\mathrm{Tr}[Q]$.
Napoved v teoriji poenotenja $SU(5)$: $\mathrm{Trace}[Q] = 0$.

KVANTNA ŠTEVILA $SU(2)_L \times U(1)_Y$

	I	I_3	Y	Q
$\begin{pmatrix}\nu_l\\ e\end{pmatrix}_L$	$\frac{1}{2}$	$\frac{1}{2}$ $-\frac{1}{2}$	-1	0 -1
e_R	0	0	-2	-1

	I	I_3	Y	Q
$\begin{pmatrix}u\\ d\end{pmatrix}_L$	$\frac{1}{2}$	$\frac{1}{2}$ $-\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$ $-\frac{1}{3}$
u_R	0	0	$\frac{4}{3}$	$\frac{2}{3}$
d_R	0	0	$-\frac{2}{3}$	$-\frac{1}{3}$

} em. (vektorski) tok

} brezmasni bosoni

Šibko int. moramo opisat z masivnimi bosoni $\Rightarrow$ v teorijo vpeljemo spontani zlom $SU(2)_L \times U(1)_Y \longrightarrow U(1)_{EM}$ (tukaj vabimo neko skalarno polje = Higgsovo seveda).

Higgsov mehanizem za $SU(2)_L \times U(1)_Y \longrightarrow U(1)_{EM}$

Naj bo Φ skalarno polje, ki je dublet v $SU(2)_L$.

$$\Phi(x) = \begin{bmatrix}\phi^+\\ \phi^0\end{bmatrix} \quad \text{in} \quad Y(\Phi) = 1 \quad \text{oz.} \quad \begin{cases} Q(\phi^+) = 1 & I_3(\phi^+) = \frac{1}{2} \\ Q(\phi^0) = 0 & I_3(\phi^0) = -\frac{1}{2} \end{cases}.$$

To si zberemo/definiramo na tak način.
ϕ^+, ϕ^0 sta kompleksni polji $\Rightarrow$ $\Phi \cong 4$ realna skalarna polja.

Umeritveno invarianten Lagrangian:

$$\mathcal{L} = \sum_i \mathcal{L}_i$$

$$\mathcal{L}_1 = \underbrace{-\tfrac{1}{4} F^i_{\mu\nu} F^{i\mu\nu}}_{SU(2)_L,\ i=1,2,3} \underbrace{- \tfrac{1}{4} B_{\mu\nu} B^{\mu\nu}}_{U(1)_Y} \quad \ldots\ldots \text{to je kinetični del}$$

$F^i_{\mu\nu} = \partial_\mu A^i_\nu - \partial_\nu A^i_\mu + g\varepsilon^{ijk} A^j_\mu A^k_\nu$ $\qquad$ $B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$

$$\mathcal{L}_2 = \bar{\Psi}_i \cdot i \not{D} \Psi_i \quad ; \qquad \Psi_i = \underbrace{\begin{bmatrix}\nu_e\\ e\end{bmatrix}_L}_{L}, e_R, \underbrace{\begin{bmatrix}u\\ d\end{bmatrix}_L}_{Q}, u_R, d_R$$

$\not{D}$:

$D_\mu \Psi = (\partial_\mu - ig \vec{T}(R) \cdot \vec{A}_\mu - ig' \frac{Y}{2} B_\mu)\Psi$

$D_\mu L = (\partial_\mu - ig(\vec{I} \cdot \vec{A}_\mu) - ig' \frac{1}{2} B_\mu) L$

$D_\mu e_R = (\partial_\mu + ig' B_\mu) e_R$

$\vec{T}(R)$ je upodobitvena matrika:
→ za singlete 0
→ za dublete $\frac{\vec{\sigma}}{2}$

$$\mathcal{L}_3 = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi) \quad \ldots\ldots \text{Higgsovega polja del}$$

$D_\mu \Phi = (\partial_\mu - ig \vec{I} \cdot \vec{A}_\mu - \frac{i}{2} g' B_\mu)\Phi$

$V(\Phi) = -\mu^2 \underbrace{\Phi^\dagger \Phi}_{|\phi^+|^2 + |\phi^0|^2} + \lambda (\Phi^\dagger \Phi)^2$

$$\mathcal{L}_4 = F^{(e)} \bar{L} \Phi e_R + F^{(d)} \bar{Q} \Phi d_R + F^{(u)} \bar{Q} \tilde{\Phi} u_R + \text{h.c.} \quad \text{---- Yukawa inter.}$$

» int. med Higgsovim in fermioni.

sklopitveni parametri

$\tilde{\Phi}_i = \varepsilon_{ij} \Phi^{j*} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Phi^* = i I_2 \Phi^*$ in $Y(\tilde{\Phi}) = -1$, zaradi Φ^*

Spontani zlom $\mu^2 > 0$ in $\lambda > 0$:

21.5.
14. teden

Mešanje nevtrinskih okusov

→ masni stanji $|\nu_1\rangle, |\nu_2\rangle \neq$ interakc. stanji $|\nu_e\rangle, |\nu_\mu\rangle$

→ Trnsf. med bazama

$$\begin{bmatrix}|\nu_e\rangle\\ |\nu_\mu\rangle\end{bmatrix} = \begin{bmatrix}\cos\theta_\nu & \sin\theta_\nu\\ -\sin & \cos\theta_\nu\end{bmatrix} \begin{bmatrix}|\nu_1\rangle\\ |\nu_2\rangle\end{bmatrix}$$

$\cos\theta_\nu \to c_\nu$, $\sin\theta_\nu \to s_\nu$

časovni razvoj $|\nu_i(t)\rangle = e^{-iEt}$

→ časovni razvoj inter. stanja:

$$|\nu_e(t)\rangle = |\nu_1(0)\rangle c_\nu e^{-iE_1 t} + |\nu_2(0)\rangle s_\nu e^{-iE_2 t}$$

hkrati ob času $t=0$:

$$|\nu_1\rangle = c_\nu |\nu_e\rangle - s_\nu |\nu_\mu\rangle$$
$$|\nu_2\rangle = s_\nu |\nu_e\rangle + c_\nu |\nu_\mu\rangle$$

$$|\nu_e(t)\rangle = c_\nu^2 e^{-iE_1 t} | + s_\nu^2 e^{-iE_2 t} |\nu_e(0)\rangle + s_\nu c_\nu |\nu_\mu(0)\rangle e^{-iE_2 t}$$
$$= |\nu_e(0)\rangle \left(c_\nu^2 e^{-iE_1 t} + s_\nu^2 e^{-iE_2 t}\right) + |\nu_\mu(0)\rangle \left(e^{-iE_2 t} - e^{-iE_1 t}\right) s_\nu c_\nu$$

Verjetnost

$$\left| \langle \nu_e(t) | \nu_\mu(0) \rangle \right|^2$$

Poglejmo:

$$\langle \nu_e(t) | \nu_\mu(0) \rangle = \tfrac{1}{2} \sin 2\theta_\nu \left(e^{+iE_2 t} - e^{iE_1 t} \right)$$

$$\langle \nu_e(t) | \nu_\mu(0) \rangle \langle \nu_\mu(0) | \nu_e(t) \rangle^* = \tfrac{1}{4} \sin^2(2\theta_\nu) \underbrace{\left(e^{iE_2 t} - e^{+iE_1 t}\right)\left(e^{-iE_2 t} - e^{-iE_1 t}\right)}_{1 - e^{-i(E_1 - E_2)t}}$$

$=$

$$P(t)_{\nu_\mu \to \nu_e} = \sin^2(2\theta_\nu)\cdot \sin^2\left(\frac{E_2-E_1}{2}t\right)$$

* $P_{\nu_\mu\to\nu_e}(t) + P_{\nu_\mu\to\nu_e}(t) = 1$
* Nevtrinske mase so majhne v primerjavi z gibalnimi (U-R limita)

$$E_\nu = \sqrt{|\vec{p}|^2 + m_\nu^2} \simeq |\vec{p}| + \frac{m_\nu^2}{2|\vec{p}|} \Rightarrow E_2 - E_1 \simeq |\vec{p}| + \frac{m_2^2}{2|\vec{p}|} - \left(|\vec{p}| + \frac{m_1^2}{2|\vec{p}|}\right) = \frac{m_2^2 - m_1^2}{2|\vec{p}|}$$

* čas potovanja od nastanka do detekcije $\Rightarrow t = L$

Also: $\sin^2\frac{E_2-E_1}{2}t \simeq \sin^2\frac{\Delta m_\nu^2}{4|\vec{p}|}L \simeq \sin^2\frac{\Delta m_\nu^2}{4E_\nu}L$

↑ oscilacijska dolžina ν

$$\boxed{P_{\nu_\mu\to\nu_e}(L) = \sin^2(2\theta_\nu)\sin^2\left(\frac{\Delta m_\nu^2}{4E_\nu}L\right)}$$

Strnemo v 2 pogoja:

* interakcijska in masna baza ne sovpadata
  oz. $\theta_\nu \neq 0$
* ~~nevtrino je so~~
  → brezmasni (vsaj 1 more bit masiven)
  → degenerirani
  oz. $\Delta m_\nu^2 \neq 0$

Posplošitev na 3 generacije:

→ 3 masna stanja, 3 interakcijska stanja

$$\nu_\alpha = \sum_i U_{\alpha i}\nu_i$$

↑ e, μ, τ

kjer je U 3×3 unitarna matrika.

→ 3 mešalni koti & CP faza (v limiti Diracovih nevtrinov)

$$\begin{bmatrix}\nu_e\\ \nu_\mu\\ \nu_\tau\end{bmatrix} = \begin{bmatrix} c_1 & c_2 s_3 & s_1 s_3 \\ -s_1 c_2 & & \end{bmatrix}\begin{bmatrix}\nu_1\\ \nu_2\\ \nu_3\end{bmatrix}$$

Primer:

$$|\nu_e(0)\rangle = c_1|\nu_1\rangle + c_1 s_3|\nu_2\rangle + s_1 s_3|\nu_3\rangle$$

Velja: $E_i^2 = |\vec{p}|^2 + m_i^2$

Po času t:

$$|\nu_e(t)\rangle = c_1 e^{-iE_1 t}|\nu_1\rangle + c_1 s_3 e^{-iE_2 t}|\nu_2\rangle + s_1 s_3 e^{-iE_3 t}|\nu_3\rangle$$

Verjetnost da po času t opazimo $|\nu_e\rangle$:

$$|\langle\nu_e(t)|\nu_e(0)\rangle|^2$$

$$|\langle\nu_e(t)|\nu_e(0)\rangle|^2 = 1 - 2c_1^2 s_1^2 c_3^2\left[1-\cos(E_1-E_2)t\right] - 2c_1^2 s_1^2 s_3^2\left[1-\cos(E_1-E_3)t\right] - 2s_1^2 s_3^2 c_3^2\left[1-\cos(E_2-E_3)t\right]$$

($E_1 - E_2 = \frac{m_1^2 - m_2^2}{2|\vec{p}|}$)

Uvedemo $L_{ij} \equiv \frac{2\pi}{E_i - E_j} = \frac{4\pi p}{|m_i^2 - m_j^2|} \simeq 2{,}5\,\mathrm{m}\left[\frac{p(\mathrm{MeV})}{|\Delta m^2|(\mathrm{eV})}\right]$

$$P_{\nu_e\to\nu_e}(d) = 1 - 2c_1 s_1 c_3\left[1-\cos\frac{2\pi d}{L_{12}}\right] - 2c_1 s_1 s_3\left[1-\cos\frac{2\pi d}{L_{13}}\right] - 2s_1 s_3 c_3\left[1-\cos\frac{2\pi d}{L_{23}}\right]$$

Zanimiva limita je $d \gg L_{ij}$: $\langle\cos\rangle = 0$

$$P_{\nu_e\to\nu_e} = 1 - 2c_1^2 s_1^2 - 2s_1^2 c_3^2$$

Posplošitev na N

N masnih stanj: $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$; $i = 1, \dots N$

$$|\nu_\alpha(t)\rangle = \sum_i U_{\alpha i}|\nu_i\rangle e^{-iE_i t}$$

Verjetnost, da v $\nu_\alpha(0)$ po času t (razdalji d) najdemo ν_β:

$$P_{\nu_\alpha\to\nu_\beta}(d) = \sum_i |U_{\alpha i}|^2 |U_{\beta i}|^2 + \sum U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}\cdot\cos\frac{2\pi d}{L_{ij}}$$

Mase nevtrinov; Dirac ali Majorana?

Spomnimo se konjugacije naboja

$$\psi^c \equiv C\gamma^0\psi^* = i\gamma^2\psi^* = C\bar{\psi}^T \quad ; \quad C = i\gamma^2\gamma^0$$

$$C^\dagger = -i\gamma^0\gamma^{2\dagger} = -i\gamma^2\gamma^0 = -C$$

$$C^* = (i\gamma^2\gamma^0)^* = -i\gamma^{2*}\gamma^0 = C$$

$$\bar{\psi}^c = \left[C\bar{\psi}^T\right]^\dagger\gamma^0 = \left[C\gamma^{0\dagger}\bar{\psi}^{\dagger T}\right]^\dagger\gamma_0 = \psi^T\gamma_0 C^\dagger\gamma_0 = \bar{\psi}^T(-i\gamma^0\gamma^2) = \psi^T(i\gamma^2\gamma^0) = \psi^T C.$$

14.5.
13. teden?

Primer: Sipanje $\nu_e e \longrightarrow \nu_e e$



pri $q^2 \ll m_{W,Z}$

$$\mathcal{M} = \frac{g^2}{8M_W^2}\left\{[\bar{\nu}_e \gamma_\mu(1-\gamma_5)e \cdot \bar{e}\gamma^\mu(1+\gamma_5)\nu_e] - \frac{1}{2}\bar{\nu}_e\right.$$

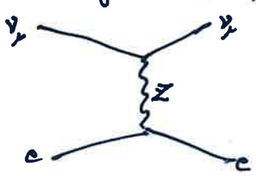
Tega ne mormo direktno sešteti, uporabimo Fierzove transf.

$$\rightarrow \bar{\psi}_1\gamma_\mu(1-\gamma_5)\psi_2\,\bar{\psi}_3\gamma^\mu(1-\gamma_5)\psi_4 =$$

$$\mathcal{M} = \frac{G_F}{\sqrt{2}}\left[\bar{\nu}_e\gamma_\mu(1-\gamma_5)\nu_e\right]$$

Primer: Sipanje $\nu_\mu e \longrightarrow \nu_\mu e$

To ni mogoče v Fermijevi (miti) teoriji



$$\mathcal{M} = \frac{g^2}{8m_W^2}\left(-\frac{1}{2}\right)\left[\bar{\nu}_\mu\gamma^\mu(1-\gamma_5)\nu_\mu\right]\left[\bar{e}\gamma_\mu(1-\gamma_5)e - \sin^2\theta_W\bar{e}\gamma_\mu e\right]$$

Replikacija fermionskih družin

→ $SU(2)_L \times U(1)_Y$

→ obstajajo 3 družine/generacije fermionov z istimi naboji (v

→ kvarki: $\begin{bmatrix} u \\ d \end{bmatrix}_L, \begin{bmatrix} c \\ s \end{bmatrix}_L, \begin{bmatrix} t \\ b \end{bmatrix}_L$, $u_R\ c_R\ t_R$, $d_R\ s_R\ b_R$

→ Lahko poskušamo pojasnit Cabibbovo mešanje nabitih šibkih tokov.

$$d_c = \cos\theta_c\, d + \sin\theta_c\, s$$
$$s_c = -\sin\theta_c\, d + \cos\theta_c\, s$$

→ Mehanizem

⇒ ločimo med dvema bazama stanj fermionov v int. prostoru.
a) Interakcijska baza (pod umeritvenimi int.)
b) masna baza

* Pred Higgsovim mehanizmom so fermioni brezmasni

* Fermionom pripišemo index dolžine/generacije

$$e \longrightarrow e'_A = [e', \mu', \tau'] \quad \Big\} \quad l_{AL} = \begin{bmatrix} \nu'_{eL} \\ e'_{AL} \end{bmatrix}$$
$$\nu_e \longrightarrow \nu'_A = [\nu'_e, \nu'_\mu, \nu'_\tau] \quad \Big\}$$
$$u \longrightarrow u'_A = [u', c', t'] \quad \Big\} \quad q_{AL} = \begin{bmatrix} u'_{AL} \\ d'_{AL} \end{bmatrix}$$
$$d \longrightarrow d'_A = [d', s', b'] \quad \Big\}$$

$$\mathcal{L}_2 = \bar{e}'_{AL}\, i\left(\partial\!\!\!/ + \frac{ig'}{2} B\!\!\!/ + \frac{ig'}{2} B\!\!\!/\right) e'_{AL}$$
$$+ \bar{e}'_{AR}\, i\left(\partial\!\!\!/ + \frac{ig'}{2} B\!\!\!/\right) e'_{AR}$$
$$+ \bar{q}'_{AL}\, i\left(\partial\!\!\!/ - \frac{ig}{2} \vec{\sigma}\cdot\vec{A}\right)$$
$$+ \bar{u}'_{AR}\, i\left(\partial\!\!\!/ - \frac{2ig'}{3} B\!\!\!/\right)$$
$$+ \bar{d}'_{AR}\, i\left(\partial\!\!\!/ + \frac{ig'}{3} B\!\!\!/ - \frac{ig_s}{2} \lambda^a G\!\!\!/^a\right) d_{AR}$$

$\mathcal{L}_4 =$

Mase dobimo po zlomu elektrošibke

$$= \frac{\eta(x) + v}{\sqrt{2}} \left[Y^{(e)}_{AB}\, \bar{e}'_{LA} e'_{RB} + Y^{(u)}_{AB}\, \bar{u}'\, u' + Y^{(d)}_{AB}\, \bar{d}'\, d' + \right]$$

Od tod definiramo „masne matrike" v int. bazi

$$M^{(q)}_{AB} = \frac{v}{\sqrt{2}} Y^{(q)}_{AB}$$

Poglejmo si lastnosti teh matrik in kako pridemo do Diracovih mas:

* $M^{(q)}_{AB}$ niso nujno (res niso) diagonalne
* Lahko jih diag. z t.i. bicunitarnimi transf.

$$S^\dagger M T = M_d \quad \text{in} \quad SS^\dagger = TT^\dagger = \mathbb{1}$$

↖ diag, pozit. lastne, realne

Mapo lahko zapišemo kot produkt hermitske in unitarne.

$$M = HV, \quad \text{kjer} \quad VV^\dagger = \mathbb{1}$$

Dokaz: $H = MV^\dagger$ ..... tako obrnemo

$MM^\dagger$ je hermitska, diagonalizabilna

$$S^\dagger (MM^\dagger) S = M_d^2$$

$$S \longrightarrow SF \; ; \quad F = \begin{bmatrix} e^{i\varphi_1} & & \\ & e^{i\varphi_2} & \\ & & e^{i\varphi_3} \end{bmatrix}$$

$$M_d \equiv \sqrt{M_d^2} = \begin{bmatrix} m_1 & & \\ & m_2 & \\ & & m_3 \end{bmatrix} \; ; \quad m_i > 0$$

Sedaj def. $H \equiv S M_d S^\dagger$ in $H^\dagger = S \underbrace{M_d^\dagger}_{M_d} S^\dagger = H$ je hermit.

Od tod lahko pokažemo da $V \equiv H^{-1} M$ je unitarna

$$V^\dagger = M^\dagger (H^{-1})^\dagger = M^\dagger H^{-1}$$

$$VV^\dagger = H^{-1} \underbrace{MM^\dagger}_{S S^\dagger M M^\dagger S S^\dagger} H^{-1} = H^{-1} S M_d^2 S^\dagger H^{-1} = H^{-1} \underbrace{S M_d S^\dagger}_{H} \underbrace{S M_d S^\dagger}_{H} H^{-1} = \mathbb{1}$$

$$\Rightarrow \underbrace{S^\dagger H S}_{MV^\dagger} = S^\dagger M \underbrace{V^\dagger S}_{T \; ; \; T^\dagger T = \mathbb{1}} = M_d$$

To uporabimo na masnem členu za fermion χ

$$\bar{\psi}'_L M \psi'_R = \underbrace{(\bar{\psi}'_L S)}_{\bar{\psi}_L} \underbrace{(S^\dagger M T)}_{M_d} \underbrace{(T^\dagger \psi'_R)}_{\psi_R}$$

$$\zeta \quad \psi'^\dagger_L \gamma_0 S = \psi^\dagger_L \gamma_0 \quad / ^\dagger$$

$$\underbrace{S^\dagger \gamma_0 \psi'_L}_{} = \gamma_0 \psi_L$$

$$\psi_L = S^\dagger \psi'_L$$

$$\Rightarrow \psi'_L = S \psi_L \quad \text{in} \quad \psi'_R = T \psi_R$$

(ψ'_L: int. baza; ψ_R: masna baza)

$$\mathcal{L}_{c.c.} = \frac{g}{\sqrt{2}} \left(J^\dagger_\mu W^{+\mu} + J^-_\mu W^{-\mu} \right)$$

$$J^+_\mu = \frac{1}{2} \bar{\nu}_A \gamma_\mu (1 - \gamma_5) e'_A + \underbrace{\frac{1}{2} \bar{u}'_A \gamma_\mu (1 - \gamma_5) d'_A}_{\bar{q}'_{AL} \gamma_\mu \frac{\sigma^+}{2} q'_{AL} = u'_L \gamma_\mu d_L} \quad \text{v int. sliki}$$

V masni bazi: $u'_L = S^{(u)} u_L \Rightarrow \bar{u}'_L = u'^\dagger_L \gamma_0 = u^\dagger_L S'^{(u)\dagger} \gamma_0 = u^\dagger_L \gamma_0 S'^{(u)\dagger} = \bar{u}_L S'^{(u)\dagger}$

$d'_L = S^{(d)} d_L$

$$\Rightarrow J^+_\mu = \bar{u}_{LB} S^{(u)\dagger}_{Bc} \gamma_\mu S^{(d)}_{cA} d_{LA} = \bar{u}_{LB} \gamma_\mu \underbrace{\left(S'^{(u)\dagger} S^{(d)} \right)_{BA}}_{(V_{CKM})_{BA}} d_{LA}$$

$V^\dagger V = \mathbb{1}$, S isto

Vpeljemo lahko generalizacijo Cabibbovega mešanja

$d_L'' = V_{CKM}\, d_L$

$Q_L'' = \begin{bmatrix} u_L \\ d_L'' \end{bmatrix} \Longrightarrow J_\mu^+ = \bar{Q}_L'' \gamma_\mu \frac{\sigma^+}{2} Q_L''$

Krsitev CP v nabitih šibkih tokovih

→ če bi bili vsi V_{CKM} realni $\Longrightarrow V_{CKM}^\dagger V_{CKM} = \mathbb{1}$

→ 3×3 ortogonalne matrike imajo 3

Splošna R 3×3 matrika: 9 parametrov

$(O^T O)_{ik} = \mathbb{1}_{ik}$ { 3 vezi za diag. el. ; 3 vezi za izvendiag. el

⇓

3 prosti par.

Splošna kompleksna 3×3 : $(3\times3)\cdot 2 = 18$ parametrov

unitarna: $U^\dagger U = \mathbb{1}$: $U_{ij}(U^\dagger)_{jk} = \delta_{ik}$

⇒ 9 parametrov (3 koti, 6 faz)

Ampak! kvarkovska polja so definirana do faze natančno.

Imamo invariance $Q_i \to Q e^{i\delta_i}$ za vsak Q_i.

$$Q_{1L}'' = \begin{bmatrix} u \\ V_{11} d + V_{12} s + V_{13} b \end{bmatrix}_L = e^{i\delta_1} \begin{bmatrix} e^{-i\delta_1} u \\ R_{11} d + V_{12}' s + V_{13}' b \end{bmatrix}$$

$V_{11} = R_{11} e^{i\delta_1}$

$$\Rightarrow J_\mu^+ = Q_{1L}'' \gamma_\mu \frac{\sigma^+}{2} Q_{1L}'' + \ldots = \bar{u}_L'' \gamma_\mu (\underbrace{R_{11}}_{\sim V_{ud}} d_L + V_{12}' s_L + V_{13} d_L)$$

Hkrati lahko absorbiramo faze V_{12}, V_{13} v s_L in b_L.

Ostane nam

$$Q_{1L}'' = \begin{bmatrix} u_L'' \\ R_{11} d_L + R_{12} s_L'' + R_{13} b'' \end{bmatrix},$$

$$Q_{2L}'' = \begin{bmatrix} c_L'' \\ R_{21} d + R_{22} e^{i\delta_2} s_L'' + R_{23} e^{i\delta_3} b_L'' \end{bmatrix}$$

$$Q_{3L}'' = \begin{bmatrix} \\ \end{bmatrix}$$

⇒ $18 - 9 - 5 = 4$: 3 koti in 1 kompl. faza

Bolj splošno (poučno):

* $n\times n$ matrike imajo $2n^2$ parametrov
* unitarnost da n^2 vezi
* $\frac{n(n-1)}{2}$ je realnih kotov (ortog. matrike)
* $2n-1$ faz lahko absorbiramo v definicijo fizikalnih polj

⇒ v 2n generacijah : 1 kot (θ_c) ; 0 faz } ohranitev CP

Degeneracije nevtrinov znotraj SM

Nabiti šibki tokovi za leptone (v int. bazi):

$J_\mu^+ = \bar{e}_{AL}' \gamma_\mu \frac{\sigma^+}{2} \nu_{AL}' = \bar{\nu}_{AL}' \gamma_\mu \nu_{AL}'$

Oz. v masni bazi:

$e_A' = S^{(e)} e_L$

$\nu_A' = S^{(\nu)} \nu_L$

$J_\mu^+ = \bar{\nu}_A \underbrace{(S^{(\nu)\dagger} S^{(e)})_{AB}}_{V = \mathbb{1} = S_{AB}} \gamma_\mu e_{BL}$

Uvedemo lahko

$\nu_{AL}'' = \begin{bmatrix} \nu_e'' \\ e \end{bmatrix}_{dg} \begin{bmatrix} \nu_\mu'' \\ \mu \end{bmatrix}_{dg} \begin{bmatrix} \nu_\tau'' \\ \tau \end{bmatrix}_{dg}$, kjer je $\begin{bmatrix} \nu_e'' \\ \nu_\mu'' \\ \nu_\tau'' \end{bmatrix} = V \begin{bmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{bmatrix}$

⇒ Teorije z enim Higgsovim in brez ν_R ohranjajo leptonska števila: $U(1)_e$, $U(\,)_\mu$, $U(\,)_\tau$.

⇒ Razpadi, ki spreminjajo leptonski okus so prepovedani do vseh redov perturbacije $\mu \to e \nu$, $\tau \to \mu \nu$, $\tau \to e \nu$ (ali $\bar{\nu}$?)

Vse to je brezpredmetno ker experimentalno $m_\nu \neq 0$.

Nevtralni šibki tok

Ohranitev okusa v nevtralnih tokovih

Nevtralni šibki tok za eno generacijo fermionov

$$J^0_\mu = \frac{1}{2}\sum_f \left[g^f_{L\alpha}\,\bar f \gamma_\mu(1-\gamma_5) f + g^{(f^c)}_{R\alpha}\,\bar f \gamma_\mu(1-\gamma_5) f \right], \quad \text{kjer } f = \nu_\alpha, e, u, d$$

$$g^f_{L,R} = T_3(f_{L,\alpha}) - Q(f)\sin^2\theta_W$$

$$J^0_\mu = g^u_L \sum_\alpha \bar u'_{L\alpha}\gamma_\mu u'_{L\alpha} + g^u_R \sum_\alpha \bar u'_{R\alpha}\gamma_\mu u'_{R\alpha}$$
$$+ g^d_L \sum_\alpha \bar d'_{L\alpha}\gamma_\mu d'_{L\alpha} + g^d_R \sum_\alpha \bar d'_{R\alpha}\gamma_\mu d'_{R\alpha}$$

Ohranitev v masni bazi ($\psi'_L = S\psi_L$, $\psi'_R = T\psi_R$)

$$\bar u'_{L\alpha}\gamma_\mu u'_{L\alpha} = \bar u_{L\alpha}\underbrace{(S^{(\dagger)})_{\alpha\beta}}\gamma_\mu (S^{(\dagger)})_{\beta c} u_{Lc} = \bar u_{L\alpha}\gamma_\mu u_{L\alpha}$$

$$(S^\dagger S)_{\alpha c} = \delta_{\alpha c}$$

$$\bar u'_{R\alpha}\gamma_\mu u'_{R\alpha} = \bar u_{R\alpha}\underbrace{(T^{(\dagger)})_{\alpha\beta}\gamma_\mu (T^{(\dagger)})_{\beta c}}_{(T^\dagger T)=\mathbb{1}} u_{Rc} = \bar u_{R\alpha}\gamma_\mu u_{R\alpha}$$

in enako za $\bar d'_L\gamma_\mu d'_L$ in $d'_R\gamma_\mu d'_R$.

POVZETEK (Zadnjih par tednov):

* $SU(2)_L \times U(1)_Y$
* $\begin{bmatrix} u_L \\ d_L \end{bmatrix} \times 3$ (N_c), $\begin{bmatrix} \nu_L \\ e_L \end{bmatrix}$, $[u_R, d_R]\times 3$, e_R
* axialne anomalije se pokrajšajo za vsako generacijo posebej.
* zaradi več družin/generacij razlikujemo med masnimi in interakcijskimi stanji okusa $\Longrightarrow$ za 3 družine kvarkov:
  3 mešalni koti + 1 CP faza (Cabibbo)

Odprta vprašanja:

1) Nevtrinska masa in 1a) ali se leptonska št. ohranja
2) Povezava med kršitvijo CP in

2a) Odsotnost kršitve CP v močnih interakcijah

$$\mathcal{L}^0_{QCD} = \frac{\alpha}{8\pi} G^{\mu\nu}\tilde G$$
$$\text{t. } \tilde G_{\mu\nu} = \frac{1}{4}\varepsilon_{\mu\nu\alpha\beta}G^{\alpha\beta}$$

$\Rightarrow d_{nevtron} \Rightarrow \theta_{exp} < 10^{-10}$

3) Hierarhična struktura fermionskih mas

Leptoni znotraj SM (nelih)

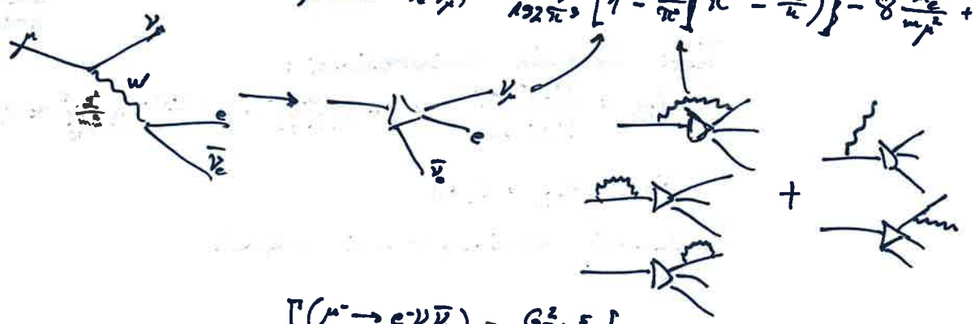
Nabiti leptoni

→ elektron: $m_e = 0{,}5$ MeV; precizni testi QED
→ mion: $m_\mu = 106$ MeV;

$$\Gamma(\mu \to e\bar\nu_e\nu_\mu) = \frac{G_F^2 m_\mu^5}{192\pi^3}$$

Določitev G_F:

$$\Gamma(\mu^- \to e^-\bar\nu_e\nu_\mu) = \frac{G_F^2 m_\mu^5}{192\pi^3}\left[1 - \frac{2}{\pi^2}\left(\pi^2 - \frac{25}{4}\right) - 8\frac{m_e^2}{m_\mu^2} + \frac{4}{5}\frac{m_e^2}{m_W^2} + \dots\right]$$



$$\Gamma(\mu^- \to e^-\nu_\mu\bar\nu_e) = \frac{G_F^2 m_\mu^5}{192\pi^3}\left[1 + 10^{-2}(-4209{,}85 - 187{,}12 + 1{,}02) + \dots\right]$$

$$\Longrightarrow G_F = 1{,}166\,37(2)\cdot 10^{-5}\,\text{GeV}^{-2}$$

Proces ohranja leptonsko

$L_e(e) = L_e(\nu_e) = 1$

$L_\mu(\mu) = L(\ ) = 1$

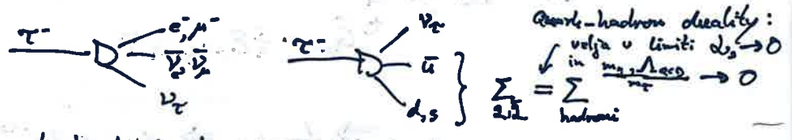
Procesi ki kršijo L_μ, L_e: $\mu^- \to e^-\nu$ in $\mu^- \to e^+e^-e^-$

Eksperimentalno: $B_n(\mu \to e\nu) = \dfrac{\Gamma(\mu\to e\nu)}{\Gamma(\mu\to e\nu\bar\nu)} < 4{,}2\cdot 10^{-13}$

$B_n(\mu \to 3e) < 10^{-12}$

→ lepton tau : $m_\tau = 1{,}78\ \mathrm{GeV}$
odkrit 1975, zelo pozno v SLAC, DESY

razpadi leptona τ:



leptonski kanal — semi-leptonski

$$J_\mu^{-(2)} = V_{ud}\,\bar d_L \gamma_\mu u + V_{cs}\,\bar s_L \gamma_\mu u_L$$

$$\Rightarrow \frac{\Gamma(\mathrm{semilept.})}{\Gamma(\mathrm{lept.})} \simeq \frac{|V_{ud}|^2 + |V_{us}|^2}{2} \cdot N_c = 1{,}5$$

$$\Rightarrow B_n(\tau \to \mu \nu \bar\nu) \simeq B_n(\tau \to e \nu \bar\nu) = \frac{1}{2+N_c} \simeq 0{,}2$$

Experimentalno: $\frac{\Gamma(\mathrm{semilept})}{\Gamma(\mathrm{lept.})} = 1{,}85 \pm 0{,}02$

Popravki QCD in QED:

$$\frac{\Gamma(\mathrm{semi})}{\Gamma(\mathrm{lept.})} = \frac{N_c}{2}\left[1 + \frac{4\alpha_s}{\pi}\log\frac{m_s}{m_\tau} + \dots\right]\cdot\left[1 + \frac{\alpha_s(m_\tau)}{\pi} + \mathcal{O}(\alpha_s^2)\right]$$

$$\simeq 1{,}84$$

$\mathcal{O}(\alpha_s^2)$ znamo; ampak grdo.

Testi leptonske univerzalnosti:

$$R_{\mu/e} = \frac{\Gamma(\tau \to \mu}{\Gamma(\tau \to e} = 1 - 8\left(\frac{m_e}{m_\tau}\right)^2 - 12\left(\frac{m_\mu}{m_\tau}\right)^4 \log\frac{m_\mu}{m_\tau} \simeq 0{,}973$$

Exp: $R_{\mu/e} = 0{,}973$

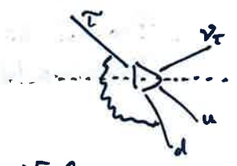
Eksklusivni semi-leptonski razpadi

$$\mathcal{M} = \frac{G_F}{\sqrt{2}} L^\mu H_\mu$$

$\langle \nu_\tau(p') | j^\mu_{\mathrm{lept.}}(0) | \tau(p) \rangle$

$\langle \mathrm{hadroni} | j^\mu_{u,d,s}(0) | 0 \rangle = \langle \mathrm{hadr.} | \cos\theta_c\, \bar d \gamma_\mu u_L + \sin\theta_c\, \bar s \gamma_\mu u_L | 0 \rangle$

$\pi^-, K^-, K^{*-}, \rho^-$



od prej vemo:

$$\langle \pi^-(q) | \bar d \gamma_\mu \gamma_5 u | 0 \rangle = -i\sqrt{2}\, f_\pi q_\mu$$
$$\langle K^-(q) | \bar s \gamma_\mu \gamma_5 u | 0 \rangle = -i\sqrt{2}\, f_K q_\mu$$
$$\langle K^{*-}(q) | \bar s \gamma_\mu u | 0 \rangle = \sqrt{2}\, g_{K^*} \varepsilon^*_\mu(q)$$

$$\Gamma(\tau \to \nu_\tau) = |V_{ud}|^2 G_F^2 m_\tau^3 \frac{\cdot}{\quad} (1 - \frac{m^2}{m_\tau^2})^2$$

$$\Gamma(\tau \to V_2\, \nu_\tau) = |V_{ud}|^2 G_F^2 m_\tau^3 \left(\frac{g_{V2}}{m_{V_2}}\right)^2 m_{V_2}^2 (1 - \frac{m^2}{m^2})^2 (1 + 2\frac{m_{V_2}^2}{m_\tau^2})^1$$

Nevtrini (najprej fenomenološko)

Če so nevtrini brezmasni, potem so degenerirani:
⇒ ni fizikalnega mešanja leptonskih okusov.
⇒ ohranj
⇒ procesi

Experimentalno poznamo pojav nevtrinskih oscilacij (Homestake, 1960):
* manjši flux solarnih nevtrinov od ~~napovedi~~ jedrskih modelov.
* potrditev: → Kamiokande (1998), Sudbury (2001)
⇒ prinanašaj mionskih nevtrinov v primerjavi z elektronskimi
→ oscilacije reaktorskih meritev
→ oscilacije pospeševalniških meritev (LSND, 1998
DUNE ~ 2030)

Mešanje nevtrinskih okusov

→ poglejmo najprej dve masni stanji $|\nu_1\rangle, |\nu_2\rangle \neq$ inter. stanji $|\nu_e\rangle, |\nu_\mu\rangle$

→ Transformacija med bazama

$$\begin{bmatrix} |\nu_e\rangle \\ |\nu_\mu\rangle \end{bmatrix} = \begin{bmatrix} \cos\theta_\nu & \sin\theta_\nu \\ -\sin\theta_\nu & \cos\theta_\nu \end{bmatrix} \begin{bmatrix} |\nu_1\rangle \\ |\nu_2\rangle \end{bmatrix}$$

↑ okrajšava: $\cos\theta_\nu \to c_\nu$, $\sin\theta_\nu \to s_\nu$

↖ časovni razvoj je $|\nu_e(t)\rangle = e^{-iEt}$

→ časovni razvoj interakcijskega stanja:

$$|\nu_e(t)\rangle = |\nu_1(0)\rangle c_\nu^* e^{-iE_1 t} + |\nu_2(0)\rangle s_\nu e^{-iE_2 t}$$

Inverzno pa ob času $t=0$:

$$|\nu_1\rangle = c_\nu |\nu_e\rangle - s_\nu |\nu_\mu\rangle$$
$$|\nu_2\rangle = s_\nu |\nu_e\rangle + c_\nu |\nu_\mu\rangle$$

$$|\nu_e(t)\rangle = c_\nu^2 e^{-iE_1 t} |\nu_e(0)\rangle - s_\nu c_\nu |\nu_\mu(0)\rangle e^{-iE_1 t} + s_\nu^2 e^{-iE_2 t} |\nu_e(0)\rangle + s_\nu c_\nu |\nu_\mu(0)\rangle e^{-iE_2 t}$$
$$= |\nu_e(0)\rangle \left(c_\nu^2 e^{-iE_1 t} + s_\nu^2 e^{-iE_2 t}\right) + |\nu_\mu(0)\rangle \left(e^{-iE_2 t} - e^{-iE_1 t}\right) s_\nu c_\nu$$

Verjetnost

$$|\langle \nu_e(t) | \nu_\mu(0)\rangle|^2$$

Poglejmo

$$\langle \nu_e(t) | \nu_\mu(0)\rangle = \frac{1}{2}\sin(2\theta_\nu)\cdot\left(e^{+iE_2 t} - e^{iE_1 t}\right)$$
$$\langle \nu_e(t) | \nu_\mu(0)\rangle \langle \nu_\mu(0) | \nu_e(t)\rangle = \frac{1}{4}\sin^2(2\theta_\nu)\underbrace{\left(e^{iE_2 t} - e^{iE_1 t}\right)\left(e^{-iE_2 t} - e^{-iE_1 t}\right)}_{1 - e^{-i(E_1 - E_2)t}}$$
$$= 2\cdot 2\sin^2\left(\frac{E_1 - E_2}{2} t\right)$$

$$P(t)_{\nu_\mu \to \nu_e} = \sin^2(2\theta_\nu)\cdot\sin^2\left(\frac{E_1 - E_2}{2} t\right)$$

* $P_{\nu_\mu \to \nu_e}(t) + P_{\nu_\mu \to \nu_e}(t) = 1$

* Nevtrinske mase so majhne v primerjavi z gibalnimi v U-R limiti

$$E_\nu = \sqrt{|\vec{p}|^2 + m_\nu^2} \simeq |\vec{p}| + \frac{m_\nu^2}{2|\vec{p}|} \implies E_2 - E_1 \simeq |\vec{p}| + \frac{m_2^2}{2|\vec{p}|} - \left(|\vec{p}| + \frac{m_1^2}{2|\vec{p}|}\right) = \frac{m_2^2 - m_1^2}{2|\vec{p}|}$$

* čas potovanja od nastanka do detekcije $\implies t = L$

Also: $\sin^2 \frac{E_2 - E_1}{2} t \simeq \sin^2 \frac{\Delta m_\nu^2}{4|\vec{p}|} L \simeq \sin^2 \frac{\Delta m^2}{4E_\nu} L$

$$\boxed{P_{\nu_\mu \to \nu_e}(L) = \sin^2(2\theta_\nu)\sin^2\left(\frac{\Delta m_\nu^2}{4E_\nu} L\right)}$$

To lahko strnemo v dva pogoja za nevtrinske oscilacije:

* interakcijska in masna baza ne sovpadata, oz. $\theta_\nu \neq 0$.

*

→ brezmasni (vsaj 1 more bit masiven)
→ degenerirani

oz. $\Delta m_\nu^2 \neq 0$.

Posplošitev na 3 generacije

Zdaj imamo torej 3 masna in 3 interakcijska stanja

$$\nu_\alpha = \sum_i U_{\alpha i} \nu_i, \quad \alpha = e, \mu, \tau$$

kjer je U 3×3 unitarna matrika.
Od tod tudi 3 mešalni koti in 1 CP faza (v limiti Diracovih nevtrinov)

$$\begin{bmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{bmatrix} = \begin{bmatrix} c_1 & c_1 s_3 & s_1 s_3 \\ -s_1 c_2 & & \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix}$$

Primer:

$$|\nu_e(0)\rangle = c_1 |\nu_1\rangle + c_1 s_3 |\nu_2\rangle + s_1 s_3 |\nu_3\rangle$$

Mjer velja: $E_i^2 = |\vec{p}|^2 + m_i^2$

Po času t imamo:

$$|\nu_e(t)\rangle = c_1 e^{-iE_1 t} |\nu_1\rangle + c_1 s_3 e^{-iE_2 t} |\nu_2\rangle + s_1 s_3 e^{-iE_3 t} |\nu_3\rangle$$

Namesto $P_{\nu_e \to \nu_2}$ si poglejmo konkretno $\nu_e \to \nu_e$:

$$|\langle \nu_e(t) | \nu_e(0)\rangle|^2 = 1 - 2c_1^2 s_1^2 c_3^2 \left[1 - \cos(E_1 - E_2)t\right] - 2c_1^2 s_1^2 s_3^2 \left[1 - \cos(E_1 - E_3)t\right] - 2s_1^2 s_3^2 c_3^2 \left[1 - \cos(E_2 - E_3)t\right]$$

$\frac{m_2 - m_1}{2|\vec{p}|}$

Uvedemo

$$L_{ij} \equiv \frac{2\pi}{E_i - E_j} = \frac{4\pi E}{|m_i^2 - m_j^2|} \simeq 2{,}5\,m \left[\frac{E(MeV)}{(eV)}\right]$$

$$P_{\nu_e \to \nu_e}(d) = 1 - 2\,c\,s\,c\,[1-\cos$$
$$- 2\,c\,s\,s\,[1-\cos$$
$$- 2\,s\,s\,c\,[1-\cos$$

Kiralni nevtrini (v Weilovi bazi)

$$(\psi_L)^c = \left[\tfrac{1}{2}(1-\gamma_5)\psi\right]^c = \tfrac{1}{2}(1-\gamma_5)\psi^c = \tfrac{1}{2}(1-\gamma_5) i\gamma^2\psi^* = i\gamma^2(1+\gamma_5)\psi^* = (\psi^c)_R$$

Majorana fermioni: $\boxed{\psi^c = \psi}$

$\rightarrow$ $C^\dagger = \dots$, $C^* = \dots$ $\Rightarrow$ $C^T = C^{-1} = -C$

$$V(x) = \sum_{\chi=\pm1}\left[a_\chi(p)\,u_\chi(p)\,e^{-ipx} + \underbrace{b_\chi^\dagger(p)\,v_\chi(p)}_{b_\chi^\dagger = a_\chi^\dagger,\quad v_\chi(p) = C(\bar{u}_\chi(p))^T = i\gamma^2 u_\chi^*(p)}\,e^{ipx}\right]$$

$$\mathcal{L}_{Mass} = A\left[\overline{(\psi_L)^c}\,\psi_L + \bar{\psi}_L(\psi_L)^c\right]$$

(1) $(\psi_L)^c = (\psi^c)_R$

(2) $\bar{\psi}_L C(\bar{\psi}_L)^T = \bar{\psi}_L i\gamma^2\gamma^0(\psi_L^T\gamma^0)^T = \bar{\psi}_L i\gamma^2\gamma^0\gamma^0(\psi_L^\dagger)^T$
$= \bar{\psi}_L \tfrac{1}{2}(1+\gamma_5) i\gamma^2\psi^*$ $\qquad \tfrac{1}{2}(1-\gamma_5)\psi^*$
$= [\tfrac{1}{2}(1-\gamma_5)\psi]^\dagger\gamma^0\tfrac{1}{2}(1+\gamma_5) i\gamma^2\psi^* = \psi^\dagger\tfrac{1}{2}(1-\gamma_5)\gamma^0\tfrac{1}{2}(1+\gamma_5) i\gamma^2\psi^*$
$= \tfrac{1}{2}\psi^{*T}\gamma^0(1+\gamma_5) i\gamma^2\psi^*$

(1) $\overline{(\psi_L)^c}\,\psi_L = (i\gamma^2\tfrac{1}{2}(1-\gamma_5)\psi^*)^\dagger\gamma^0\tfrac{1}{2}(1-\gamma_5)\psi$
$= \psi^T\tfrac{1}{2}(1-\gamma_5)(-i\gamma^2\gamma^0)\tfrac{1}{2}(i\gamma_5)\psi$
$= \psi^T(-i\gamma^2\gamma^0)\tfrac{1}{2}(1-\gamma_5)\psi$

$\psi' = e^{i\alpha_M}\psi$

$\Rightarrow \psi'^T i\gamma^0\gamma^2\tfrac{1}{2}(1-\gamma_5)\psi' = e^{2i\alpha_M}\psi^T i\gamma^0\gamma^2(1-\gamma_5)\tfrac{1}{2}\psi$

Majorana ni

Kombinacija diracovih in Maj.

Diracov fermion : $\Psi = \Psi_L + \Psi_R$

$$\Rightarrow \mathcal{L}_D = D\left(\overline{\Psi_L}\Psi_R + \overline{\Psi_R}\Psi_L\right) \equiv D\,\overline{\Psi}\Psi$$

$$\mathcal{L}_{MD} = B\left(\overline{(\Psi_R)^c}\Psi_R + \overline{\Psi_R}(\Psi_R)^c\right)$$

$$\chi \equiv \Psi_L + (\Psi_L)^c$$
$$\omega \equiv \Psi_R + (\Psi_R)^c$$

$$\Rightarrow \mathcal{L}_M = A\,\overline{\chi}\chi \quad \text{in} \quad \mathcal{L}_{MD} = B\,\overline{\omega}\omega.$$

medtem ko $\Psi_L = \frac{1}{2}(1-\gamma_5)\Psi = \frac{1}{2}(1-\gamma_5)\chi$ in $\Psi_L^c = \frac{1}{2}(1+\gamma_5)\Psi^c = \frac{1}{2}(1+\gamma_5)\chi$

$\Psi_R = \frac{1}{2}(1+\gamma_5)\omega$ te $\Psi_R^c = \frac{1}{2}(1-\gamma_5)\omega$

$$\Rightarrow \gamma_5\begin{bmatrix}\Psi\\ \chi\\ \omega\end{bmatrix} = \begin{bmatrix}\Psi'\\ \chi'\\ \omega'\end{bmatrix} = \begin{bmatrix}-\Psi_L + \Psi_R\\ -\Psi_L + (\Psi_L)^c\\ \Psi_R - (\Psi_R)^c\end{bmatrix}$$

$$\mathcal{L}_{DM} = D\,\overline{\Psi_L}\Psi_R + A\,\overline{(\Psi_L)^c}\Psi_L + B\,\overline{(\Psi_R)^c}\Psi_R + \text{h.c.}$$
$$= \frac{1}{2}D\left(\overline{\Psi}\omega + \overline{\omega}\chi\right) + A\,\overline{\chi}\chi + B\,\overline{\omega}\omega$$
$$= \left[\overline{\chi}, \overline{\omega}\right]\cdot\underbrace{\begin{bmatrix}A & \frac{1}{2}D\\ \frac{1}{2}D & B\end{bmatrix}}\begin{bmatrix}\chi\\ \omega\end{bmatrix}$$

$$\det\begin{bmatrix}A-\lambda & \frac{1}{2}D\\ \frac{1}{2}D & B-\lambda\end{bmatrix} = (A-\lambda)(B-\lambda) - \frac{1}{4}D^2 = 0$$

!

$$\lambda_{1,2} = \frac{1}{2}(A+B) \pm \sqrt{\frac{1}{4}(A+B)^2 - AB + \frac{1}{4}D^2}$$
$$= \frac{1}{2}\left[(A+B) \pm \sqrt{(A-B)^2 + D^2}\right]$$

Lastna stanja v spinorskem

$$\eta_1 = \cos\theta_\chi - \sin\theta_\omega$$
$$\eta_2 = \sin\theta_\chi + \cos\theta_\omega$$
$$\begin{bmatrix}c & -s\\ s & c\end{bmatrix}\begin{bmatrix}A & \frac{1}{2}D\\ \frac{1}{2}D & B\end{bmatrix}\begin{bmatrix}c & s\\ -s & c\end{bmatrix} = \begin{bmatrix}\lambda_1 & \\ & \lambda_2\end{bmatrix}$$

$$\Rightarrow \tan(2\theta) = \frac{D}{B-A} \quad ; \quad D = (\lambda_1 - \lambda_2)\sin 2\theta$$
$$A = \lambda_1\cos^2\theta + \lambda_2\sin^2\theta$$
$$B = \lambda_1\sin^2\theta + \lambda_2\cos^2\theta$$

$A = B = 0 \Rightarrow \theta = \frac{\pi}{4}$

Lastni stanji $\underline{(\chi \pm \omega)\frac{1}{\sqrt{2}}}$, lastni vrednosti: $\lambda_{1,2} = \pm\frac{D}{2}$

$$\frac{1}{\sqrt{2}}(\chi+\omega) = \frac{1}{\sqrt{2}}(\Psi_L + \Psi_L^c + \Psi_R + \Psi_R^c) \equiv \xi_1$$
$$\frac{1}{\sqrt{2}}(\chi-\omega) = \frac{1}{\sqrt{2}}(-\Psi_L + \Psi_L^c - \Psi_R + \Psi_R^c) \equiv \xi_2$$